

MCP Specification

MX6432_DB, 254FBGA, 11.5x13x1.0mmt
64GB e.MMC +
32Gb(16Gb*2) DDP LPDDR4X SDRAM

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Datasheet

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Table Of Contents**MCP Specification**

1.0 FEATURES	5
2.0 GENERAL DESCRIPTION	6
3.0 PIN CONFIGURATION	7
4.0 PIN DESCRIPTION	8
5.0 ORDERING INFORMATION	9
6.0 FUNCTIONAL BLOCK DIAGRAM	10
7.0 PACKAGE DIMENSION	11

e.MMC 64GB

1.0 PRODUCT SPECIFICATION	13
1.1 Product Architecture	13
1.2 Operating Conditions	14
1.2.1 Power Supply	14
1.2.2 Temperature	14
1.3 Power Mode	14
1.3.1 Auto Power Saving Mode	14
1.4 Performance	14
2.0 HS400 MODE	15
3.0 NEW eMMC 5.1 FEATURES	16
3.1 Overview	16
3.2 Command Queuing	16
3.2.1 Command Set Description	16
3.2.2 New Response : QSR (Queue Status Register)	16
3.2.3 Send Status : CMD13	16
3.2.4 Mechanism of CMD Queue Operation	17
3.2.5 CMD Queue Register Description	17
3.3 Enhanced Strobe Mode	17
3.4 RPMB Throughput Improve	18
3.5 Secure Write Protection	18
4.0 TECHNICAL NOTES	19
4.1 S/W Algorithm	19
4.1.1 Partition Management	19
4.1.1.1 Enhanced Partition (Area)	19
4.1.2 Boot Operation	20
4.1.3 User Density	21
5.0 REGISTER VALUE	22
5.1 OCR Register	22
5.2 CID Register	22
5.2.1 Product Name Table (In CID Register)	22
5.3 CSD Register	23
5.4 Extended CSD Register	24
6.0 DC PARAMETER	28
6.1 Power Consumption	28
6.1.1 Active Power Consumption During Operation	28
6.1.2 Standby Power Consumption in Auto Power Saving Mode and Standby State	28
6.1.3 Sleep Power Consumption in Sleep State	28
7.0 AC PARAMETER	29
7.1 Timing Parameter	29
7.2 Previous Bus Timing Parameters for DDR52 and HS200 Mode are defined by JEDEC Standard	29
7.3 Bus Timing Specification in HS400 Mode	30
7.3.1 HS400 Device Input Timing	30
7.3.2 HS400 Device Output Timing	31
7.4 Bus Signal Line Load	32
7.5 Bus Signal Levels	33
7.5.1 Open-Drain Mode Bus Signal Level	33
7.5.2 Push-Pull Mode Bus Signal Level eMMC	33

LPDDR4X 32Gb DDP

1.0 LPDDR4X PAD DEFINITION AND DESCRIPTION	35
1.1 Dual channel per die device	35
2.0 FUNCTIONAL DESCRIPTION	36
2.1 LPDDR4X SDRAM Addressing	36
2.2 Simplified LPDDR4X State Diagram	37
2.3 Mode Register Definition	39
2.3.1 Mode Register Assignment and Definition in LPDDR4X SDRAM	39
3.0 TRUTH TABLES	61

3.1	CKE Truth Tables	63
3.2	State Truth Table	64
4.0	ABSOLUTE MAXIMUM DC RATINGS	66
5.0	AC & DC OPERATING CONDITIONS	67
5.1	Recommended DC Operating Conditions	67
5.2	Input Leakage Current	67
5.3	Input/Output Leakage Current	67
5.4	Operating Temperature Range	67
6.0	AC AND DC INPUT/OUTPUT MEASUREMENT LEVELS	68
6.1	1.1V High speed LVCMOS (HS_LLVC MOS)	68
6.1.1	Standard specifications	68
6.1.2	DC electrical characteristics	68
6.1.2.1	LPDDR4X Input Level for CKE	68
6.1.2.2	LPDDR4X Input Level for Reset_n and ODT_CA	68
6.1.3	AC Over/Undershoot	69
6.1.3.1	LPDDR4X AC Over/Undershoot	69
6.2	Differential Input Voltage	70
6.2.1	Differential Input Voltage for CK	70
6.2.2	Peak voltage calculation method	71
6.2.3	Single-Ended Input Voltage for Clock	72
6.2.4	Differential Input Slew Rate Definition for Clock	73
6.2.5	Differential Input Cross Point Voltage for Clock	74
6.2.6	Differential Input Voltage for DQS	75
6.2.7	Peak voltage calculation method	76
6.2.8	Single-Ended Input Voltage for DQS	77
6.2.9	Differential Input Slew Rate Definition for DQS	78
6.3	Differential Input Cross Point Voltage for DQS	79
6.4	Single Ended Output Slew Rate	80
6.5	Differential Output Slew Rate	81
6.6	Overshoot and Undershoot for LVSTL	82
6.7	LPDDR4X Driver Output Timing Reference Load	83
6.8	LVSTL (Low Voltage Swing Terminated Logic) IO System	84
7.0	INPUT/OUTPUT CAPACITANCE	86
8.0	IDD SPECIFICATION PARAMETERS AND TEST CONDITIONS	87
8.1	IDD Measurement Conditions	87
8.2	IDD Specifications	103
8.3	IDD Spec Table	106
9.0	AC AND DC OUTPUT MEASUREMENT LEVELS	108
9.1	Single Ended AC and DC Output Levels	108
9.2	Pull Up/Pull Down Driver Characteristics and Calibration	109
10.0	ELECTRICAL CHARACTERISTICS AND AC TIMING	110
10.1	Clock Specification	110
10.1.1	Definition for tCK(avg) and nCK	110
10.1.2	Definition for tCK(abs)	110
10.1.3	Definition for tCH(avg) and tCL(avg)	110
10.1.4	Definition for tCH(abs) and tCL(abs)	110
10.1.5	Definition for tJIT(per)	110
10.1.6	Definition for tJIT(cc)	111
10.1.7	Definition for tERR(nper)	111
10.1.8	Definition for duty cycle jitter tJIT(duty)	111
10.1.9	Definition for tCK(abs), tCH(abs) and tCL(abs)	111
10.2	Period Clock Jitter	112
10.2.1	Clock period jitter effects on core timing parameters	112
10.2.1.1	Cycle time de-rating for core timing parameters	112
10.2.1.2	Clock Cycle de-rating for core timing parameters	112
10.2.2	Clock jitter effects on Command/Address timing parameters	112
10.2.3	Clock jitter effects on Read timing parameters	113
10.2.3.1	tRPRE	113
10.2.3.2	tLZ(DQ), tHZ(DQ), tDQSCK, tLZ(DQS), tHZ(DQS)	113
10.2.3.3	tQSH, tQSL	113
10.2.3.4	tRPST	113
10.2.4	Clock jitter effects on Write timing parameters	113
10.2.4.1	tDS, tDH	113
10.2.4.2	tDSS, tDSH	113
10.2.4.3	tDQSS	114
10.3	LPDDR4X Refresh Requirement	114
10.4	AC Timing	115
10.5	CA Rx Voltage and Timing	121
10.6	DRAM Data Timing	124
10.7	DQ Rx Voltage And Timing	127

1.0 FEATURES

<Common>

- Operating Temperature : -25°C ~ 85°C
- Package : 254Ball FBGA Type - 11.5mm x 13mm x 1.0mm
0.5mm ball pitch

<eMMC>

- embedded MultiMediaCard Ver. 5.1 compatible.
- SAMSUNG eMMC supports features of eMMC5.1 which are defined in JEDEC Standard.
 - Major Supported Features: HS400, Field Firmware Update, Cache, Command Queuing, Enhanced Strobe Mode, Secure Write Protection, Partition types.
 - Non-supported Features: Large Sector Size (4KB).
- Backward compatibility with previous MultiMediaCard system specification (1bit data bus, multi-eMMC systems).
- Data bus width: 1bit (Default), 4bit and 8bit.
- MMC I/F Clock Frequency: 0 ~ 200MHz
MMC I/F Boot Frequency: 0 ~ 52MHz
- Temperature: Operation (-25°C ~ 85°C), Storage without operation (-40°C ~ 85°C)
- Power: Memory power → V_{CC} (2.7V ~ 3.6V), Interface power → V_{CCQ} (1.70V ~ 1.95V or 2.7V ~ 3.6V)

<LPDDR4X SDRAM>

- Double-data rate architecture; two data transfers per clock cycle
- Bidirectional data strobes (DQS_t, DQS_c), These are transmitted/received with data to be used in capturing data at the receiver
- Differential clock inputs (CK_t and CK_c)
- Differential data strobes (DQS_t and DQS_c)
- Commands & addresses entered positive CK edges; data and data mask referenced to both edges of DQS
- 2channel composition per die
- 8 internal banks for each channel
- DMI Pin : DBI (Data Bus Inversion) when normal write and read operation, Data mask (DM) for masked write when DBI off
 - Counting # of DQ's 1 for masked write when DBI on
- Burst Length: 16, 32 (OTF)
- Burst Type: Sequential
- Read & Write latency : Refer to Table 13.4 AC Timing
- Auto Precharge option for each burst access
- Configurable Drive Strength
- Refresh and Self Refresh Modes
- Partial Array Self Refresh and Temperature Compensated Self Refresh
- Write Leveling
- CA Calibration
- Internal VREF and VREF training
- FIFO based write/read training
- MPC (Multi Purpose Command)
- LVSTLE (Low Voltage Swing Terminated Logic Extension) IO
- VDD1/VDD2/VDDQ : 1.8V/1.1V/0.6V
- VSSQ Termination
- No DLL : CK to DQS is not synchronized
- Edge aligned data output, write training for data input center align
- Refresh rate : 3.9us

2.0 GENERAL DESCRIPTION

The KMDC6001DM is a Multi Chip Package Memory which combines 64GB eMMC and 32Gb(16Gb*2) DDP LPDDR4X SDRAM.

SAMSUNG eMMC is an embedded MMC solution designed in a BGA package form. eMMC operation is identical to a MMC device and therefore is a simple read and write to memory using MMC protocol v5.1 which is a industry standard.

eMMC consists of NAND flash and a MMC controller. 3V supply voltage is required for the NAND area (V_{CC}) whereas 1.8V or 3V dual supply voltage (V_{CCQ}) is supported for the MMC controller. SAMSUNG eMMC supports HS400 in order to improve sequential bandwidth, especially sequential read performance.

There are several advantages of using eMMC. It is easy to use as the MMC interface allows easy integration with any microprocessor with MMC host.

Any revision or amendment of NAND is invisible to the host as the embedded MMC controller insulates NAND technology from the host.

This leads to faster product development as well as faster times to market.

The embedded flash management software or FTL(Flash Transition Layer) of eMMC manages Wear Leveling, Bad Block Management and ECC.

The FTL supports all features of the Samsung NAND flash and achieves optimal performance.

LPDDR4X-SDRAM is a high-speed synchronous DRAM device internally configured with 2 channels. Dual channel is comprised of 8-banks with from 2Gb to 16Gb per channel density. The configuration for channel density that is greater than 16Gb is still TBD.

This device contains the following number of bits:

Dual-channel SDRAM devices contain the following number of bits:

16Gb has 17,179,869,184 bits

LPDDR4X devices use a 2 or 4 clocks architecture on the Command/Address (CA) bus to reduce the number of input pins in the system. The 6-bit CA bus contains command, address, and bank information. Each command uses 1, 2 or 4 clock cycle, during which command information is transferred on the positive edge of the clock. See command truth table for details.

These devices use a double data rate architecture on the DQ pins to achieve high speed operation. The double data rate architecture is essentially an 16n prefetch architecture with an interface designed to transfer two data bits per DQ every clock cycle at the I/O pins. A single read or write access for the LPDDR4X SDRAM effectively consists of a single 16n-bit wide, one clock cycle data transfer at the internal DRAM core and eight corresponding n-bit wide, one-half-clock-cycle data transfers at the I/O pins. Read and write accesses to the LPDDR4X SDRAMs are burst oriented; accesses start at a selected location and continue for a programmed number of locations in a programmed sequence. Accesses begin with the registration of an Activate command, which is then followed by a Read, Write or Mask Write command. The address and BA bits registered coincident with the Activate command are used to select the row and the Bank to be accessed. The address bits registered coincident with the Read, Write or Mask Write command are used to select the Bank and the starting column location for the burst access.

Prior to normal operation, the LPDDR4X SDRAM must be initialized. The following section provides detailed information covering device initialization, register definition, command description and device operation.

The KMDC6001DM suitable for use in data memory of mobile communication system to reduce not only mount area but also power consumption. This device is available in 254-ball FBGA Type.

3.0 PIN CONFIGURATION

254Ball																		
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
A	DNU	DNU	DQ0_A	VDD1	VDD2	VDDQ	VDDQ	VDD2	VDD1				VDDQ	VDDQ	VDD1	VDD1	DNU	DNU
B	DNU		DQ1_A	VSS	VDDQ	VSS	DQ4_A	VSS	VDD2				VDD2	VDD2	VDD1	ZQ0_A		DNU
C			DQ2_A	VSS	VSS	DQ5_A	VSS	DQ7_A	DQS0_t_A				CA2_A	VSS	CA5_A	ZQ1_A		
D			DQ3_A	VSS	DMI0_A	VSS	DQ6_A	VSS	DQS0_c_A				CA3_A	VSS	VSS	NC		
E													CA4_A	VSS	CS0_A	CKE0_A		
F													CA1_A	VSS	CS1_A	CKE1_A		
G			DQ13_A	VSS	VSS	VSS	VDD2	VDD2	VDD2				VSS	CA0_A	VSS	CK_c_A		
H			DMI1_A	VSS	VDDQ	DQ14_A	VSS	DQ15_A	VDDQ				VSS	NC	VSS	CK_t_A		
J			DQ11_A	VDDQ	VDDQ	VSS	DQ12_A	VDDQ	DQS1_c_A				ODT_A ^{1), 2)}	NC	VCCQ	VCCQ	VCCQ	
K		VDD2	DQ10_A	VSS	DQ8_A	DQ9_A	VSS	VSS	DQS1_t_A				VSS_m	VSS_m	VCCQ	VSS_m	NC	
L							VDD2	VDD2	VDD2				VSS_m	DAT7	DAT6	VSS_m	VSS_m	VDDI
M			VSF1	VSF3	VSF5	VSF7	VSF9	VSS_m	CMD				DS	VSS_m	VSS_m	DAT1	DAT4	VCC
N			VSF2	VSF4	VSF6	VSF8	NC	VSS_m	RST_n				VSS_m	DAT2	DAT5	VSS_m	VSS_m	VCC
P							VDD2	VDD2	VDD2				CLK	VSS_m	VSS_m	DAT3	DAT0	VCC
R		VDD2	DQ10_B	VSS	DQ8_B	DQ9_B	VSS	VSS	DQS1_t_B				VCCQ	VCCQ	VSS_m	VSS_m	VSS_m	
T			DQ11_B	VDDQ	VDDQ	VSS	DQ12_B	VDDQ	DQS1_c_B				ODT_B ^{1), 2)}	NC	VCCQ	VCCQ	NC	
U			DMI1_B	VSS	VDDQ	DQ14_B	VSS	DQ15_B	VDDQ				VSS	NC	VSS	CK_t_B		
V			DQ13_B	VSS	VSS	VSS	VDD2	VDD2	VDD2				VSS	CA0_B	VSS	CK_c_B		
W													CA1_B	VSS	CS1_B	CKE1_B		
Y													CA4_B	VSS	CS0_B	CKE0_B		
AA			DQ3_B	VSS	DMI0_B	VSS	DQ6_B	VSS	DQS0_c_B				CA3_B	VSS	VSS	RESET_n		
AB			DQ2_B	VSS	VSS	DQ5_B	VSS	DQ7_B	DQS0_t_B				CA2_B	VSS	CA5_B	NC		
AC	DNU		DQ1_B	VSS	VDDQ	VSS	DQ4_B	VSS	VDD2				VDD2	VDD2	VDD1	NC		DNU
AD	DNU	DNU	DQ0_B	VDD1	VDD2	VDDQ	VDDQ	VDD2	VDD1				VDDQ	VDDQ	VDD1	VDD1	DNU	DNU

254FBGA: Top View (Ball Down)

	LPDDR4X Channel A		LPDDR4X Channel B
	DRAM Power		e.MMC
	ODT		Ground
	ZQ		Reset, NC, RFU DNU, VSF

NOTE :

- 1) ODT(CA)_x balls are wired to ODT(CA)_x pads of Rank 0 DRAM die. ODT(CA)_x pads for other ranks (if present) are disabled in the package.
 2) For the LPDDR4X, ODT(CA)_Bond pad is ignored. ODT pin should be connect to VDD2 or VSS.

4.0 PIN DESCRIPTION

Pin Name	Pin Function (eMMC)
DAT0 ~ DAT7	Data Inputs/Outputs
CLK	Clock
DS	Data Strobe
CMD	Command
RST_n	Reset
VCC	Power Supply for Flash
VCCQ	Power Supply for Controller
VDDI	External Capacitance for internal power stability
VSS_m	Ground for Controller / Flash

Pin Name	Pin Function
NC	Not Connected
DNU	Do Not Use
VSF	Vendor Specific Function
RFU	Reserved For Use

Pin Name	Pin Function (LPDDR4X)
CK_c, CK_t	System Differential Clock
CKE[1:0]	Clock Enable
CS[1:0]	Chip Select
CA[5:0]	DDR Command / Address Inputs
DMI[1:0]	Input Data Inversion
DQS[1:0]_t	Data Strobe Bi-directional
DQS[1:0]_c	Data Strobe Complementary
DQ[15:0]	Data Inputs / Outputs
VDD1	Core Power Supply 1
VDD2	Core Power Supply 2
VDDQ	I/O Power Supply
VSS	Ground
ZQ[1:0]	Reference Pin for Output Drive Strength Calibration
ODT	On die termination
RESET_n	RESET

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5.0 ORDERING INFORMATION

KM D C 6 0 0 1D M - B 625

KM : Memory eMCP

Device Type

D : eMMC + LPDDR4X + LPDDR4X

eMMC Density & VCC & Org.

C : 64GB, 128G*4, MLC, 3.3V, 1.8V~3.3V, x8

LSI

6 : Vincent

NOR Flash Density & VCC & Org.

0 : NONE

625 :
e.MMC = 5ns(4)
LPDDR4X = 0.468ns(2)

Package
B : FBGA(OSP)

Generation
M : 1'st Generation

DRAM I/F & Density & VCC & Org.
1D : LPDDR4X*2, 16Gb*2, 1.8V, 0.6V, x32, 2CS/2CKE

NAND/OneNAND Density & VCC & Org
0 : NONE

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6.0 FUNCTIONAL BLOCK DIAGRAM

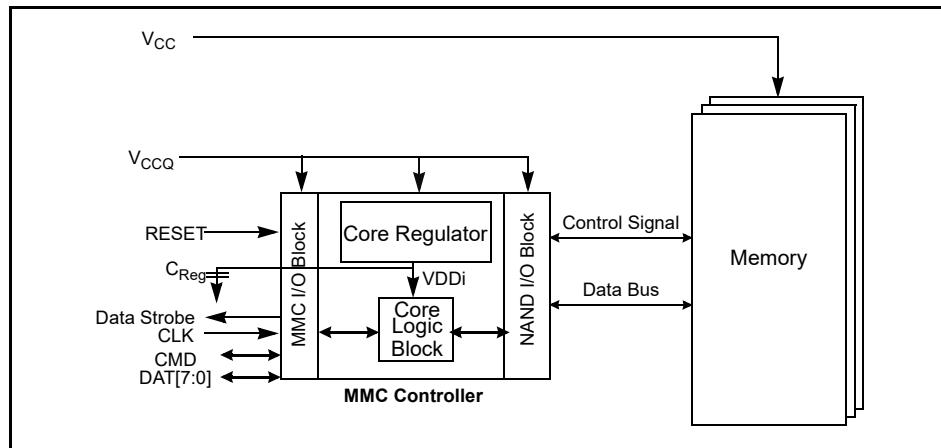


Figure 1. eMMC Block Diagram

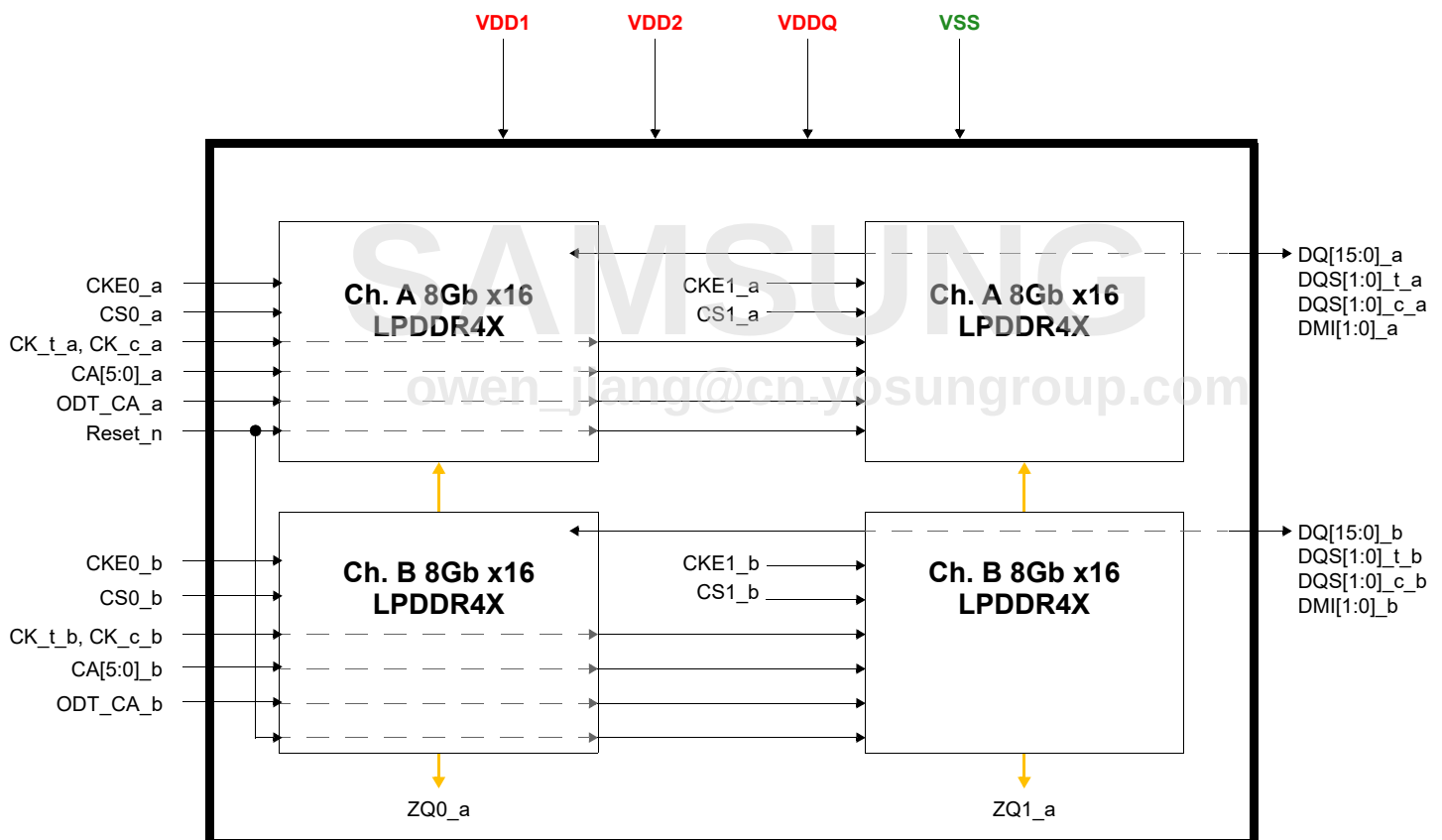
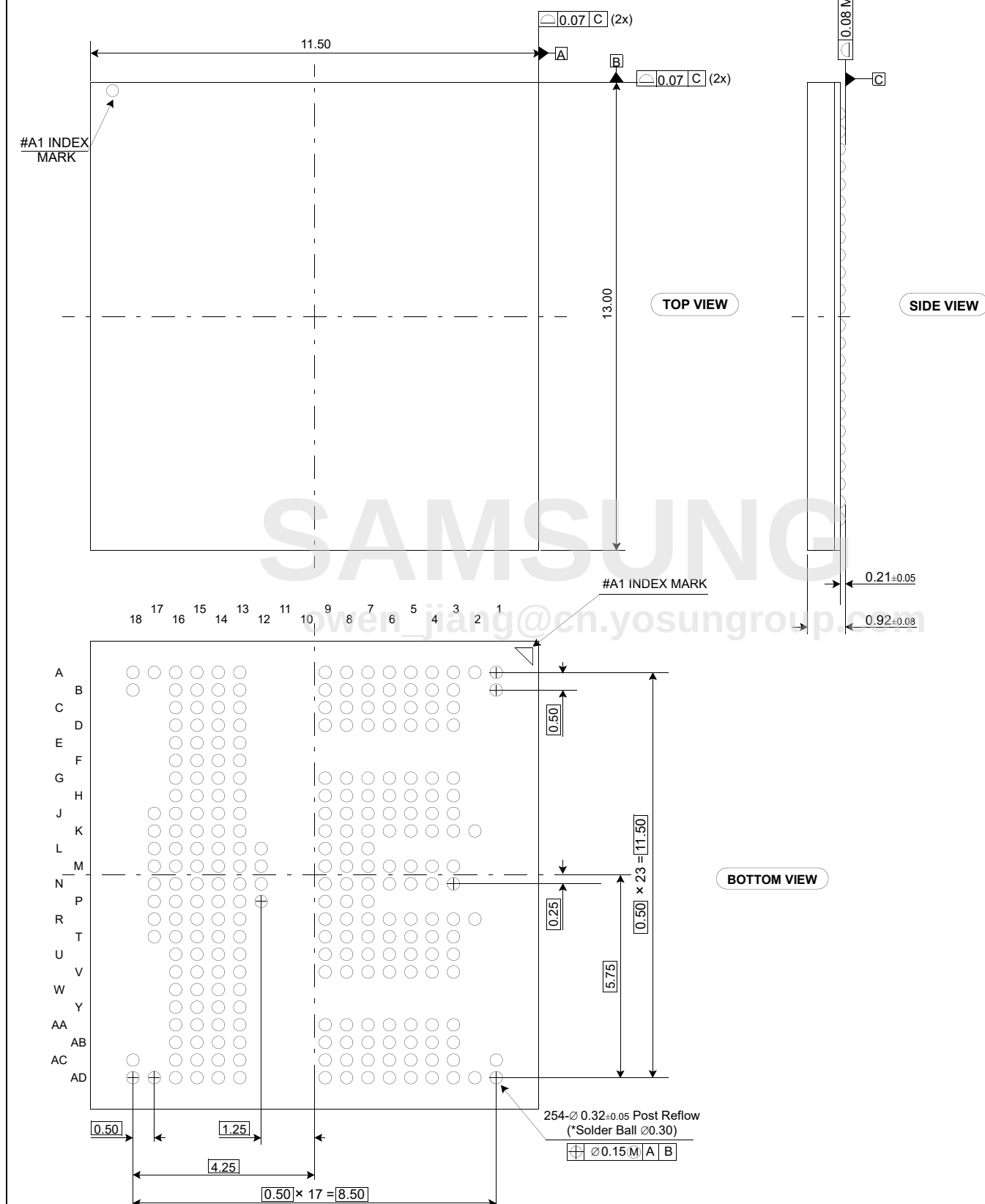


Figure 2. LPDDR4X Block Diagram

7.0 PACKAGE DIMENSION

Units:millimeters



CONFIDENTIAL

e.MMC 64GB

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1.0 PRODUCT SPECIFICATION

1.1 Product Architecture

eMMC consists of NAND Flash and Controller. V_{CCQ} is for Controller power and V_{CC} is for NAND Flash power.

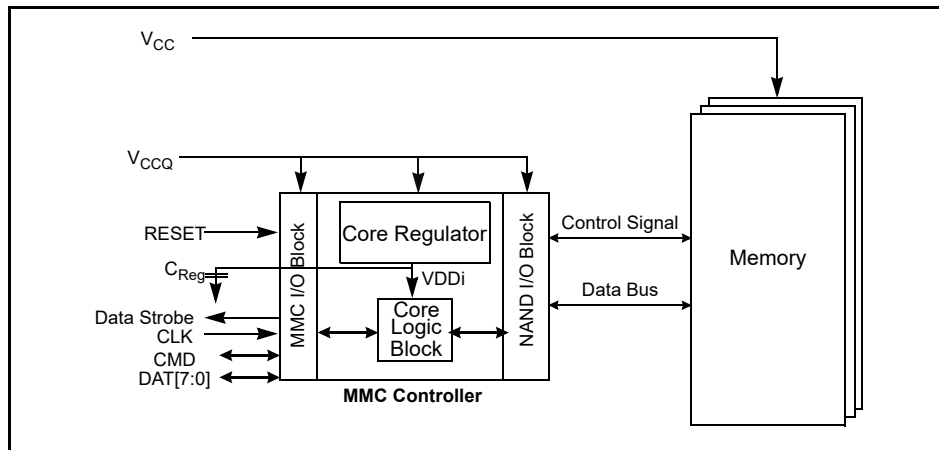


Figure 1. eMMC Block Diagram

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1.2 Operating Conditions

1.2.1 Power Supply

[Table 1] Power Supply Parameters

Parameter	Symbol	Min	Max	Unit
VCC Supply Voltage	V_{CC}	2.7	3.6	V
VCCQ Supply Voltage	V_{CCQ}	1.70	1.95	V
		2.7	3.6	V
VSS Supply Voltage	V_{SS}	-0.5	0.5	V

1.2.2 Temperature

The operating temperature is the eMMC case surface temperature on the center of top side of the case.

[Table 2] Temperature Condition

Condition	Min	Max	Unit
Operation	-25	85	°C
Storage without Operation	-40	85	°C

1.3 Power Mode

1.3.1 Auto Power Saving Mode

If host does not issue any command during a certain duration (1ms), after previously issued command is completed, the device enters "Power Saving mode" to reduce power consumption.

At this time, commands arriving at the device while it is in power saving mode will be serviced in normal fashion.

[Table 3] Auto Power Saving Mode and Exit

Mode	Enter Condition	Escape Condition
Auto Power Saving Mode	When previous operation which came from Host is completed and no command is issued during a certain time.	If Host issues any command

[Table 4] Auto Power Saving Mode and Sleep Mode

Mode	Auto Power Saving Mode	Sleep Mode
VCC Power	ON	OFF
Time to Sleep	< 1ms	< 1ms

1.4 Performance

[Table 5] Performance

Density	Sequential Read	Sequential Write	Unit
64GB	330	200	MB/s

NOTE :

Test Condition : Bus width x8, HS400, 512KB data transfer, w/o file system overhead, measured on Samsung's internal board.

2.0 HS400 MODE

eMMC 5.1 product supports high speed DDR interface timing mode up to 400MB/s at 200MHz with 1.8V I/O supply.

HS400 mode supports the following features:

- DDR Data sampling method
- CLK frequency up to 200MHz DDR (up to 400Mbps)
- Only 8-bits bus width available
- Signaling levels of 1.8V
- Five selectable Drive Strength (refer to the table below)

[Table 6] I/O Driver Strength Types

Driver Type Values	HS200 & HS400 Support	Nominal Impedance	Approximated Driving Capability Compared to Type-0	Remark
0	Mandatory	50Ω	x1	Default Driver Type. Supports up to 200MHz operation.
1	Optional	33Ω	x1.5	Supports up to 200MHz Operation.
2	Optional	66Ω	x0.75	The weakest driver that supports up to 200MHz operation.
3	Optional	100Ω	x0.5	For low noise and low EMI systems. Maximal operating frequency is decided by Host design.
4	Optional	40Ω	x1.2	Supports up to 200MHz DDR operation.

NOTE :

Support of Driver Type-0 is default for HS200 & HS400 Device, while supporting Driver types 1~ 4 are optional for HS200 & HS400 Device.

[Table 7] Device Types (EXT_CSD Register : DEVICE_TYPE [196])

Bit	Device Type	Support
7	HS400 Dual Data Rate eMMC @ 200 MHz - 1.2V I/O	No
6	HS400 Dual Data Rate eMMC @ 200 MHz - 1.8V I/O	Yes
5	HS200 Single Data Rate eMMC @ 200 MHz - 1.2V I/O	No
4	HS200 Single Data Rate eMMC @ 200 MHz - 1.8V I/O	Yes
3	High-Speed Dual Data Rate eMMC @ 52MHz - 1.2V I/O	No
2	High-Speed Dual Data Rate eMMC @ 52MHz - 1.8V or 3V I/O	Yes
1	High-Speed eMMC @ 52MHz - at rated device voltage(s)	Yes
0	High-Speed eMMC @ 26MHz - at rated device voltage(s)	Yes

[Table 8] Extended CSD Revisions (EXT_CSD Register : EXT_CSD_REV [192])

EXT_CSD_REV	Extended CSD Revision	Value
255-8	Reserved	
8	Revision 1.8 (for MMC V5.1)	0x08
7	Revision 1.7 (for MMC V5.0)	
6	Revision 1.6 (for MMC V4.5, V4.51)	
5	Revision 1.5 (for MMC V4.41)	
4	Revision 1.4 (Obsolete)	
3	Revision 1.3 (for MMC V4.3)	
2	Revision 1.2 (for MMC V4.2)	
1	Revision 1.1 (for MMC V4.1)	
0	Revision 1.0 (for MMC V4.0)	

[Table 9] High Speed Timing Interface Values (EXT_CSD Register : HS_TIMING [185])

Value	Timing Interface	Support
0x0	Selecting backwards compatibility interface timing	Yes
0x1	High Speed	Yes
0x2	HS200	Yes
0x3	HS400	Yes

3.0 NEW eMMC 5.1 FEATURES

3.1 Overview

[Table 10] New Features List for Device Types

Function	JEDEC Standard	Support
Cache Flushing Report	Mandatory	Yes
Background operation control	Mandatory	Yes
Command Queuing	Optional	Yes
Enhanced Strobe	Optional	Yes
Cache Barrier	Optional	No
RPMB Throughput improve	Optional	Yes
Secure Write Protection	Optional	Yes

3.2 Command Queuing

To facilitate command queuing in eMMC, the device manages an internal task queue that the host can queue during data transfer tasks.

Every task is issued by the host and initially queued as pending. The device works to prepare pending tasks for execution. When a task is ready for execution, its state changes to "ready for execution".

The host tracks the state of all queued tasks and may order the execution of any task, marked as "ready for execution", by sending a command indicating its Task ID. The device executes the data transfer transaction after receiving the execute command (CMD46/CMD47).

3.2.1 Command Set Description

[Table 11] Command Queue

CMD	Type	Argument	Resp	Abbreviation	Command Description
CMD44	ac	[31] Reliable Write Request [30] DAT_DIR - "0" write / "1" read [29] tag request [28:25] context ID [24] forced programming [23] Priority: "0" simple / "1" high [20:16] Task ID [15:0] number of blocks	R1	QUEUED_TASK_PARAMS	Define direction of operation (Read or Write) and Set high priority CMD Queue with Task ID.
CMD45	ac	[31:0] Start block address	R1	QUEUED_TASK_ADDRESS	Indicate data address for Queued CMD.
CMD46	adtc	[20:16] Task ID	R1	EXECUTE_READ_TASK	(Read) Transmit the requested number of data blocks.
CMD47	adtc	[20:16] Task ID	R1	EXECUTE_WRITE_TASK	(Write) Transmit the requested number of data blocks.
CMD48	ac	[20:16] Task ID [3:0] TM op-code	R1b	CMDQ_TASK_MGMT	Reset a specific task or entire queue. [20:16] when TM op-code = 2h these bits represent Task ID. When TM op-code = 1h these bits are reserved.

3.2.2 New Response : QSR (Queue Status Register)

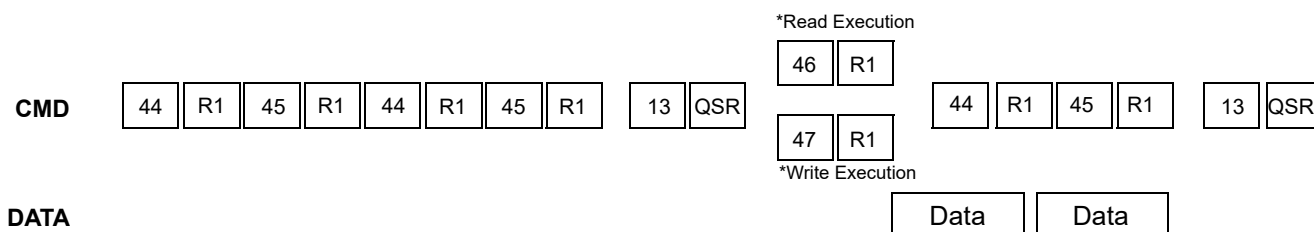
The 32-bit Queue Status Register (QSR) carries the state of tasks in the queue at a specific point in time. The host has read access to this register through device response to SEND_STATUS command (CMD13 with bit[15]="1", R1's argument will be the 32-bit Queue Status Register (QSR). Every bit in the QSR represents the task whose ID corresponds to the bit index. If bit QSR[i] = "0", then the queued task with a Task ID i is not ready for execution. The task may be queued and pending, or the Task ID is unused. If bit QSR[i] = "1", then the queued task with Task ID i is ready for execution.

3.2.3 Send Status : CMD13

CMD13 for reading the Queue Status Register (QSR) by the host. If bit[15] in CMD13's argument is set to 1, then the device shall send an R1 Response with the QSR instead of the Device Status. There is still legacy CMD13 with R1 response.

3.2.4 Mechanism of CMD Queue Operation

Host issues CMD44 with Task ID number, Sector, Count, Direction, Priority to the device followed by CMD45 and host checks the Queue Status check with CMD13 [15]bits to 1. After that host issues CMD46 for Read or CMD47 for write During CMD queue operation, CMD44/CMD45 is able to be issued at anytime when the CMD line is not in use.



3.2.5 CMD Queue Register Description

Configuration and capability structures shall be added to the EXT_CSD register, as described below.

[Table 12] CMD Queuing Support (EXT_CSD Register : CMDQ_SUPPORT [308])

Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
Reserved							CMDQ_SUPPORT

This field indicates whether the device supports command queuing or not.

Bit[7:1]: Reserved

Bit[0]: Command queuing support

- 0x0: CMD Queue function is not supported.
- 0x1: CMD Queue function is supported.

[Table 13] Command Queue Mode Enable (EXT_CSD Register : CMDQ_MODE_EN [15])

Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
Reserved							CMDQ_MODE_EN

This field is used by the host enable command queuing.

Bit[7:1]: Reserved

Bit[0]: Command queuing enable

- 0x0: Queue function is not enabled.
- 0x1: Queue function is enabled.

[Table 14] CMD Queuing Depth (EXT_CSD Register : CMDQ_DEPTH [307])

Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
Reserved				N			

This field is used to calculate the depth of the queue supported by the device.

Bit[7:5]: Reserved

Bit[4:0]: N, a parameter used to calculate the Queue Depth of task queue in the device.

- Queue Depth = N+1.

3.3 Enhanced Strobe Mode

This product supports Enhanced Strobe in HS400 mode and refer to the details as described in eMMC5.1 JEDEC standard section 6.15.9.

3.4 RPMB Throughput Improve

[Table 15] Enhanced RPMB Reliable Write (Related in EXT_CSD Register : WR_REL_PARAM [166])

Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
Reserved			EN_RPMB_REL_WR	Reserved	EN_REL_WR	Reserved	HS_CTRL_REL

Bit[4]: EN_RPMB_REL_WR(R)

- **0x0**: RPMB transfer size is either 256B (single 512B frame) or 512B (Two 512B frame).
- **0x1**: RPMB transfer size is either 256B (single 512B frame), 512B (Two 512B frame) or 8KB (Thirthy two 512B frames).

3.5 Secure Write Protection

Configuration and capability structures shall be added to the EXT_CSD register and Authenticated Device Configuration Area as described below.

[Table 16] Secure Write Protection (EXT_CSD : SECURE_WP_INFO [211])

Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
Reserved						SECURE_WP_EN_STATUS	SECURE_WP_SUPPORT

Bit[7:2]: Reserved

Bit[1]: SECURE_WP_EN_STATUS(R)

- **0x0**: Legacy Write Protection mode.
- **0x1**: Secure Write Protection mode.

Bit[0]: SECURE_WP_SUPPORT(R)

- **0x0**: Secure Write Protection is NOT supported by this device.
- **0x1**: Secure Write Protection is supported by this device.

[Table 17] Authenticated Device Configuration Area[1] : SECURE_WP_MODE_ENABLE

Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
Reserved							SECURE_WP_EN

Bit[7:1]: Reserved

Bit[0]: SECURE_WP_EN (R/W/E)

The default value of this field is 0x0.

- **0x0**: Legacy Write Protection mode, i.e., TMP_WRITE_PROTECT[12], PERM_WRITE_PROTECT[13] is updated by CMD27. USER_WP[171], BOOT_WP[173] and BOOT_WP_STATUS[174] are updated by CMD6.
- **0x1**: Secure Write Protection mode. The access to the write protection related EXT_CSD and CSD fields depends on the value of SECURE_WP_P_MASK bit in SECURE_WP_MODE_CONFIG field.

[Table 18] Authenticated Device Configuration Area[2] : SECURE_WP_MODE_CONFIG

Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
Reserved							SECURE_WP_MASK

Bit[7:1]: Reserved

Bit[0]: SECURE_WP_MASK (R/W/E_P)

The default value of this field is 0x0.

- **0x0**: Disabling updating WP related EXT_CSD and CSD fields. CMD27 (Program CSD) will generate generic error for setting TMP_WRITE_PROTECT[12], PERM_WRITE_PROTECT[13]. CMD6 for updating USER_WP[171], BOOT_WP[173] and BOOT_WP_STATUS[174] generates SWITCH_ERROR. If a force erase command is issued, the command will fail (Device stays locked) and the LOCK_UNLOCK_FAILED error bit will be set in the status register. If CMD28 or CMD29 is issued, then generic error will be occurred. Power-on Write Protected boot partitions will keep protected mode after power failure, H/W reset assertion and any CMD0 reset. The device keeps the current value of BOOT_WP_STATUS in the EXT_CSD register to be same after power cycle, H/W reset assertion, and any CMD0 reset.
- **0x1**: Enabling updating WP related EXT_CSD and CSD fields. I.e TMP_WRITE_PROTECT[12], PERM_WRITE_PROTECT[13], USER_WP[171], BOOT_WP[173] and BOOT_WP_STATUS[174] are accessed using CMD6, CMD8 and CMD27. If a force erase command is issued and accepted, then ALL THE DEVICE CONTENT WILL BE ERASED including the PWD and PWD_LEN register content and the locked Device will get unlocked. If a force erase command is issued and power-on protected or a permanently-write-protected write protect groups exist on the device, the command will fail (Device stays locked) and the LOCK_UNLOCK_FAILED error bit will be set in the status register. An attempt to force erase on an unlocked Device will fail and LOCK_UNLOCK_FAILED error bit will be set in the status register. Write Protection is applied to the WPG indicated by CMD28 with the WP type indicated by the bit[2] and bit[0] of USER_WP[171]. All temporary WP Groups and power-on Write Protected boot partitions become writable/erasable temporarily which means write protect type is not changed. All power-on and permanent WP Groups in user area will not become writable/erasable temporarily. Those temporarily writable/erasable area will become write protected when this bit is cleared to 0x0 by the host or when there is power failure, H/W reset assertion and any CMD0 reset. The device keeps the current value of BOOT_WP_STATUS in the EXT_CSD register to be same after power cycle, H/W reset assertion, and any CMD0 reset.

4.0 TECHNICAL NOTES

4.1 S/W Algorithm

4.1.1 Partition Management

The device initially consists of two Boot Partitions and RPMB Partition and User Data Area.

The User Data Area can be divided into four General Purpose Area Partitions and User Data Area partition. Each of the General Purpose Area partitions and a section of User Data Area partition can be configured as enhanced partition.

4.1.1.1 Enhanced Partition (Area)

SAMSUNG eMMC adopts Enhanced User Data Area as SLC Mode. Therefore when master adopts some portion as enhanced user data area in User Data Area, that area occupies double size of original set up size.

(ex> if master set 1MB for enhanced mode, total 3MB user data area is needed to generate 1MB enhanced area).

Max Enhanced User Data Area size is defined as $(MAX_ENH_SIZE_MULT \times HC_WP_GRP_SIZE \times HC_ERASE_GRP_SIZE \times 512K \text{ Bytes})$.

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4.1.2 Boot Operation

Device supports not only boot mode but also alternative boot mode. Device supports high speed timing and dual data rate during boot.

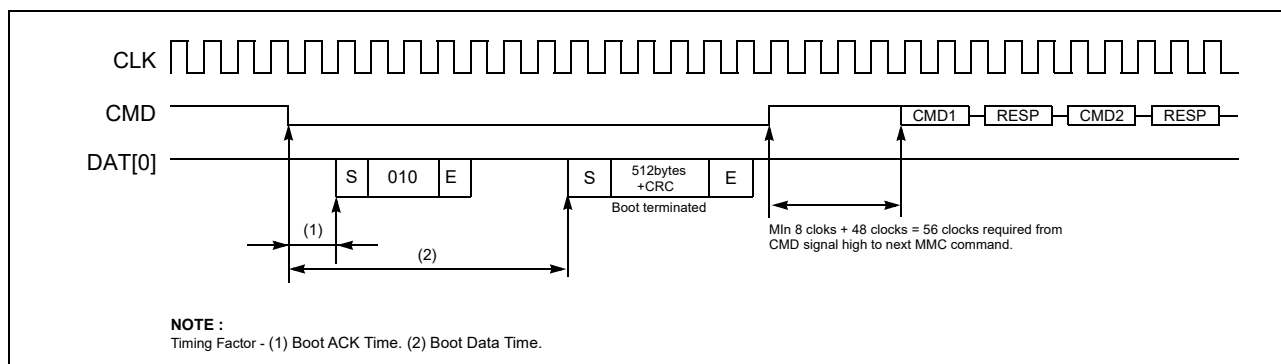


Figure 2. embedded MultiMediaCard State Diagram (Boot Mode)

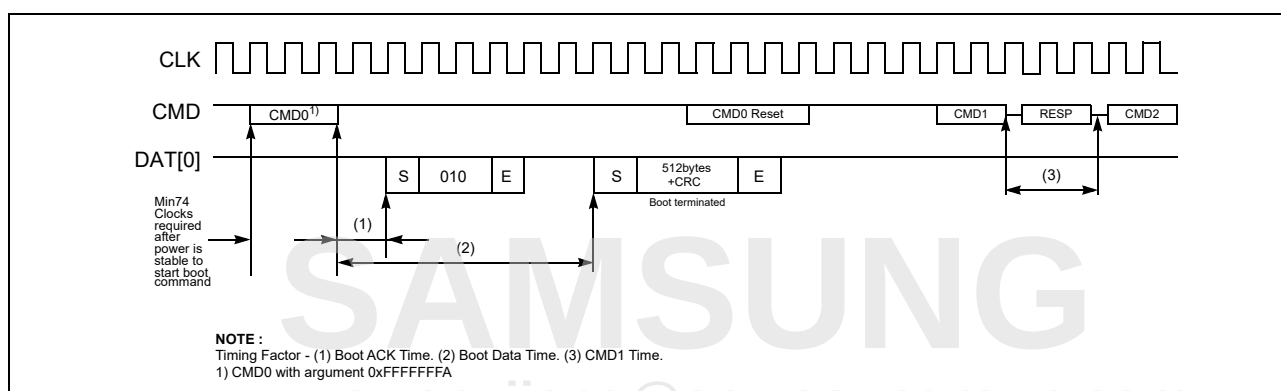


Figure 3. embedded MultiMediaCard State Diagram (Alternative Boot Mode)

[Table 19] Boot ACK, Boot Data and Initialization Time

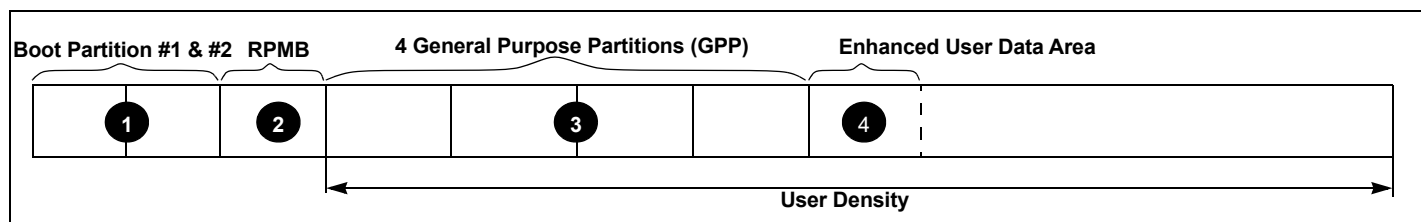
Timing Factor	Value
(1) Boot ACK Time	< 50 ms
(2) Boot Data Time	< 100 ms
(3) Initialization Time ¹⁾	< 3 secs

NOTE :

1) This initialization time includes partition setting, Please refer to INI_TIMEOUT_AP in Chapter 5.4 Extended CSD Register.
Normal initialization time (without partition setting) is completed within 1sec.

4.1.3 User Density

Total User Density depends on device type. For example, 32MB in the SLC Mode requires 64MB in MLC. This results in decreasing of user density.



[Table 20] Capacity According to Partition Size

Density	Boot Partition 1	Boot Partition 2	RPMB	Unit
64GB	4,096	4,096	4,096	KB

[Table 21] Maximum Enhanced Partition Size

Density	Enhanced Partition Size (Max.)	Unit
64GB	31,264,342,016	Byte

[Table 22] User Density Size

Density	User Density Size	Unit
64GB	62,537,072,640	Byte

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5.0 REGISTER VALUE

5.1 OCR Register

The 32-bit operation conditions register stores the V_{CCQ} voltage profile of the eMMC. In addition, this register includes a status information bit. This status bit is set if the eMMC power up procedure has been finished. The OCR register shall be implemented by all eMMC.

[Table 23] OCR Register

OCR Bit	VCCQ Voltage Window ¹⁾	Register Value
[6:0]	Reserved	000 0000b
[7]	1.70 - 1.95V	1b
[14:8]	2.0 - 2.6V	000 0000b
[23:15]	2.7 - 3.6V	1 1111 1111b
[28:24]	Reserved	0 0000b
[30:29]	Access Mode	00b (byte mode), 10b (sector mode) - [*Higher than 2GB only]
[31]	eMMC power up status bit (busy) ²⁾	

NOTE :

1) The voltage for internal flash memory(V_{CC}) should be 2.7-3.6V regardless of OCR Register value.

2) This bit is set to LOW if the eMMC has not finished the power up routine.

5.2 CID Register

[Table 24] CID Register

Name	Field	Size (Byte)	CID Slice	CID Value
Manufacturer ID	MID	8	[127:120]	0x15
Reserved		6	[119:114]	0x00
Card/BGA	CBX	2	[113:112]	0x01
OEM / Application ID	OID	8	[111:104]	1)
Product Name	PNM	48	[103:56]	See Product Name Table
Product Revision	PRV	8	[55:48]	2)
Product Serial Number	PSN	32	[47:16]	3)
Manufacturing Date	MDT	8	[15:8]	4)
CRC7 Checksum	CRC	7	[7:1]	5)
Not Used, Always '1'		1	[0:0]	0x01

NOTE :

1),4),5) description are same as eMMC JEDEC standard.

2) PRV is composed of the revision count of controller and the revision count of F/W patch.

3) A 32 bits unsigned binary integer. (Random Number).

5.2.1 Product Name Table (In CID Register)

[Table 25] Product Name Table

Density	Product ID	Product Name in CID Register (PNM)
64GB	KMDC6001DM-B625	0 x 444336444D42

5.3 CSD Register

The Card-Specific Data register provides information on how to access the eMMC contents. The CSD defines the data format, error correction type, maximum data access time, data transfer speed, whether the DSR register can be used etc. The programmable part of the register (entries marked by W or E, see below) can be changed by CMD27.

The type of the entries in the table below is coded as follows:

- **R**: Read only.
- **W**: One time programmable and not readable.
- **R/W**: One time programmable and readable.
- **W/E**: Multiple writable with value kept after power failure, H/W reset assertion and any CMD0 reset and not readable.
- **R/W/E**: Multiple writable with value kept after power failure, H/W reset assertion and any CMD0 reset and readable.
- **R/W/C_P**: Writable after value cleared by power failure and HW/ rest assertion (the value not cleared by CMD0 reset) and readable.
- **R/W/E_P**: Multiple writable with value reset after power failure, H/W reset assertion and any CMD0 reset and readable.
- **W/E_P**: Multiple writable with value reset after power failure, H/W reset assertion and any CMD0 reset and not readable.

[Table 26] CSD Register

Name	Field	Size (Byte)	Cell Type	CSD Slice	CSD Value
CSD Structure	CSD_STRUCTURE	2	R	[127:126]	0x03
System Specification Version	SPEC_VERS	4	R	[125:122]	0x04
Reserved		2	R	[121:120]	0x00
Data Read Access-Time 1	TAAC	8	R	[119:112]	0x27
Data Read Access-Time 2 in CLK Cycles (NSAC*100)	NSAC	8	R	[111:104]	0x01
Max. Bus Clock Frequency	TRAN_SPEED	8	R	[103:96]	0x32
Device Command Classes	CCC	12	R	[95:84]	0xF5
Max. Read Data Block Length	READ_BL_LEN	4	R	[83:80]	0x09
Partial Blocks for Read Allowed	READ_BL_PARTIAL	1	R	[79:79]	0x00
Write Block Misalignment	WRITE_BLK_MISALIGN	1	R	[78:78]	0x00
Read Block Misalignment	READ_BLK_MISALIGN	1	R	[77:77]	0x00
DSR Implemented	DSR_IMP	1	R	[76:76]	0x00
Reserved		2	R	[75:74]	0x00
Device Size	C_SIZE	12	R	[73:62]	0xFFF
Max. Read Current @ VDD Min	VDD_R_CURR_MIN	3	R	[61:59]	0x06
Max. Read Current @ VDD Max	VDD_R_CURR_MAX	3	R	[58:56]	0x06
Max. Write Current @ VDD Min	VDD_W_CURR_MIN	3	R	[55:53]	0x06
Max. Write Current @ VDD Max	VDD_W_CURR_MAX	3	R	[52:50]	0x06
Device Size Multiplier	C_SIZE_MULT	3	R	[49:47]	0x07
Erase Group Size	ERASE_GRP_SIZE	5	R	[46:42]	0x1F
Erase Group Size Multiplier	ERASE_GRP_MULT	5	R	[41:37]	0x1F
Write Protect Group Size	WP_GRP_SIZE	5	R	[36:32]	0x0F
Write Protect Group Enable	WP_GRP_ENABLE	1	R	[31:31]	0x01
Manufacturer Default ECC	DEFAULT_ECC	2	R	[30:29]	0x00
Write Speed Factor	R2W_FACTOR	3	R	[28:26]	0x03
Max. Write Data Block Length	WRITE_BL_LEN	4	R	[25:22]	0x09
Partial Blocks for Write Allowed	WRITE_BL_PARTIAL	1	R	[21:21]	0x00
Reserved		4	R	[20:17]	0x00
Content Protection Application	CONTENT_PROT_APP	1	R	[16:16]	0x00
File Format Group	FILE_FORMAT_GRP	1	R/W	[15:15]	0x00
Copy Flag (OTP)	COPY	1	R/W	[14:14]	0x01
Permanent Write Protection	PERM_WRITE_PROTECT	1	R/W	[13:13]	0x00
Temporary Write Protection	TMP_WRITE_PROTECT	1	R/W/E	[12:12]	0x00
File Format	FILE_FORMAT	2	R/W	[11:10]	0x00
ECC Code	ECC	2	R/W/E	[9:8]	0x00
CRC	CRC	7	R/W/E	[7:1]	0x06
Not used, Always '1'		1		[0:0]	0x01

IF THERE IS ANY OTHER OPERATION TO IMPLEMENT IN ADDITION TO SPECIFICATION IN THE DATASHEET OR JEDEC STANDARD, PLEASE CONTACT EACH BRANCH OFFICE OR HEADQUARTERS OF SAMSUNG ELECTRONICS.

Rev. 1.0

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5.4 Extended CSD Register

The Extended CSD register defines the eMMC properties and selected modes. It is 512 bytes long.

The most significant 320 bytes are the Properties segment, which defines the eMMC capabilities and cannot be modified by the host. The lower 192 bytes are the Modes segment, which defines the configuration the eMMC is working in. These modes can be changed by the host by means of the SWITCH command.

The type of the entries in the table below is coded as follows:

- **R**: Read only.
- **W**: One time programmable and not readable.
- **R/W**: One time programmable and readable.
- **W/E**: Multiple writable with value kept after power failure, H/W reset assertion and any CMD0 reset and not readable.
- **R/W/E**: Multiple writable with value kept after power failure, H/W reset assertion and any CMD0 reset and readable.
- **R/W/C_P**: Writable after value cleared by power failure and HW/ rest assertion (the value not cleared by CMD0 reset) and readable.
- **R/W/E_P**: Multiple writable with value reset after power failure, H/W reset assertion and any CMD0 reset and readable.
- **W/E_P**: Multiple writable with value reset after power failure, H/W reset assertion and any CMD0 reset and not readable.

[Table 27] Extended CSD Register

Name	Field	Size (Byte)	Cell Type	CSD Slice	CSD Value
Properties Segment					
Reserved		6		[511:506]	0x00
Extended Security Commands Error	EXT_SECURITY_ERR	1	R	[505]	0x00
Supported Command Sets	S_CMD_SET	1	R	[504]	0x01
HPI Features	HPI_FEATURES	1	R	[503]	0x01
Background Operations Support	BKOPS_SUPPORT	1	R	[502]	0x01
Max Packed Read Commands	MAX_PACKED_READS	1	R	[501]	0x3F
Max Packed Write Commands	MAX_PACKED_WRITES	1	R	[500]	0x3F
Data Tag Support	DATA_TAG_SUPPORT	1	R	[499]	0x01
Tag Unit Size	TAG_UNIT_SIZE	1	R	[498]	0x02
Tag Resources Size	TAG_RES_SIZE	1	R	[497]	0x00
Context Management Capabilities	CONTEXT_CAPABILITIES	1	R	[496]	0x05
Large Unit Size	LARGE_UNIT_SIZE_M1	1	R	[495]	0x07
Extended Partitions Attribute Support	EXT_SUPPORT	1	R	[494]	0x03
Supported Modes	SUPPORTED_MODES	1	R	[493]	0x03
FFU Features	FFU_FEATURES	1	R	[492]	0x00
Operation Codes Timeout	OPERATION_CODE_TIMEOUT	1	R	[491]	0x00
FFU Argument	FFU_ARG	4	R	[490:487]	0xC7810000
Barrier Support	BARRIER_SUPPORT	1	R	[486]	0x00
Reserved		177		[485:309]	0x00
CMD Queuing Support	CMDQ_SUPPORT	1	R	[308]	0x01
CMD Queuing Depth	CMDQ_DEPTH	1	R	[307]	0x0F
Reserved		1		[306]	0x00
Number of FW Sectors Correctly Programmed	NUMBER_OF_FW_SECTORS_CORRECTLY_PROGRAMMED	4	R	[305:302]	0x00
Vendor Proprietary Health Report	VENDOR_PROPRIETARY_HEALTH_REPORT	32	R	[301:270]	0x00
Device Life Time Estimation Type B	DEVICE_LIFE_TIME_EST_TYP_B	1	R	[269]	0x01
Device Life Time Estimation Type A	DEVICE_LIFE_TIME_EST_TYP_A	1	R	[268]	0x01
Pre EOL Information	PRE_EOL_INFO	1	R	[267]	0x01
Optimal Read Size	OPTIMAL_READ_SIZE	1	R	[266]	0x00
Optimal Write Size	OPTIMAL_WRITE_SIZE	1	R	[265]	0x20
Optimal Trim Unit Size	OPTIMAL_TRIM_UNIT_SIZE	1	R	[264]	0x01
Device Version	DEVICE_VERSION	2	R	[263:262]	0x00
Firmware Version	FIRMWARE_VERSION	8	R	[261:254]	FW Version

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Name	Field	Size (Byte)	Cell Type	CSD Slice	CSD Value
Power Class for 200MHz, DDR at VCC=3.6V	PWR_CL_DDR_200_360	1	R	[253]	0x00
Cache Size	CACHE_SIZE	4	R	[252:249]	0x10000
Generic CMD6 Timeout	GENERIC_CMD6_TIME	1	R	[248]	0x0A
Power Off Notification (Long) Timeout	POWER_OFF_LONG_TIME	1	R	[247]	0x3C
Background Operations Status	BKOPS_STATUS	1	R	[246]	0x00
Number of Correctly Programmed Sectors	CORRECTLY_PRG_SECTORS_NUM	4	R	[245:242]	0x00
1st Initialization Time after Partitioning	INI_TIMEOUT_AP	1	R	[241]	0x1E
Cache Flush Policy	CACHE_FLUSH_POLICY	1	R	[240]	0x00
Power Class for 52MHz, DDR at 3.6V	PWR_CL_DDR_52_360	1	R	[239]	0x00
Power Class for 52MHz, DDR at 1.95V	PWR_CL_DDR_52_195	1	R	[238]	0x00
Power Class for 200MHz at V _{CCQ} =1.95V, V _{CC} =3.6V	PWR_CL_200_360	1	R	[237]	0x00
Power Class for 200MHz, at V _{CCQ} =1.3V, V _{CC} =3.6V	PWR_CL_200_195	1	R	[236]	0x00
Minimum Write Performance for 8bit at 52MHz in DDR Mode	MIN_PERF_DDR_W_8_52	1	R	[235]	0x00
Minimum Read Performance for 8bit at 52MHz in DDR Mode	MIN_PERF_DDR_R_8_52	1	R	[234]	0x00
Reserved		1		[233]	0x00
Trim Multiplier	TRIM_MULT	1	R	[232]	0x02
Secure Feature Support	SEC_FEATURE_SUPPORT	1	R	[231]	0x55
Secure Erase Multiplier	SEC_ERASE_MULT	1	R	[230]	0x1B
Secure Trim Multiplier	SEC_TRIM_MULT	1	R	[229]	0x11
Boot Information	BOOT_INFO	1	R	[228]	0x07
Reserved		1		[227]	0x00
Boot Partition Size	BOOT_SIZE_MULT	1	R	[226]	0x20
Access Size	ACC_SIZE	1	R	[225]	0x07
High-Capacity Erase Unit Size	HC_ERASE_GRP_SIZE	1	R	[224]	0x01
High-Capacity Erase Timeout	ERASE_TIMEOUT_MULT	1	R	[223]	0x01
Reliable Write Sector Count	REL_WR_SEC_C	1	R	[222]	0x01
High-Capacity Write Protect Group Size	HC_WP_GRP_SIZE	1	R	[221]	0x10
Sleep Current (V _{CC})	S_C_VCC	1	R	[220]	0x07
Sleep Current (V _{CCQ})	S_C_VCCQ	1	R	[219]	0x07
Production State Awareness Timeout	PRODUCTION_STATE_AWARENESS_TIMEOUT	1	R	[218]	0x00
Sleep / Awake Timeout	S_A_TIMEOUT	1	R	[217]	0x11
Sleep Notification Timeout	SLEEP_NOTIFICATION_TIME	1	R	[216]	0x07
Sector Count	SEC_COUNT	4	R	[215:212]	0x747C000
Secure Write Protect Information	SECURE_WP_INFO	1	R	[211]	0x01
Minimum Write Performance for 8bit at 52MHz	MIN_PERF_W_8_52	1	R	[210]	0x00
Minimum Read Performance for 8bit at 52MHz	MIN_PERF_R_8_52	1	R	[209]	0x00
Minimum Write Performance for 8bit at 26MHz, for 4bit at 52MHz	MIN_PERF_W_8_26_4_52	1	R	[208]	0x00
Minimum Read Performance for 8bit at 26MHz, for 4bit at 52MHz	MIN_PERF_R_8_26_4_52	1	R	[207]	0x00
Minimum Write Performance for 4bit at 26MHz	MIN_PERF_W_4_26	1	R	[206]	0x00
Minimum Read Performance for 4bit at 26MHz	MIN_PERF_R_4_26	1	R	[205]	0x00
Reserved		1		[204]	0x00
Power Class for 26MHz at 3.6V 1 R	PWR_CL_26_360	1	R	[203]	0x00
Power Class for 52MHz at 3.6V 1 R	PWR_CL_52_360	1	R	[202]	0x00

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Name	Field	Size (Byte)	Cell Type	CSD Slice	CSD Value
Power Class for 26MHz at 1.95V 1 R	PWR_CL_26_195	1	R	[201]	0x00
Power Class for 52MHz at 1.95V 1 R	PWR_CL_52_195	1	R	[200]	0x00
Partition Switching Timing	PARTITION_SWITCH_TIME	1	R	[199]	0x02
Out-of-Interrupt Busy Timing	OUT_OF_INTERRUPT_TIME	1	R	[198]	0x0A
I/O Driver Strength	DRIVER_STRENGTH	1	R	[197]	0x1F
Device Type	DEVICE_TYPE	1	R	[196]	0x57
Reserved		1		[195]	0x00
CSD Structure	CSD_STRUCTURE	1	R	[194]	0x02
Reserved		1		[193]	0x00
Extended CSD Revision	EXT_CSD_REV	1	R	[192]	0x08
Modes Segment					
Command Set	CMD_SET	1	R/W/E_P	[191]	0x00
Reserved		1		[190]	0x00
Command Set Revision	CMD_SET_REV	1	R	[189]	0x00
Reserved		1		[188]	0x00
Power Class	POWER_CLASS	1	R/W/E_P	[187]	0x00
Reserved		1		[186]	0x00
High-Speed Interface Timing	HS_TIMING	1	R/W/E_P	[185]	0x00
Strobe Support	STROBE_SUPPORT	1	R	[184]	0x01
Bus Width Mode	BUS_WIDTH	1	W/E_P	[183]	0x00
Reserved		1		[182]	0x00
Erased Memory Content	ERASED_MEM_CONT	1	R	[181]	0x00
Reserved		1		[180]	0x00
Partition Configuration	PARTITION_CONFIG	1	R/W/E & R/W/E_P	[179]	0x00
Boot Config Protection	BOOT_CONFIG_PROT	1	R/W & R/W/C_P	[178]	0x00
Boot Bus Conditions	BOOT_BUS_CONDITIONS	1	R/W/E	[177]	0x00
Reserved		1		[176]	0x00
High-Density Erase Group Definition	ERASE_GROUP_DEF	1	R/W/E_P	[175]	0x00
Boot Write Protection Status Registers	BOOT_WP_STATUS	1	R	[174]	0x00
Boot Area Write Protection Register	BOOT_WP	1	R/W & R/W/C_P	[173]	0x00
Reserved		1		[172]	0x00
User Area Write Protection Register	USER_WP	1	R/W, R/W/C_P & R/W/E_P	[171]	0x00
Reserved		1		[170]	0x00
FW Configuration	FW_CONFIG	1	R/W	[169]	0x00
RPMB Size	RPMB_SIZE_MULT	1	R	[168]	0x20
Write Reliability Setting Register	WR_REL_SET	1	R/W	[167]	0x1F
Write Reliability Parameter Register	WR_REL_PARAM	1	R	[166]	0x14
Start Sanitize Operation	SANITIZE_START	1	W/E_P	[165]	0x00
Manually Start Background Operations	BKOPS_START	1	W/E_P	[164]	0x00
Enable Background Operations Handshake	BKOPS_EN	1	R/W&R/W/E	[163]	0x00
H/W Reset Function	RST_n_FUNCTION	1	R/W	[162]	0x00
HPI Management	HPI_MGMT	1	R/W/E_P	[161]	0x00
Partitioning Support	PARTITIONING_SUPPORT	1	R	[160]	0x07
Max Enhanced Area Size	MAX_ENH_SIZE_MULT	3	R	[159:157]	0xE8F
Partitions Attribute	PARTITIONS_ATTRIBUTE	1	R/W	[156]	0x00

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Name	Field	Size (Byte)	Cell Type	CSD Slice	CSD Value
Partitioning Setting	PARTITION_SETTING_COMPLETED	1	R/W	[155]	0x00
General Purpose Partition Size	GP_SIZE_MULT	12	R/W	[154:143]	0x00
Enhanced User Data Area Size	ENH_SIZE_MULT	3	R/W	[142:140]	0x00
Enhanced User Data Start Address	ENH_START_ADDR	4	R/W	[139:136]	0x00
Reserved		1		[135]	0x00
Bad Block Management Mode	SEC_BAD_BLK_MGMNT	1	R/W	[134]	0x00
Production State Awareness	PRODUCTION_STATE_AWARENESS	1	R/W/E	[133]	0x00
Package Case Temperature is Controlled	TCASE_SUPPORT	1	W/E_P	[132]	0x00
Periodic Wake-Up	PERIODIC_WAKEUP	1	R/W/E	[131]	0x00
Program CID / CSD in DDR Mode Support	PROGRAM_CID_CSD_DDR_SUPPORT	1	R	[130]	0x01
Reserved		2		[129:128]	0x00
Vendor Specific Fields	VENDOR_SPECIFIC_FIELD	64	Vendor Specific	[127:64]	Vendor Specific
Native Sector Size	NATIVE_SECTOR_SIZE	1	R	[63]	0x00
Sector Size Emulation	USE_NATIVE_SECTOR	1	R/W	[62]	0x00
Sector Size	DATA_SECTOR_SIZE	1	R	[61]	0x00
1st Initialization after Disabling Sector Size Emulation	INI_TIMEOUT_EMU	1	R	[60]	0x00
Class 6 Commands Control	CLASS_6_CTRL	1	R/W/E_P	[59]	0x00
Number of Addressed Group to be Released	DYNCAP_NEEDED	1	R	[58]	0x00
Exception Events Control	EXCEPTION_EVENTS_CTRL	2	R/W/E_P	[57:56]	0x00
Exception Events Status	EXCEPTION_EVENTS_STATUS	2	R	[55:54]	0x00
Extended Partitions Attribute	EXT_PARTITIONS_ATTRIBUTE	2	R/W	[53:52]	0x00
Context Configuration	CONTEXT_CONF	15	R/W/E_P	[51:37]	0x00
Packed Command Status	PACKED_COMMAND_STATUS	1	R	[36]	0x00
Packed Command Failure Index	PACKED_FAILURE_INDEX	1	R	[35]	0x00
Power Off Notification	POWER_OFF_NOTIFICATION	1	R/W/E_P	[34]	0x00
Control to Turn the Cache On / Off	CACHE_CTRL	1	R/W/E_P	[33]	0x00
Flushing of the Cache	FLUSH_CACHE	1	W/E_P	[32]	0x00
Control to Turn the Barrier On / Off	BARRIER_CTRL	1	R/W	[31]	0x00
Mode Config	MODE_CONFIG	1	R/W/E_P	[30]	0x00
Mode Operation Codes	MODE_OPERATION_CODES	1	W/E_P	[29]	0x00
Reserved		2		[28:27]	0x00
FFU Status	FFU_STATUS	1	R	[26]	0x00
Pre Loading Data Size	PRE_LOADING_DATA_SIZE	4	R/W/E_P	[25:22]	0x00
Max Pre Loading Data Size	MAX_PRE_LOADING_DATA_SIZE	4	R	[21:18]	0x00
Product State Awareness Enablement	PRODUCT_STATE_AWARENESS_ENABLEMENT	1	R/W/E & R	[17]	0x00
Secure Removal Type	SECURE_REMOVAL_TYPE	1	R/W & R	[16]	0x39
Command Queue Mode Enable	CMDQ_MODE_EN	1	R/W/E_P	[15]	0x00
Reserved		15		[14:0]	0x00

6.0 DC PARAMETER

6.1 Power Consumption

6.1.1 Active Power Consumption During Operation

[Table 28] Active Power Consumption During Operation

Density	V _{CC}	V _{CCQ}	Unit
64GB	200	180	mA

NOTE :

Power Measurement conditions: Bus configuration =x8 @HS400.

The measurement for max RMS current is the average RMS current consumption over a period of 100ms.

6.1.2 Standby Power Consumption in Auto Power Saving Mode and Standby State

[Table 29] Standby Power Consumption in Auto Power Saving Mode and Standby State

Density	V _{CC}	V _{CCQ}	Unit
64GB	70	120	uA

NOTE :

Power Measurement conditions: Bus configuration =x8, No CLK.

Typical value is measured at V_{CC}=3.3V, TA=25°C. Not 100% tested.

6.1.3 Sleep Power Consumption in Sleep State

[Table 30] Sleep Power Consumption in Sleep State

Density	V _{CC}	V _{CCQ}	Unit
64GB	0 ¹⁾	120	uA

NOTE :

Power Measurement conditions: Bus configuration =x8, No CLK.

1) In auto power saving mode, V_{CC} power supply shall be turned off. However in sleep mode V_{CC} power may be turned off.

7.0 AC PARAMETER

7.1 Timing Parameter

[Table 31] Timing Parameter

Timing Parameter		Value (Max.)	Unit
Initialization Time (tINIT)	Normal ¹⁾	1	s
	After Partition Setting ²⁾	3	s
Read Timeout		100	ms
Write Timeout		350	ms
Erase Timeout		20	ms
Force Erase Timeout		3	min
Secure Erase Timeout		8	s
Secure Trim Step1 Timeout		5	s
Secure Trim Step2 Timeout		3	s
Trim Timeout		600	ms
Partition Switching Timeout (after Init)		1	ms
Power Off Notification (Short) Timeout		100	ms
Power Off Notification (Long) Timeout		600	ms

NOTE :

Be advised Timeout Values specified in Table above are for testing purposes under Samsung test pattern only and actual timeout situations may vary. EXCEPTION_EVENT may occur and the actual timeout values may vary due to user environment.

1) Normal Initialization Time without partition setting.

2) Initialization Time after partition setting, refer to INI_TIMEOUT_AP in Chapter 5.4 EXT_CSD register.

7.2 Previous Bus Timing Parameters for DDR52 and HS200 Mode are defined by JEDEC Standard

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7.3 Bus Timing Specification in HS400 Mode

7.3.1 HS400 Device Input Timing

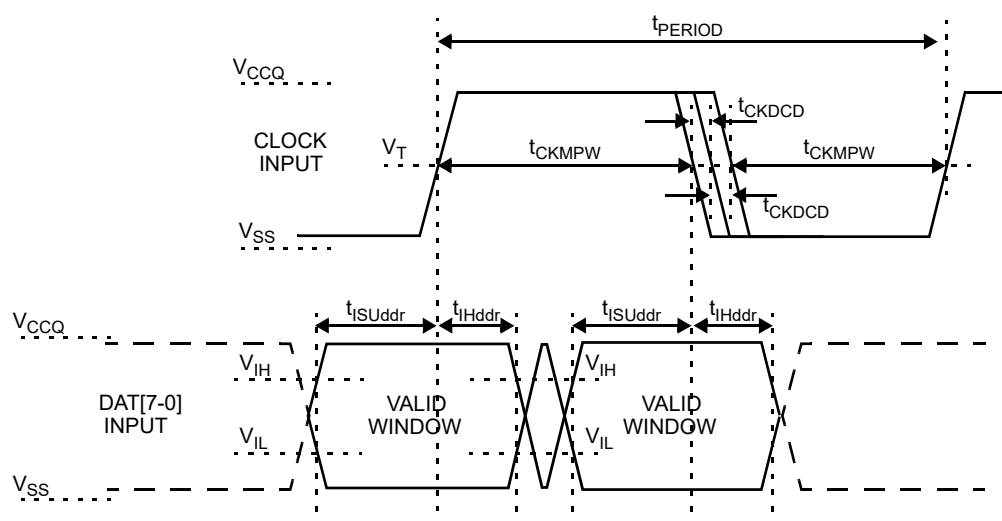


Figure 4. HS400 Device Input Timing

NOTE :

t_{ISU} and t_{IH} are measured at $V_{IL}(\max)$ and $V_{IH}(\min)$.
 V_{IH} denotes $V_{IH}(\min)$ and V_{IL} denotes $V_{IL}(\max)$.

[Table 32] HS400 Device Input Timing

Parameter	Symbol	Min	Max	Unit
Input CLK				
Cycle Time Data Transfer Mode	t_{PERIOD}	5		ns
Slew Rate	SR	1.125		V/ns
Duty Cycle Distortion	t_{CKDCD}	0.0	0.3	ns
Minimum Pulse Width	t_{CKMPW}	2.2		ns
Input Dat (Referenced to CLK)				
Input Set-Up Time	t_{ISUddr}	0.4		ns
Input Hold Time	t_{IHddr}	0.4		ns
Slew Rate	SR	1.125		V/ns

7.3.2 HS400 Device Output Timing

Data Strobe is used to read data (data read and CRC status response read) in HS400 mode.

The device output value of Data Strobe is "High-Z" when the device is not in outputting data (data read, CRC status response).

Data Strobe is toggled only during data read period.

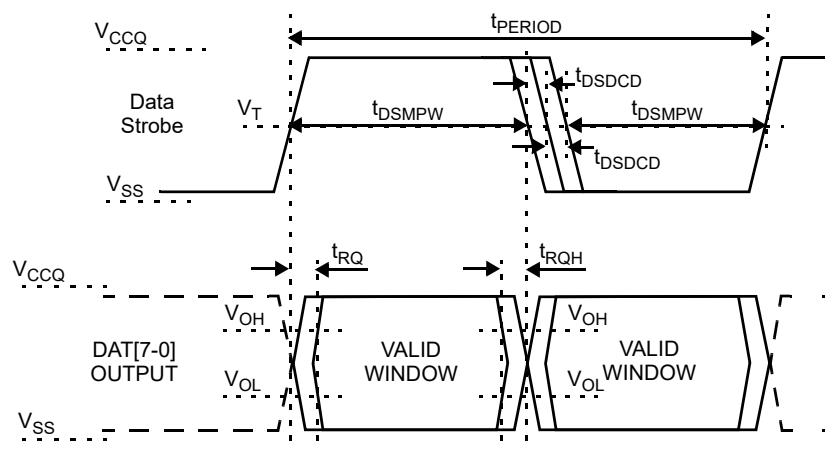


Figure 5. HS400 Device Output Timing

NOTE :

V_{OH} denotes $V_{OH(min)}$ and V_{OL} denotes $V_{OL(max)}$.

[Table 33] HS400 Device Output Timing

Parameter	Symbol	Min	Max	Unit
Data Strobe				
Cycle Time Data Transfer Mode	t_{PERIOD}	5		ns
Slew Rate	SR	1.125		V/ns
Duty Cycle Distortion	t_{DSDCD}	0.0	0.2	ns
Minimum Pulse Width	t_{DSMPW}	2.0		ns
Read Pre-Amble	t_{RPRE}	0.4		t_{PERIOD}
Read Post-Amble	t_{RPST}	0.4		t_{PERIOD}
Output DAT (Referenced to Data Strobe)				
Output Skew	t_{RQ}		0.4	ns
Output Hold Skew	t_{RQH}		0.4	ns
Slew Rate	SR	1.125		V/ns

7.4 Bus Signal Line Load

The total capacitance C_L of each line of the eMMC bus is the sum of the bus master capacitance C_{HOST} , the bus capacitance C_{BUS} itself and the capacitance C_{DEVICE} of the eMMC connected to this line:

$$C_L = C_{HOST} + C_{BUS} + C_{DEVICE}$$

The sum of the host and bus capacitances should be under 20pF.

[Table 34] Bus Signal Line Load

Parameter	Symbol	Min	Max	Unit	Remark
Pull-Up Resistance for CMD	R_{CMD}	4.7	100	KOhm	to prevent bus floating
Pull-Up Resistance for DAT0-DAT7	R_{DAT}	10	100	KOhm	to prevent bus floating
Internal Pull Up Resistance DAT1-DAT7	R_{int}	10	150	KOhm	to prevent unconnected lines floating
Single Device Capacitance	C_{DEVICE}		12	pF	
Maximum Signal Line Inductance			16	nH	$f_{PP} \leq 52$ MHz

[Table 35] Capacitance and Resistance for HS400 Mode

Parameter	Symbol	Min	Max	Unit	Remark
Bus Signal Line Capacitance	C_L		13	pF	Single Device
Single Device Capacitance	C_{DEVICE}		6	pF	
Pull-Down Resistance for Data Strobe	$R_{Data\ Strobe}$	10	100	KOhm	

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7.5 Bus Signal Levels

As the bus can be supplied with a variable supply voltage, all signal levels are related to the supply voltage.

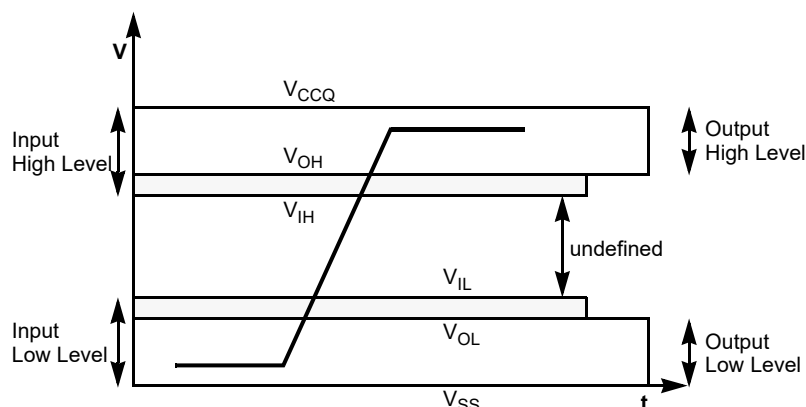


Figure 6. Bus Signal Levels

7.5.1 Open-Drain Mode Bus Signal Level

[Table 36] Open-Drain Bus Signal Level

Parameter	Symbol	Min	Max	Unit	Condition
Output High Voltage	V_{OH}	$V_{CCQ} - 0.2$		V	1)
Output Low Voltage	V_{OL}		0.3	V	$I_{OL} = 2 \text{ mA}$

NOTE :

1) Because V_{OH} depends on external resistance value (including outside the package), this value does not apply as device specification.

Host is responsible to choose the external pull-up and open drain resistance value to meet V_{OH} Min value.

7.5.2 Push-Pull Mode Bus Signal Level eMMC

The device input and output voltages shall be within the following specified ranges for any V_{CCQ} of the allowed voltage range.

[Table 37] Push-Pull Signal Level—High-Voltage eMMC

Parameter	Symbol	Min	Max	Unit	Condition
Output High Voltage	V_{OH}	$0.75 \cdot V_{CCQ}$		V	$I_{OH} = -100 \text{ uA} @ V_{CCQ} \text{ min}$
Output Low Voltage	V_{OL}		$0.125 \cdot V_{CCQ}$	V	$I_{OL} = 100 \text{ uA} @ V_{CCQ} \text{ min}$
Input High Voltage	V_{IH}	$0.625 \cdot V_{CCQ}$	$V_{CCQ} + 0.3$	V	
Input Low Voltage	V_{IL}	$V_{SS} - 0.3$	$0.25 \cdot V_{CCQ}$	V	

[Table 38] Push-Pull Signal Level—1.70 - 1.95 V_{CCQ} Voltage Range

Parameter	Symbol	Min	Max	Unit	Condition
Output High Voltage	V_{OH}	$V_{CCQ} - 0.45V$		V	$I_{OH} = -2mA$
Output Low Voltage	V_{OL}		0.45V	V	$I_{OL} = 2mA$
Input High Voltage	V_{IH}	$0.65 \cdot V_{CCQ}^{1)}$	$V_{CCQ} + 0.3$	V	
Input Low Voltage	V_{IL}	$V_{SS} - 0.3$	$0.35 \cdot V_{CCQ}^{2)}$	V	

NOTE :

1) $0.7 \cdot V_{CCQ}$ for MMC4.3 and older revisions.

2) $0.3 \cdot V_{CCQ}$ for MMC4.3 and older revisions.

LPDDR4X 32Gb DDP

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LPDDR4X SDRAM SPECIFICATION**32Gb DDP = 128M x16DQ x8banks x2channels****1.0 LPDDR4X PAD DEFINITION AND DESCRIPTION****1.1 Dual channel per die device****[Table 1] Pad Definition and Description for Dual channel**

Symbol	Type	Description
CK_t_A CK_c_A CK_t_B CK_c_B	Input	Clock: CK_t and CK_c are differential clock inputs. All address, command, and control input signals are sampled on the crossing of the positive edge of CK_t and the negative edge of CK_c. AC timings for CA parameters are referenced to CK. Each channel (A & B) has its own clock pair.
CKE_A CKE_B	Input	Clock Enable: CKE HIGH activates and CKE LOW deactivates the internal clock circuits, input buffers, and output drivers. Power-saving modes are entered and exited via CKE transitions. CKE is part of the command code. Each channel (A & B) has its own CKE signal.
CS_A, CS_B	Input	Chip Select: CS is part of the command code. Each channel (A & B) has its own CS signal.
CA[5:0]_A CA[5:0]_B	Input	Command/Address Inputs: CA signals provide the Command and Address inputs according to the Command Truth Table. Each channel (A&B) has its own CA signals.
ODT_CA_A ODT_CA_B	Input	CA ODT Control: The ODT_CA pin is used in conjunction with the Mode Register to turn on/off the On-Die-Termination for CA pins.
DQ[15:0]_A DQ[15:0]_B	I/O	Data Inputs/Outputs: Bi-direction data bus
DQS[1:0]_t_A DQS[1:0]_c_A DQS[1:0]_t_B DQS[1:0]_c_B	I/O	Data Strobe: DQS_t and DQS_c are bi-directional differential output clock signals used to strobe data during a READ or WRITE. The Data Strobe is generated by the DRAM for a READ and is edge-aligned with Data. The Data Strobe is generated by the Memory Controller for a WRITE and must arrive prior to Data. Each byte of data has a Data Strobe signal pair. Each channel (A & B) has its own DQS strobes.
DMI[1:0]_A DMI[1:0]_B	I/O	Data Mask Inversion: DMI is a bi-directional signal which is driven HIGH when the data on the data bus is inverted, or driven LOW when the data is in its normal state. Data Inversion can be disabled via a mode register setting. Each byte of data has a DMI signal. Each channel (A & B) has its own DMI signals. This signal is also used along with the DQ signals to provide write data masking information to the DRAM. The DMI pin function - Data Inversion or Data mask - depends on Mode Register setting.
ZQ	Reference	Calibration Reference: Used to calibrate the output drive strength and the termination resistance. There is one ZQ pin per die. The ZQ pin shall be connected to VDDQ through a 240Ω ± 1% resistor.
VDDQ, VDD1, VDD2	Supply	Power Supplies: Isolated on the die for improved noise immunity.
V _{SS} , V _{SSQ}	GND	Ground Reference: Power supply ground reference.
RESET_n	Input	RESET: When asserted LOW, the RESET_n signal resets all channels of the die. There is one RESET_n pad per die.

NOTE :

1) "_A" and "_B" indicate DRAM channel "_A" pads are present in all devices. "_B" pads are present in dual channel SDRAM devices only.

2.0 FUNCTIONAL DESCRIPTION

LPDDR4X-SDRAM is a high-speed synchronous DRAM device internally configured with 2 channels. Dual channel is comprised of 8-banks with from 2Gb to 16Gb per channel density. The configuration for channel density that is greater than 16Gb is still TBD¹⁾.

This device contains the following number of bits:

Dual-channel SDRAM devices contain the following number of bits:

16Gb has 17,179,869,184 bits

LPDDR4X devices use a 2 or 4 clocks architecture on the Command/Address (CA) bus to reduce the number of input pins in the system. The 6-bit CA bus contains command, address, and bank information. Each command uses 1, 2 or 4 clock cycle, during which command information is transferred on the positive edge of the clock. See command truth table for details.

These devices use a double data rate architecture on the DQ pins to achieve high speed operation. The double data rate architecture is essentially an 16n prefetch architecture with an interface designed to transfer two data bits per DQ every clock cycle at the I/O pins. A single read or write access for the LPDDR4X SDRAM effectively consists of a single 16n-bit wide, one clock cycle data transfer at the internal DRAM core and eight corresponding n-bit wide, one-half-clock-cycle data transfers at the I/O pins. Read and write accesses to the LPDDR4X SDRAMs are burst oriented; accesses start at a selected location and continue for a programmed number of locations in a programmed sequence. Accesses begin with the registration of an Activate command, which is then followed by a Read, Write or Mask Write command. The address and BA bits registered coincident with the Activate command are used to select the row and the Bank to be accessed. The address bits registered coincident with the Read, Write or Mask Write command are used to select the Bank and the starting column location for the burst access.

Prior to normal operation, the LPDDR4X SDRAM must be initialized. The following section provides detailed information covering device initialization, register definition, command description and device operation.

2.1 LPDDR4X SDRAM Addressing

[Table 2] LPDDR4X SDRAM x16 mode Addressing for Dual Channel SDRAM Device

Memory Density (per Die)		16Gb
Memory Density (per x16 channel)		8Gb
Configuration		64Mb x 16DQ x 8 banks x 2 channels
Number of Channels (per die)		2
Number of Banks (per channel)		8
Array Pre-Fetch (bits, per channel)		256
Number of Rows (per Channel)		65,536
Number of Columns (fetch boundaries)		64
Page Size (Bytes)		2048
Channel Density (Bits per channel)		8,589,934,592
Total Density (Bits per die)		17,179,869,184
Bank Addresses		BA0-BA2
x16	Row Addresses ²⁾	R0 - R15
	Column Addresses ^{1), 2)}	C0-C9
Burst Starting Address Boundary		64 - bit

NOTE :

1) The lower two column addresses (C0-C1) are assumed to be "zero" and are not transmitted on the CA bus.

2) Row and Column Address values on the CA bus that are not used for a particular density is required to at valid logic levels.

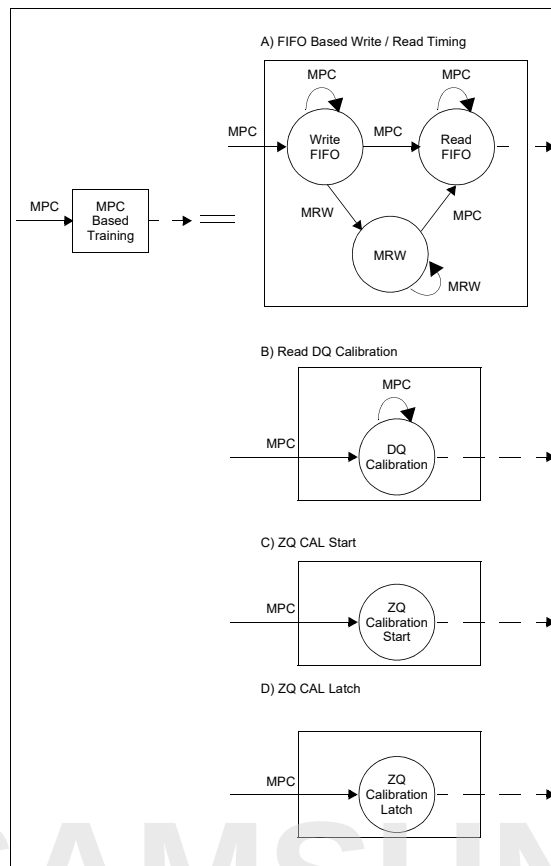


Figure 2. LPDDR4X: Simplified Bus Interface State Diagram -2

NOTE :

- 1) From the Self-Refresh state the device can enter Power-Down, MRR, MRW, or MPC states. See the section on Self-Refresh for more information.
- 2) In IDLE state, all banks are precharged.
- 3) In the case of a MRW command to enter a training mode, the state machine will not automatically return to the IDLE state at the conclusion of training. See the applicable training section for more information.
- 4) In the case of a MPC command to enter a training mode, the state machine may not automatically return to the IDLE state at the conclusion of training. See the applicable training section for more information.
- 5) This simplified State Diagram is intended to provide an overview of the possible state transitions and the commands to control them. In particular, situations involving more than one bank, the enabling or disabling of on-die termination, and some other events are not captured in full detail.
- 6) States that have an "automatic return" and can be accessed from more than one prior state (Ex. MRW from either Idle or Active states) will return to the state from when they were initiated (Ex. MRW from Idle will return to Idle).
- 7) The RESET_n pin can be asserted from any state, and will cause the SDRAM to go to the Reset State. The diagram shows RESET applied from the Power-On as an example, but the Diagram should not be construed as a restriction on RESET_n.

2.3 Mode Register Definition

2.3.1 Mode Register Assignment and Definition in LPDDR4X SDRAM

Table 3 shows the mode registers for LPDDR4X SDRAM. Each register is denoted as “R” if it can be read but not written, “W” if it can be written but not read, and “R/W” if it can be read and written. A Mode Register Read command is used to read a mode register. A Mode Register Write command is used to write a mode register.

[Table 3] Mode Register Assignment in LPDDR4X SDRAM

MR#	MA <7:0>	Function	Access	OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
0	00 _H	Device Info.	R	CATR	(RFU)	Single ended mode	RZQI		(RFU)		Refresh mode
1	01 _H	Device Feature 1	W	RPST0	nWR0			RD-PRE0	WR-PRE0	BL	
				RPST1	nWR1			RD-PRE1	WR-PRE1		
2	02 _H	Device Feature 2	W	WR Lev	WL Select0	WL0			RL0		
					WL Select1	WL1			RL1		
3	03 _H	I/O Configuration-1	W	DBI-WR0	DBI-RD0	PDDS0			PPR Protection	WR PST	PU-CAL0
				DBI-WR1	DBI-RD1	PDDS1				WR PST	PU-CAL1
4	04 _H	Refresh Rate	R/W	TUF	Thermal Offset		PPRE	SR Abort	Refresh Rate		
5	05 _H	Basic Configuration-1	R	LPDDR4X Manufacturer ID							
6	06 _H	Basic Configuration-2	R	Revision ID-1							
7	07 _H	Basic Configuration-3	R	Revision ID-2							
8	08 _H	Basic Configuration-4	R	I/O width		Density				Type	
9	09 _H	Test Mode	W	Vendor Specific Test Register							
10	0A _H	IO Calibration	W	(RFU)							ZQ-RESET
11	0B _H	ODT Feature	W	(RFU)	CA ODT0			(RFU)	DQ ODT0		
					CA ODT1				DQ ODT1		
12	0C _H	VREF(ca) Setting/Range	R/W	(RFU)	VR-CA0	VREF0(ca)					
					VR-CA1	VREF1(ca)					
13	0D _H	CBT,RPT,VRO,VRCG, RRO, DM_DIS,FSP-WR,FSP-OP	W	FSP-OP	FSP-WR	DM_DIS	RRO	VRCG	VRO	RPT	CBT
14	0E _H	VREF(dq) Setting/Range	R/W	(RFU)	VR-DQ0	VREF0(DQ)					
				(RFU)	VR-DQ1	VREF1(DQ)					
15	0F _H	Lower-Byte Invert for DQ Calibration	W	Lower-Byte Invert Register for DQ Calibration							
16	10 _H	PASR_Bank	W	PASR Bank Mask							
17	11 _H	PASR_Segment	W	PASR Segment Mask							
18	12 _H	IT-LSB	R	DQS Oscillator Count-LSB							
19	13 _H	IT-MSB	R	DQS Oscillator Count-MSB							
20	14 _H	Upper-Byte Invert for DQ Calibration	W	Upper-Byte Invert Register for DQ Calibration							
21	15 _H	Low Speed CA Buffer	W	(RFU)		Low Speed CA buffer	(RFU)				

[Table 3] Mode Register Assignment in LPDDR4X SDRAM

MR#	MA <7:0>	Function	Access	OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
22	16 _H	ODT Feature	W	(RFU)		ODT-CA0	ODT-CS0	ODT-CK0	SoC ODT0		
						ODT-CA1	ODT-CS1	ODT-CK1	SoC ODT1		
23	17 _H	DQS interval timer run time	W	DQS interval timer run time setting							
24	18 _H	RFU	N/A	(RFU)							
25	19 _H	PPR Resource	R	Bank 7	Bank 6	Bank 5	Bank 4	Bank 3	Bank 2	Bank 1	Bank 0
26:29	1A _H : 1D _H	RFU	N/A	Reserved for Future Use							
30	1E _H	Reserved for Testing	N/A	Reserved for Testing-SDRAM will ignore							
31	1F _H	RFU	N/A	Bytemode Vref Selection		Reserved for Future Use					
32	20 _H	DQ Calibration Pattern A	W	DQ Calibration Pattern "A" (default = 5AH)							
33:38	21 _H ~26 _H	(Do Not Use)	NA	Do Not Use							
39	27 _H	Reserved for Testing	N/A	Reserved for Testing-SDRAM will ignore							
40	28 _H	DQ Calibration Pattern B	W	DQ Calibration Pattern "B" (default = 3CH)							
41:47	29 _H ~2F _H	(Do Not Use)	NA	Do Not Use							
48:50	30 _H ~32 _H	RFU	NA	(RFU)							
51	33 _H	Single Ended RDQS, WDQS, CLK	W	(RFU)				Single ended Clock	Single ended WDQS	Single ended RDQS	(RFU)
52:63	34 _H ~3F _H	RFU	NA	(RFU)							

	Applied when FSP = 0
	Applied when FSP = 1

NOTE :

- 1) RFU bits shall be set to '0' during writes.
- 2) RFU bits shall be read as '0' during reads.
- 3) All mode registers that are specified as RFU or write-only shall return undefined data when read and DQS_t, DQS_c shall be toggled.
- 4) All mode registers that are specified as RFU shall not be written.
- 5) Writes to read-only registers shall have no impact on the functionality of the device.

MR0_Device Information (MA<7:0> = 00_H) :

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
CATR	(RFU)	Single ended mode	RZQI		(RFU)		Refresh mode

Function	Register Type	Operand	Data	Notes
Refresh mode	Read-only	OP[0]	0 _B : Both legacy & modified refresh mode supported 1 _B : Only modified refresh mode supported	
RZQI (Built-in Self-Test for RZQ)		OP[4:3]	00 _B : RZQ self-test not supported 01 _B : ZQ-pin may connect to V _{SSQ} or float 10 _B : ZQ-pin may short to V _{DDQ} 11 _B : ZQ-pin self test completed, no error condition detected (ZQ-pin may not connect to V _{SSQ} or float, nor short to V _{DDQ})	1,2,3,4
Single ended mode		OP[5]	0 _B : No support for Single ended mode 1 _B : Supports for Single ended mode	6
CATR (CA Terminating Rank)		OP[7]	0 _B : CA for this rank is not terminated 1 _B : Vendor specific	5

NOTE :

- 1) RZQI MR value, if supported, will be valid after the following sequence:
 - a. Completion of MPC ZQCAL Start command to either channel.
 - b. Completion of MPC ZQCAL Latch command to either channel then t_{ZQLAT} is satisfied.
RZQI value will be lost after Reset.
- 2) If the ZQ-pin is connected to V_{SSQ} to set default calibration, OP[4:3] shall be set to 01_B. If the ZQ-pin is not connected to V_{SSQ}, either OP[4:3] = 01_B or OP[4:3] = 10_B might indicate a ZQ-pin assembly error. It is recommended that the assembly error is corrected.
- 3) In the case of possible assembly error, the LPDDR4X-SDRAM device will default to factory trim settings for RON, and will ignore ZQ calibration commands. In either case, the device may not function as intended.
- 4) In ZQ self-test returns OP[4:3] = 11_B, the device has detected a resistor connected to the ZQ pin. However, this result cannot be used to validate the ZQ resistor value or that the ZQ resistor tolerance meets the specified limits (i.e., 240-Ω +/- 1%).
- 5) CATR functionality is Vendor specific. CATR can either indicate the connection status of the ODTCA pad for the die or whether CA for the rank is terminated. Consult the vendor device datasheet for details.
- 6) Support for Single Ended Mode is optional. If supported, Single Ended Write DQS, Read DQS and CK can be enabled in MR51.

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MR1_Device Feature 1 (MA<7:0> = 01_H) :

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
RPST	nWR (for AP)			RD-PRE	WR-PRE	BL	

Function	Register Type	Operand	Data	Notes
BL (Burst Length)	Write-only	OP[1:0]	00 _B : BL=16 Sequential (default) 01 _B : BL=32 Sequential 10 _B : BL=16 or 32 Sequential (on-the-fly) All others: Reserved	1,7
WR-PRE (WR Pre-amble Length)		OP[2]	0 _B : Reserved 1 _B : WR Pre-amble = 2×tCK	5,6
RD-PRE (RD Pre-amble Type)		OP[3]	0 _B : RD Pre-amble = Static (default) 1 _B : RD Pre-amble = Toggle	3,5,6
nWR (Write-Recovery for Auto-Precharge commands)		OP[6:4]	000 _B : nWR = 6 (default) 001 _B : nWR = 10 010 _B : nWR = 16 011 _B : nWR = 20 100 _B : nWR = 24 101 _B : nWR = 30 110 _B : nWR = 34 111 _B : nWR = 40	2,5,6
RPST (RD Post-Amble Length)		OP[7]	0 _B : RD Post-amble = 0.5×tCK (default) 1 _B : RD Post-amble = 1.5×tCK	4,5,6

NOTE :

- 1) Burst length on-the-fly can be set to either BL=16 or BL=32 by setting the "BL" bit in the command operands. See the Command Truth Table.
- 2) The programmed value of nWR is the number of clock cycles the LPDDR4X-SDRAM device uses to determine the starting point of an internal Precharge operation after a Write burst with AP (auto-precharge) enabled. See "Read and Write Latencies" later in this section.
- 3) For Read operations this bit must be set to select between a "toggling" pre-amble and a "Non-toggling" pre-amble. See the Read Preamble and Postamble section in Operation timing for a drawing of each type of pre-amble.
- 4) OP[7] provides an optional READ post-amble with an additional rising and falling edge of DQS_t. The optional postamble cycle is provided for the benefit of certain memory controllers.
- 5) There are two physical registers assigned to each bit of this MR parameter, designated set point 0 and set point 1. Only the registers for the set point determined by the state of the FSP-WR bit (MR13 OP[6]) will be read from with an MRR command to this MR address.
- 6) There are two physical registers assigned to each bit of this MR parameter, designated set point 0 and set point 1. The device will operate only according to the values stored in the registers for the active set point, i.e., the set point determined by the state of the FSP-OP bit (MR13 OP[7]). The values in the registers for the inactive set point will be ignored by the device, and may be changed without affecting device operation.
- 7) Supporting the two physical registers for Burst Length: MR1 OP[1:0] as optional feature. Applications requiring support of both vendor options shall assure that both FSP-OP[0] and FSP-OP[1] are set to the same code. Refer to vendor datasheets for detail.

[Table 4] Read and Write Latencies for x16 mode

Read Latency [nCK]		Write Latency [nCK]		nWR [nCK]	nRTP [nCK]	Lower Clock Frequency Limit [MHz] (Greater than)	Upper Clock Frequency Limit [MHz] (Same or less than)	Notes
No DBI	w/ DBI	Set "A"	Set "B"					
6	6	4	4	6	8	10	266	1,2,3,4,5,6
10	12	6	8	10	8	266	533	
14	16	8	12	16	8	533	800	
20	22	10	18	20	8	800	1066	
24	28	12	22	24	10	1066	1333	
28	32	14	26	30	12	1333	1600	
32	36	16	30	34	14	1600	1866	
36	40	18	34	40	16	1866	2133	

NOTE :

- 1) The LPDDR4X-SDRAM device should not be operated at a frequency above the Upper Frequency Limit, or below the Lower Frequency Limit, shown for each RL, WL, nRTP, or nWR value.
- 2) DBI for Read operations is enabled in MR3 OP[6]. When MR3 OP[6]=0_B, then the "No DBI" column should be used for Read Latency. When MR3 OP[6]=1_B, then the "w/DBI" column should be used for Read Latency.
- 3) Write Latency Set "A" and Set "B" is determined by MR2 OP[6]. When MR2 OP[6]=0_B, then Write Latency Set "A" should be used. When MR2 OP[6]=1_B, then Write Latency Set "B" should be used.
- 4) The programmed value of nWR is the number of clock cycles the LPDDR4X-SDRAM device uses to determine the starting point of an internal Precharge operation after a Write burst with AP (auto precharge). It is determined by RU(tWR/tCK).
- 5) The programmed value of nRTP is the number of clock cycles the LPDDR4X-SDRAM device uses to determine the starting point of an internal Precharge operation after a Read burst with AP (auto precharge). It is determined by RU(tRTP/tCK).
- 6) nRTP shown in this table is valid for BL16 only. For BL32, the SDRAM will add 8 clocks to the nRTP value before starting a precharge.

NOTE :

- 1) C0-C1 are assumed to be '0', and are not transmitted on the command bus.
- 2) The starting burst address is on 64-bit (4n) boundaries.

[Table 6] Burst Sequence for Write

NOTE :

- 1) C0-C1 are assumed to be '0', and are not transmitted on the command bus.
- 2) The starting address is on 256-bit (16n) boundaries for Burst length 16.
- 3) The starting address is on 512-bit (32n) boundaries for Burst length 32.
- 4) C2-C3 shall be set to '0' for all Write operations.

MR2_Device Feature 2 (MA<7:0> = 02_H):

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
WR Lev	WLS	WL			RL		

Function	Register Type	Operand	Data	Notes
RL (Read latency)	Write-only	OP[2:0]	RL & nRTP for DBI-RD Disabled (MR3 OP[6]=0_B) 000_B: RL=6, nRTP=8 (Default) 001_B: RL=10, nRTP=8 010_B: RL=14, nRTP=8 011_B: RL=20, nRTP=8 100_B: RL=24, nRTP=10 101_B: RL=28, nRTP=12 110_B: RL=32, nRTP=14 111_B: RL=36, nRTP=16 RL & nRTP for DBI-RD Enabled (MR3 OP[6]=1_B) 000_B: RL= 6, nRTP=8 001_B: RL= 12, nRTP=8 010_B: RL= 16, nRTP=8 011_B: RL= 22, nRTP=8 100_B: RL= 28, nRTP=10 101_B: RL= 32, nRTP=12 110_B: RL= 36, nRTP=14 111_B: RL= 40, nRTP=16	1,3,4
WL (Write latency)		OP[5:3]	WL Set "A" (MR2 OP[6]=0_B) 000_B: WL=4 (Default) 001_B: WL=6 010_B: WL=8 011_B: WL=10 100_B: WL=12 101_B: WL=14 110_B: WL=16 111_B: WL=18 WL Set "B" (MR2 OP[6]=1_B) 000_B: WL=4 001_B: WL=8 010_B: WL=12 011_B: WL=18 100_B: WL=22 101_B: WL=26 110_B: WL=30 111_B: WL=34	1,3,4
WLS (Write latency set)		OP[6]	0_B: WL Set "A" (default) 1_B: WL Set "B"	1,3,4
WR Leveling		OP[7]	0_B: Disabled (default) 1_B: Enabled	2

- NOTE :**
- 1) See Latency Code Frequency Table for allowable frequency ranges for RL/WL/nWR/nRTP.
 - 2) After a MRW to set the Write Leveling Enable bit (OP[7]=1_B), the LPDDR4X-SDRAM device remains in the MRW state until another MRW command clears the bit (OP[7]=0_B). No other commands are allowed until the Write Leveling Enable bit is cleared.
 - 3) There are two physical registers assigned to each bit of this MR parameter, designated set point 0 and set point 1. Only the registers for the set point determined by the state of the FSP-WR bit (MR13 OP[6]) will be written to with an MRW command to this MR address, or read from with an MRR command to this address.
 - 4) There are two physical registers assigned to each bit of this MR parameter, designated set point 0 and set point 1. The device will operate only according to the values stored in the registers for the active set point, i.e., the set point determined by the state of the FSP-OP bit (MR13 OP[7]). The values in the registers for the inactive set point will be ignored by the device, and may be changed without affecting device operation.

MR3_I/O Configuration 1 (MA<7:0> = 03_H):

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
DBI-WR	DBI-RD	PDDS			PPRP	WR PST	PU-CAL

Function	Register Type	Operand	Data	Notes
PU-Cal (Pull-up Calibration Point)	Write-only	OP[0]	0_B : V _{DDQ} ×0.6 1_B : V _{DDQ} ×0.5 (default)	1,4
WR PST (WR Post-Amble Length)		OP[1]	0_B : WR Post-amble = 0.5×tCK (default) 1_B : WR Post-amble = 1.5×tCK (Vendor specific function)	2,3,5
Post Package Repair Protection		OP[2]	0_B : PPR protection disabled (default) 1_B : PPR protection enabled	6
PDDS (Pull-Down Drive Strength)		OP[5:3]	000_B : RFU 001_B : RZQ/1 010_B : RZQ/2 011_B : RZQ/3 100_B : RZQ/4 101_B : RZQ/5 110_B : RZQ/6 (default) 111_B : Reserved	1,2,3
DBI-RD (DBI-Read Enable)		OP[6]	0_B : Disabled (default) 1_B : Enabled	2,3
DBI-WR (DBI-Write Enable)		OP[7]	0_B : Disabled (default) 1_B : Enabled	2,3

NOTE :

- 1) All values are "typical". The actual value after calibration will be within the specified tolerance for a given voltage and temperature. Re-calibration may be required as voltage and temperature vary.
- 2) There are two physical registers assigned to each bit of this MR parameter, designated set point 0 and set point 1. Only the registers for the set point determined by the state of the FSP-WR bit (MR13 OP[6]) will be written to with an MRW command to this MR address, or read from with an MRR command to this address.
- 3) There are two physical registers assigned to each bit of this MR parameter, designated set point 0 and set point 1. The device will operate only according to the values stored in the registers for the active set point, i.e., the set point determined by the state of the FSP-OP bit (MR13 OP[7]). The values in the registers for the inactive set point will be ignored by the device, and may be changed without affecting device operation.
- 4) For dual channel devices, PU-CAL setting is required as the same value for both Ch.A and Ch.B before issuing ZQ Cal start command.
- 5) Refer to the supplier data sheet for vendor specific function. 1.5×tCK apply > 1.6GHz clock.
- 6) If MR3 OP[2] is set to 1b then PPR protection mode is enabled. The PPR Protection bit is a sticky bit and can only be set to 0b by a power on reset.
MR4 OP[4] controls entry to PPR Mode. If PPR protection is enabled then DRAM will not allow writing of 1 to MR4 OP[4].

MR4_Refresh rate (MA<7:0> = 04_H)

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
TUF	Thermal Offset		PPRE	SR Abort	Refresh Rate		

Function	Register Type	Operand	Data	Notes
Refresh Rate	Read-only	OP[2:0]	000_B : SDRAM Low temperature operating limit exceeded 001_B : 4x refresh 010_B : 2x refresh 011_B : 1x refresh (default) 100_B : 0.5x refresh 101_B : 0.25x refresh, no de-rating 110_B : 0.25x refresh, with de-rating 111_B : SDRAM High temperature operating limit exceeded	1,2,3,4 7,8,9
SR Abort (Self Refresh Abort)	Write-only	OP[3]	0_B : Disable (default) 1_B : Enable	9,11
PPRE (Post-package repair entry/exit)	Write-only	OP[4]	0_B : Exit PPR mode (default) 1_B : Enter PPR mode	5,9
Thermal Offset (Vender Specific Function)	Write-only	OP[6:5]	00_B : No offset, 0~5°C gradient (default) 01_B : 5°C offset, 5~10°C gradient 10_B : 10°C offset, 10~15°C gradient 11_B : Reserved	10
TUF (Temperature Update Flag)	Read-only	OP[7]	0_B : No change in OP[2:0] since last MR4 read (default) 1_B : Change in OP[2:0] since last MR4 read	6,7,8

NOTE :

- 1) The refresh rate for each MR4 OP[2:0] setting applies to tREFI, tREFIpb and tREFW. OP[2:0]=011_B corresponds to a device temperature of 85°C. Other values require either a longer (2x, 4x) refresh interval at lower temperatures, or a shorter (0.5x, 0.25x) refresh interval at higher temperatures. If OP[2]=1_B, the device temperature is greater than 85°C.
- 2) At higher temperatures (>85°C), AC timing derating may be required. If derating is required the LPDDR4X-SDRAM will set OP[2:0]=110_B. See derating timing requirements in the AC Timing section.
- 3) DRAM vendors may or may not report all of the possible settings over the operating temperature range of the device. Each vendor guarantees that their device will work at any temperature within the range using the refresh interval requested by their device.
- 4) The device may not operate properly when OP[2:0]=000_B or 111_B.
- 5) Post-package repair can be entered or exited by writing to OP[4].
- 6) When OP[7]=1_B, the refresh rate reported in OP[2:0] has changed since the last MR4 read. A mode register read from MR4 will reset OP[7] to '0'.
- 7) OP[7]=0_B at power-up. OP[2:0] bits are valid after initialization sequence (Te).
- 8) See the section on "Temperature Sensor" for information on the recommended frequency of reading MR4.
- 9) OP[6:3] bits that can be written in this register. All other bits will be ignored by the DRAM during a MRW to this register.
- 10) Refer to the supplier data sheet for vender specific function.
- 11) Self refresh abort feature is available for higher density devices starting with 12Gb device.

MR5_Basic Configuration 1 (MA<7:0> = 05_H):

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
LPDDR4X Manufacturer ID							

Function	Register Type	Operand	Data	Notes
LPDDR4X Manufacturer ID	Read-only	OP[7:0]	0000 0001_B : Samsung	

MR6_Basic Configuration 2 (MA<7:0> = 06_H):

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
Revision ID-1							

Function	Register Type	Operand	Data	Notes
LPDDR4X Revision ID-1	Read-only	OP[7:0]	0000 1000_B : I-version	1

NOTE :

- 1) MR6 is vendor specific.

MR7_Basic Configuration 3 (MA<7:0> = 07_H):

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
Revision ID-2							

Function	Register Type	Operand	Data	Notes
LPDDR4X Revision ID-2	Read-only	OP[7:0]	0000 0000 _B	1

NOTE :

1) MR7 is vendor specific.

MR8_Basic Configuration 4 (MA<7:0> = 08_H) :

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
I/O width		Density				Type	

Function	Register Type	Operand	Data	Notes
Type	Read-only	OP[1:0]	00 _B : LPDDR4 S16 SDRAM (16n pre-fetch) Standard VDDQ(1.1V) only 10 _B : LPDDR4X S16 SDRAM (16n pre-fetch) LOW VDDQ(0.6V) support only All Others: Reserved	
Density		OP[5:2]	0000 _B : 4Gb dual channel die / 2Gb single channel die 0001 _B : 6Gb dual channel die / 3Gb single channel die 0010 _B : 8Gb dual channel die / 4Gb single channel die 0011 _B : 12Gb dual channel die / 6Gb single channel die 0100 _B : 16Gb dual channel die / 8Gb single channel die 0101 _B : 24Gb dual channel die / 12Gb single channel die 0110 _B : 32Gb dual channel die / 16Gb single channel die All Others: Reserved	
I/O width		OP[7:6]	00 _B : x16 (per channel) All Others : Reserved	

MR9_Test Mode (MA<7:0> = 09_H):

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
Vendor-specific Test Register							

NOTE :1) Only 00_H should be written to this register.**MR10_IO Calibration (MA<7:0> = 0A_H):**

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
(RFU)							ZQ-Reset

Function	Register Type	Operand	Data	Notes
ZQ-Reset	Write-only	OP[0]	0 _B : Normal Operation (Default) 1 _B : ZQ Reset	1,2

NOTE :

1) See ZQCal Timing Parameters for calibration latency and timing in AC Timing table.

2) If the ZQ-pin is connected to V_{DDQ} through RZQ, either the ZQ calibration function or default calibration (via ZQ-Reset) is supported. If the ZQ-pin is connected to V_{SS}, the device operates with default calibration, and ZQ calibration commands are ignored. In both cases, the ZQ connection shall not change after power is applied to the device.

MR11_ODT Feature (MA<7:0> = 0B_H):

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
(RFU)	CA ODT			(RFU)	DQ ODT		

Function	Register Type	Operand	Data	Notes
DQ ODT (DQ Bus Receiver On-Die-Termination)	Write-only	OP[2:0]	000_B : Disable (Default) 001_B : RZQ/1 010_B : RZQ/2 011_B : RZQ/3 100_B : RZQ/4 101_B : RZQ/5 110_B : RZQ/6 111_B : RFU	1,2,3
CA ODT (CA Bus Receiver On-Die-Termination)		OP[6:4]	000_B : Disable (Default) 001_B : RZQ/1 010_B : RZQ/2 011_B : RZQ/3 100_B : RZQ/4 101_B : RZQ/5 110_B : RZQ/6 111_B : RFU	1,2,3

NOTE :

- 1) All values are "typical". The actual value after calibration will be within the specified tolerance for a given voltage and temperature. Re-calibration may be required as voltage and temperature vary.
- 2) There are two physical registers assigned to each bit of this MR parameter, designated set point 0 and set point 1. Only the registers for the set point determined by the state of the FSP-WR bit (MR13 OP[6]) will be written to with an MRW command to this MR address, or read from with an MRR command to this address.
- 3) There are two physical registers assigned to each bit of this MR parameter, designated set point 0 and set point 1. The device will operate only according to the values stored in the registers for the active set point, i.e., the set point determined by the state of the FSP-OP bit (MR13 OP[7]). The values in the registers for the inactive set point will be ignored by the device, and may be changed without affecting device operation.

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MR12_VREF(CA) Setting/Range (MA<7:0> = 0C_H):

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
(RFU)	VR-CA	V _{REF(CA)}					

Function	Register Type	Operand	Data	Notes
V _{REF(CA)} (V _{REF(CA)} Setting)	Read/Write	OP[5:0]	000000 _B : -- Thru . 110010 _B : See table below All Others: Reserved	1,2,3,5, 6
VR-CA (V _{REF(CA)} Range)		OP[6]	0 _B : V _{REF(CA)} Range[0] enabled 1 _B : V _{REF(CA)} Range[1] enabled (default)	1,2,4,5, 6

- NOTE :**
- 1) This register controls the V_{REF(CA)} levels. Refer to Table 12, VREF Settings for Range[0] and Range[1].
 - 2) A read to this register places the contents of OP[7:0] on DQ[7:0]. Any RFU bits and unused DQ's shall be set to '0'. See the section on MRR Operation.
 - 3) A write to OP[5:0] sets the internal V_{REF(CA)} level for FSP[0] when MR13 OP[6]=0_B, or sets FSP[1] when MR13 OP[6]=1_B. The time required for V_{REF(CA)} to reach the set level depends on the step size from the current level to the new level. See the section on V_{REF(CA)} training for more information.
 - 4) A write to OP[6] switches the LPDDR4X-SDRAM between two internal V_{REF(CA)} ranges. The range (Range[0] or Range[1]) must be selected when setting the V_{REF(CA)} register. The value, once set, will be retained until overwritten, or until the next power-on or RESET event.
 - 5) There are two physical registers assigned to each bit of this MR parameter, designated set point 0 and set point 1. Only the registers for the set point determined by the state of the FSP-WR bit (MR13 OP[6]) will be written to with an MRW command to this MR address, or read from with an MRR command to this address.
 - 6) There are two physical registers assigned to each bit of this MR parameter, designated set point 0 and set point 1. The device will operate only according to the values stored in the registers for the active set point, i.e., the set point determined by the state of the FSP-OP bit (MR13 OP[7]). The values in the registers for the inactive set point will be ignored by the device, and may be changed without affecting device operation.

[Table 7] VREF Settings for Range[0] and Range[1]

Function	Operand	Range[0] Values (%of V _{DDQ})		Range[1] Values (%of V _{DDQ})		Notes
V _{REF} Settings for MR12	OP[5:0]	000000 _B : 15.0%	011010 _B : 30.5%	000000 _B : 32.9%	011010 _B : 48.5%	1,2,3
		000001 _B : 15.6%	011011 _B : 31.1%	000001 _B : 33.5%	011011 _B : 49.1%	
		000010 _B : 16.2%	011100 _B : 31.7%	000010 _B : 34.1%	011100 _B : 49.7%	
		000011 _B : 16.8%	011101 _B : 32.3%	000011 _B : 34.7%	011101 _B : 50.3% (Default)	
		000100 _B : 17.4%	011110 _B : 32.9%	000100 _B : 35.3%	011110 _B : 50.9%	
		000101 _B : 18.0%	011111 _B : 33.5%	000101 _B : 35.9%	011111 _B : 51.5%	
		000110 _B : 18.6%	100000 _B : 34.1%	000110 _B : 36.5%	100000 _B : 52.1%	
		000111 _B : 19.2%	100001 _B : 34.7%	000111 _B : 37.1%	100001 _B : 52.7%	
		001000 _B : 19.8%	100010 _B : 35.3%	001000 _B : 37.7%	100010 _B : 53.3%	
		001001 _B : 20.4%	100011 _B : 35.9%	001001 _B : 38.3%	100011 _B : 53.9%	
		001010 _B : 21.0%	100100 _B : 36.5%	001010 _B : 38.9%	100100 _B : 54.5%	
		001011 _B : 21.6%	100101 _B : 37.1%	001011 _B : 39.5%	100101 _B : 55.1%	
		001100 _B : 22.2%	100110 _B : 37.7%	001100 _B : 40.1%	100110 _B : 55.7%	
		001101 _B : 22.8%	100111 _B : 38.3%	001101 _B : 40.7%	100111 _B : 56.3%	
		001110 _B : 23.4%	101000 _B : 38.9%	001110 _B : 41.3%	101000 _B : 56.9%	
		001111 _B : 24.0%	101001 _B : 39.5%	001111 _B : 41.9%	101001 _B : 57.5%	
		010000 _B : 24.6%	101010 _B : 40.1%	010000 _B : 42.5%	101010 _B : 58.1%	
		010001 _B : 25.1%	101011 _B : 40.7%	010001 _B : 43.1%	101011 _B : 58.7%	
		010010 _B : 25.7%	101100 _B : 41.3%	010010 _B : 43.7%	101100 _B : 59.3%	
		010011 _B : 26.3%	101101 _B : 41.9%	010011 _B : 44.3%	101101 _B : 59.9%	
		010100 _B : 26.9%	101110 _B : 42.5%	010100 _B : 44.9%	101110 _B : 60.5%	
		010101 _B : 27.5%	101111 _B : 43.1%	010101 _B : 45.5%	101111 _B : 61.1%	
		010110 _B : 28.1%	110000 _B : 43.7%	010110 _B : 46.1%	110000 _B : 61.7%	
		010111 _B : 28.7%	110001 _B : 44.3%	010111 _B : 46.7%	110001 _B : 62.3%	
		011000 _B : 29.3%	110010 _B : 44.9%	011000 _B : 47.3%	110010 _B : 62.9%	
		011001 _B : 29.9%	All Others: Reserved	011001 _B : 47.9%	All Others: Reserved	

- NOTE:**
- 1) These values may be used for MR12 OP[5:0] to set the V_{REF(CA)} levels in the LPDDR4X-SDRAM.
 - 2) The range may be selected in the MR12 register by setting OP[6] appropriately.
 - 3) The MR12 registers represents either FSP[0] or FSP[1]. Two frequency-set-points each for CA and DQ are provided to allow for faster switching between terminated and un-terminated operation, or between different high-frequency setting which may use different terminations values.

MR13_CBT,RPT,VRO,VRCG,RRO,DM_DIS,FSP-WR, FSP-OP (MA<7:0> = 0D_H):

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
FSP-OP	FSP-WR	DMD	RRO	VRCG	VRO	RPT	CBT

Function	Register Type	Operand	Data	Notes
CBT (Command Bus Training)	Write-only	OP[0]	0_B : Normal Operation (default) 1_B : Command Bus Training Mode Enabled	1
RPT (Read Preamble Training Mode)		OP[1]	0_B : Disable (default) 1_B : Enable	
VRO (V _{REF} Output)		OP[2]	0_B : Normal operation (default) 1_B : Output the V _{REF(CA)} and V _{REF(DQ)} values on DQ bits	2
VRCG (V _{REF} Current Generator)		OP[3]	0_B : Normal operation (default) 1_B : V _{REF} fast response (high current) mode	3
RRO (Refresh Rate Option)		OP[4]	0_B : Disable codes 001 and 010 in MR4 OP[2:0] 1_B : Enable all codes in MR4 OP[2:0]	4,5
DMD (Data Mask Disable)		OP[5]	0_B : Data Mask Operation Enabled (default) 1_B : Data Mask Operation Disabled	6
FSP-WR (Frequency Set Point Write/Read)		OP[6]	0_B : Frequency-Set-Point [0] (default) 1_B : Frequency-Set-Point [1]	7
FSP-OP (Frequency Set Point Operation Mode)		OP[7]	0_B : Frequency-Set-Point [0] (default) 1_B : Frequency-Set-Point [1]	8

NOTE :

- 1) A write to set OP[0]=1_B causes the LPDDR4X-SDRAM to enter the Command Bus training mode. When OP[0]=1_B and CKE goes LOW, commands are ignored and the contents of CA[5:0] are mapped to the DQ bus. CKE must be brought HIGH before doing a MRW to clear this bit (OP[0]=0_B) and return to normal operation. See the Command Bus Training section for more information.
- 2) When set, the LPDDR4X-SDRAM will output the V_{REF(CA)} and V_{REF(DQ)} voltages on DQ pins. Only the "active" frequency-set-point, as defined by MR13 OP[7], will be output on the DQ pins. This function allows an external test system to measure the internal VREF levels. The DQ pins used for V_{REF} output are vendor specific.
- 3) When OP[3]=1_B, the V_{REF} circuit uses a high-current mode to improve V_{REF} settling time.
- 4) MR13 OP[4] RRO bit is valid only when MR0 OP[0]=1_B. For LPDDR4X devices with MR0 OP[0]=0_B, MR4 OP[2:0] bits are not dependent on MR13 OP4.
- 5) When OP[4]=0_B, only 001_B and 010_B in MR4 OP[2:0] are disabled. LPDDR4X devices must report 011_B instead of 001_B or 010_B in this case. Controller should follow the refresh mode reported by MR4 OP[2:0], regardless of RRO setting. TCSR function does not depend on RRO setting.
- 6) When enabled (OP[5]=0_B) data masking is enabled for the device. When disabled (OP[5]=1_B), masked write command is illegal. See LPDDR4X Data Mask (DM) and Data Bus Inversion (DBI_{dc}) Function in operation timing datasheet.
- 7) FSP-WR determines which frequency-set-point registers are accessed with MRW commands for the following functions such as V_{REF(CA)} Setting, V_{REF(CA)} Range, V_{REF(DQ)} Setting, V_{REF(DQ)} Range. For more information, refer to Frequency Set Point section in operations and timing spec.
- 8) FSP-OP determines which frequency-set-point register values are currently used to specify device operation for the following functions such as V_{REF(CA)} Setting, V_{REF(CA)} Range, V_{REF(DQ)} Setting, V_{REF(DQ)} Range. For more information, refer to Frequency Set Point section in operations and timing spec.

MR14_VREF(DQ) Setting/Range (MA<7:0> = 0E_H):

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
(RFU)	VR(DQ)	V _{REF} (DQ)					

Function	Register Type	Operand	Data	Notes
V _{REF} (DQ) (V _{REF} (DQ) Setting)	Read/Write	OP[5:0]	000000 _B : -- Thru . 110010 _B : See table below All Others: Reserved	1,2,3,5,6
V _{REF} (DQ) (V _{REF} (DQ) Range)		OP[6]	0 _B : V _{REF} (DQ) Range [0] enabled 1 _B : V _{REF} (DQ) Range [1] enabled (default)	1,2,4,5,6

- NOTE :**
- 1) This register controls the V_{REF}(DQ) levels for Frequency-Set-Point[1:0]. Values from either VR(DQ)[0] or VR(DQ)[1] may be selected by setting OP[6] appropriately.
 - 2) A read (MRR) to this register places the contents of OP[7:0] on DQ[7:0]. Any RFU bits and unused DQ's shall be set to '0'. See the section on MRR Operation.
 - 3) A write to OP[5:0] sets the internal V_{REF}(DQ) level for FSP[0] when MR13 OP[6]=0_B, or sets FSP[1] when MR13 OP[6]=1_B. The time required for V_{REF}(DQ) to reach the set level depends on the step size from the current level to the new level. See the section on V_{REF}(DQ) training for more information.
 - 4) A write to OP[6] switches the LPDDR4X-SDRAM between two internal V_{REF}(DQ) ranges. The range (Range[0] or Range[1]) must be selected when setting the V_{REF}(DQ) register. The value, once set, will be retained until overwritten, or until the next power-on or RESET event.
 - 5) There are two physical registers assigned to each bit of this MR parameter, designated set point 0 and set point 1. Only the registers for the set point determined by the state of the FSP-WR bit (MR13 OP[6]) will be written to with an MRW command to this MR address, or read from with an MRR command to this address.
 - 6) There are two physical registers assigned to each bit of this MR parameter, designated set point 0 and set point 1. The device will operate only according to the values stored in the registers for the active set point, i.e., the set point determined by the state of the FSP-OP bit (MR13 OP[7]). The values in the registers for the inactive set point will be ignored by the device, and may be changed without affecting device operation.

[Table 8] VREF Settings for Range[0] and Range[1]

Function	Operand	Range[0] Values (%of VDDQ)		Range[1] Values (%of VDDQ)		Notes
VREF Settings for MR14	OP[5:0]	000000 _B : 15.0%	011010 _B : 30.5%	000000 _B : 32.9%	011010 _B : 48.5%	1,2,3
		000001 _B : 15.6%	011011 _B : 31.1%	000001 _B : 33.5%	011011 _B : 49.1%	
		000010 _B : 16.2%	011100 _B : 31.7%	000010 _B : 34.1%	011100 _B : 49.7%	
		000011 _B : 16.8%	011101 _B : 32.3%	000011 _B : 34.7%	011101 _B : 50.3% (Default)	
		000100 _B : 17.4%	011110 _B : 32.9%	000100 _B : 35.3%	011110 _B : 50.9%	
		000101 _B : 18.0%	011111 _B : 33.5%	000101 _B : 35.9%	011111 _B : 51.5%	
		000110 _B : 18.6%	100000 _B : 34.1%	000110 _B : 36.5%	100000 _B : 52.1%	
		000111 _B : 19.2%	100001 _B : 34.7%	000111 _B : 37.1%	100001 _B : 52.7%	
		001000 _B : 19.8%	100010 _B : 35.3%	001000 _B : 37.7%	100010 _B : 53.3%	
		001001 _B : 20.4%	100011 _B : 35.9%	001001 _B : 38.3%	100011 _B : 53.9%	
		001010 _B : 21.0%	100100 _B : 36.5%	001010 _B : 38.9%	100100 _B : 54.5%	
		001011 _B : 21.6%	100101 _B : 37.1%	001011 _B : 39.5%	100101 _B : 55.1%	
		001100 _B : 22.2%	100110 _B : 37.7%	001100 _B : 40.1%	100110 _B : 55.7%	
		001101 _B : 22.8%	100111 _B : 38.3%	001101 _B : 40.7%	100111 _B : 56.3%	
		001110 _B : 23.4%	101000 _B : 38.9%	001110 _B : 41.3%	101000 _B : 56.9%	
		001111 _B : 24.0%	101001 _B : 39.5%	001111 _B : 41.9%	101001 _B : 57.5%	
		010000 _B : 24.6%	101010 _B : 40.1%	010000 _B : 42.5%	101010 _B : 58.1%	
		010001 _B : 25.1%	101011 _B : 40.7%	010001 _B : 43.1%	101011 _B : 58.7%	
		010010 _B : 25.7%	101100 _B : 41.3%	010010 _B : 43.7%	101100 _B : 59.3%	
		010011 _B : 26.3%	101101 _B : 41.9%	010011 _B : 44.3%	101101 _B : 59.9%	
		010100 _B : 26.9%	101110 _B : 42.5%	010100 _B : 44.9%	101110 _B : 60.5%	
		010101 _B : 27.5%	101111 _B : 43.1%	010101 _B : 45.5%	101111 _B : 61.1%	
		010110 _B : 28.1%	110000 _B : 43.7%	010110 _B : 46.1%	110000 _B : 61.7%	
		010111 _B : 28.7%	110001 _B : 44.3%	010111 _B : 46.7%	110001 _B : 62.3%	
		011000 _B : 29.3%	110010 _B : 44.9%	011000 _B : 47.3%	110010 _B : 62.9%	
		011001 _B : 29.9%	All Others: Reserved	011001 _B : 47.9%	All Others: Reserved	

- NOTE:**
- 1) These values may be used for MR14 OP[5:0] to set the V_{REF}(DQ) levels in the LPDDR4X-SDRAM.
 - 2) The range may be selected in the MR14 register by setting OP[6] appropriately.
 - 3) The MR14 registers represents either FSP[0] or FSP[1]. Two frequency-set-points each for CA and DQ are provided to allow for faster switching between terminated and un-terminated operation, or between different high-frequency setting which may use different terminations values.

MR15_Lower-Byte Invert for DQ Calibration (MA<7:0> = 0F_H):

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
Lower-Byte Invert Register for DQ Calibration							

Function	Register Type	Operand	Data	Notes
Lower-Byte Invert for DQ Calibration	Write-only	OP[7:0]	The following values may be written for any operand OP[7:0], and will be applied to the corresponding DQ locations DQ[7:0] within a byte lane: 0_B: Do not invert 1_B: Invert the DQ Calibration patterns in MR32 and MR40 Default value for OP[7:0]=55_H	1,2,3

NOTE :

- 1) This register will invert the DQ Calibration pattern found in MR32 and MR40 for any single DQ, or any combination of DQ's. Example: If MR15 OP[7:0]=00010101_B, then the DQ Calibration patterns transmitted on DQ[7,6,5,3,1] will not be inverted, but the DQ Calibration patterns transmitted on DQ[4,2,0] will be inverted.
- 2) DMI[0] is not inverted, and always transmits the "true" data contained in MR32/MR40.
- 3) No Data Bus Inversion (DBI) function is enacted during DQ Read Calibration, even if DBI is enabled in MR3-OP[6].

[Table 9] MR15 Invert Register Pin Mapping

PIN	DQ0	DQ1	DQ2	DQ3	DMI0	DQ4	DQ5	DQ6	DQ7
MR15	OP0	OP1	OP2	OP3	NO-Invert	OP4	OP5	OP6	OP7

MR16_PASR_Bank Mask (MA<7:0> = 010_H):

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
PASR Bank Mask							

Function	Register Type	Operand	Data	Notes
Bank [7:0] Mask	Write-only	OP[7:0]	0_B: Bank Refresh Enabled (default) : Unmasked 1_B: Bank Refresh disabled : Masked	1

OP	Bank Mask	8-Bank SDRAM
0	XXXXXXX1	Bank 0
1	XXXXXX1X	Bank 1
2	XXXXX1XX	Bank 2
3	XXXX1XXX	Bank 3
4	XXX1XXXX	Bank 4
5	XX1XXXXX	Bank 5
6	X1XXXXXX	Bank 6
7	1XXXXXXX	Bank 7

NOTE :

- 1) When a mask bit is asserted (OP[n]=1), refresh to that bank is disabled.
- 2) PASR bank-masking is on a per-channel basis. The two channels on the die may have different bank masking in dual channel devices.

MR17_PASR Segment Mask (MA<7:0> = 011_H):

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
PASR Segment Mask							

Function	Register Type	Operand	Data	Notes
PASR Segment Mask	Write-only	OP[7:0]	0 _B : Segment Refresh enabled (default) 1 _B : Segment Refresh disabled	

Segment	OP	Segment Mask	2Gb per channel	3Gb per channel	4Gb per channel	6Gb per channel	8Gb per channel	12Gb per channel	16Gb per channel
			R13:11	R14:12	R14:12	R15:13	R15:13	R16:14	R16:14
0	0	XXXXXXXX1	000 _B						
1	1	XXXXXXXX1X	001 _B						
2	2	XXXXX1XX	010 _B						
3	3	XXXX1XXX	011 _B						
4	4	XXX1XXXX	100 _B						
5	5	XX1XXXXX	101 _B						
6	6	X1XXXXXX	110 _B	Not Allowed	110 _B	Not Allowed	110 _B	Not Allowed	110 _B
7	7	1XXXXXXX	111 _B		111 _B		111 _B		111 _B

NOTE :

- 1) This table indicates the range of row addresses in each masked segment. "X" is don't care for a particular segment.
- 2) PASR segment-masking is on a per-channel basis. The two channels on the die may have different segment masking in dual channel devices.
- 3) For 3Gb, 6Gb and 12Gb per channel densities, OP[7:6] must always be LOW (=00_B).

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MR18_IT-LSB (MA<7:0> = 12_H) :

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
DQS Oscillator Count-LSB							

Function	Register Type	Operand	Data	Notes
DQS Oscillator (WR Training DQS Oscillator)	Read-only	OP[7:0]	0-255 LSB DRAM DQS Oscillator Count	1,2,3

NOTE :

- 1) MR18 reports the LSB bits of the DRAM DQS Oscillator count. The DRAM DQS Oscillator count value is used to train DQS to the DQ data valid window. The value reported by the DRAM in this mode register can be used by the memory controller to periodically adjust the phase of DQS relative to DQ.
- 2) Both MR18 and MR19 must be read (MRR) and combined to get the value of the DQS Oscillator count.
- 3) A new MPC [Start DQS Oscillator] should be issued to reset the contents of MR18/MR19.

MR19_IT-MSB (MA<7:0> = 13_H) :

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
DQS Oscillator Count-MSB							

Function	Register Type	Operand	Data	Notes
DQS Oscillator (WR Training DQS Oscillator)	Read-only	OP[7:0]	0-255 MSB DRAM DQS Oscillator Count	1,2,3

NOTE :

- 1) MR19 reports the MSB bits of the DRAM DQS Oscillator count. The DRAM DQS Oscillator count value is used to train DQS to the DQ data valid window. The value reported by the DRAM in this mode register can be used by the memory controller to periodically adjust the phase of DQS relative to DQ.
- 2) Both MR18 and MR19 must be read (MRR) and combined to get the value of the DQS Oscillator count.
- 3) A new MPC [Start DQS Oscillator] should be issued to reset the contents of MR18/MR19.

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MR20_Upper-Byte Invert for DQ Calibration (MA<7:0> = 14_H):

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
Upper-Byte Invert Register for DQ Calibration							

Function	Register Type	Operand	Data	Notes
Upper-Byte Invert for DQ Calibration	Write-only	OP[7:0]	The following values may be written for any operand OP[7:0], and will be applied to the corresponding DQ locations DQ[15:8] within a byte lane: 0_B: Do not invert 1_B: Invert the DQ Calibration patterns in MR32 and MR40 Default value for OP[7:0] = 55_H	1,2

NOTE:

- 1) This register will invert the DQ Calibration pattern found in MR32 and MR40 for any single DQ, or any combination of DQ's. Example: If MR20 OP[7:0]=00010101B, then the DQ Calibration patterns transmitted on DQ[15,14,13,11,9] will not be inverted, but the DQ Calibration patterns transmitted on DQ[12,10,8] will be inverted.
- 2) DMI[1] is not inverted, and always transmits the "true" data contained in MR32/MR40.
- 3) No Data Bus Inversion (DBI) function is enacted during DQ Read Calibration, even if DBI is enabled in MR3-OP[6].

[Table 10] MR20 Invert Register Pin Mapping

PIN	DQ8	DQ9	DQ10	DQ11	DMI1	DQ12	DQ13	DQ14	DQ15
MR20	OP0	OP1	OP2	OP3	NO-Invert	OP4	OP5	OP6	OP7

MR21_Low Speed CA Buffer (MA<7:0> = 015_H):

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
(RFU)		Low Speed CA buffer	(RFU)				

Function	Register Type	Operand	Data	Notes
Low Speed CA Buffer	Write-only	OP[5]	0_B: Normal CA Buffer (Default) 1_B: Low Speed CA Buffer	1,2,3,4,5,6,7

NOTE :

- 1) Support for the Low Speed CA Buffer feature enabled by MR21 OP[5] is optional. Refer to manufacturer data sheet for availability.
- 2) Low Speed CA Buffer feature can enable lower power for some manufacturers' designs. The maximum clock speed for this mode is vendor-specific, but is not above 800 MHz. Refer to manufacturer data sheet for details.
- 3) There are two physical registers assigned to each bit of this MR parameter, designated set point 0 and set point 1. Only the registers for the set point determined by the state of the FSP-WR bit (MR13 OP[6]) will be written to with an MRW command to this MR address.
- 4) There are two physical registers assigned to each bit of this MR parameter, designated set point 0 and set point 1. The device will operate only according to the values stored in the registers for the active set point, i.e., the set point determined by the state of the FSP-OP bit (MR13 OP[7]). The values in the registers for the inactive set point will be ignored by the device, and may be changed without affecting device operation.
- 5) After completing all the other MRW commands to set the values besides MR21 OP[5] setting, MR21 OP[5] can be enabled to "High". Low Power CA Buffer cannot be enabled prior to full device initialization (completion of Step 9 in power up sequence).
- 6) Low speed CA buffer is allowed to be enabled only when CA ODT is disabled.
- 7) SDRAM not supporting Low Speed CA Buffer will ignore MR21 OP[5] setting.

MR22_ODT Feature (MA<7:0> = 16_H):

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
(RFU)		ODT-CA	ODT-CS	ODT-CK	SoC ODT		

Function	Register Type	Operand	Data	Notes
SoC ODT (Controller ODT Value for VOH calibration)	Write-only	OP[2:0]	000_B : Disable (Default) 001_B : RZQ/1 (illegal if MR3 OP0=0 _B) 010_B : RZQ/2 011_B : RZQ/3 (illegal if MR3 OP0=0 _B) 100_B : RZQ/4 101_B : RZQ/5 (illegal if MR3 OP0=0 _B) 110_B : RZQ/6 (illegal if MR3 OP0=0 _B) 111_B : RFU	1,2,3
ODT-CK		OP[3]	ODT bond PAD is ignored 0_B : ODT-CK Enable (Default) 1_B : ODT-CK Disable	2,3,4
ODT-CS		OP[4]	ODT bond PAD is ignored 0_B : ODT-CS Enable (Default) 1_B : ODT-CS Disable	2,3,4
ODT-CA		OP[5]	ODT bond PAD is ignored 0_B : ODT-CA Enable (default) 1_B : ODT-CA Disabled	2,3,4

NOTE :

- 1) All values are "typical".
- 2) There are two physical registers assigned to each bit of this MR parameter, designated set point 0 and set point 1. Only the registers for the set point determined by the state of the FSP-WR bit (MR13 OP[6]) will be written to with an MRW command to this MR address, or read from with an MRR command to this address.
- 3) There are two physical registers assigned to each bit of this MR parameter, designated set point 0 and set point 1. The device will operate only according to the values stored in the registers for the active set point, i.e., the set point determined by the state of the FSP-OP bit (MR13 OP[7]). The values in the registers for the inactive set point will be ignored by the device, and may be changed without affecting device operation.
- 4) LPDDR4X device ignore ODT bond PAD.

[Table 11] DRAM PU strength & SoC ODT relation

VOH CAL (MR3 OP0)		SoC ODT Value (MR22 OP[2:0])							
		000 _B	001 _B	010 _B	011 _B	100 _B	101 _B	110 _B	111 _B
0_B (VOH=VDDQ*3/5)	SoC ODT	Disable	illegal	RZQ/ 2=120ohm	illegal	RZQ/ 4=60ohm	illegal	illegal	RFU
	DRAM Pull-up	Default	N/A	RZQ/ 3=80ohm	N/A	RZQ/ 6=40ohm	N/A	N/A	RFU
1_B (VOH=VDDQ/2)	SoC ODT	Disable	RZQ/ 1=240ohm	RZQ/ 2=120ohm	RZQ/ 3=80ohm	RZQ/ 4=60ohm	RZQ/ 5=48ohm	RZQ/ 6=40ohm	RFU
	DRAM Pull-up	Default	RZQ/ 1=240ohm	RZQ/ 2=120ohm	RZQ/ 3=80ohm	RZQ/ 4=60ohm	RZQ/ 5=48ohm	RZQ/ 6=40ohm	RFU

A) There is no corresponding RZQ/x value for 001B,011B,101B,110B when MR3 OP0 = 0B to support 3/5 VDDQ VOH value.

MR23_DQS Interval Timer Run Time (MA<7:0> = 17_H):

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
DQS interval timer run time setting							

Function	Register Type	Operand	Data	Notes
DQS interval timer run time	Write-only	OP[7:0]	00000000_B : DQS interval timer stop via MPC Command (Default) 00000001_B : DQS timer stops automatically at 16 th clocks after timer start 00000010_B : DQS timer stops automatically at 32 nd clocks after timer start 00000011_B : DQS timer stops automatically at 48 th clocks after timer start 00000100_B : DQS timer stops automatically at 64 th clocks after timer start ----- Thru ----- 00111111_B : DQS timer stops automatically at (63X16) th clocks after timer start 01XXXXXX_B : DQS timer stops automatically at 2048 th clocks after timer start 10XXXXXX_B : DQS timer stops automatically at 4096 th clocks after timer start 11XXXXXX_B : DQS timer stops automatically at 8192 nd clocks after timer start	1,2

NOTE :

- 1) MPC command with OP[6:0]=1001101_B (Stop DQS Interval Oscillator) stops DQS interval timer in case of MR23 OP[7:0] = 00000000_B.
 2) MPC command with OP[6:0]=1001101_B (Stop DQS Interval Oscillator) is illegal with non-zero values in MR23 OP[7:0].

MR24_(RFU) (MA<7:0> = 18_H):

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MR25_PPR Resources (MA<7:0> = 19_H):

Mode Register 25 contains one bit of readout per bank indicating that at least one resource is available for Post Package Repair programming.

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
Bank 7	Bank 6	Bank 5	Bank 4	Bank 3	Bank 2	Bank 1	Bank 0

Function	Register Type	Operand	Data	Notes
PPR Resource	Read-only	OP[7:0]	0_B : PPR Resource is not available 1_B : PPR Resource is available	

MR26-29_(RFU) (MA<7:0> = 1A_H - 1D_H):**MR30_Reserved for Testing (MA<7:0> = 1E_H):**

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
Valid 0 or 1							

Function	Register Type	Operand	Data	Notes
SDRAM will ignore	Write-only	OP[7:0]	Don't care	1

NOTE :

1) This register is reserved for testing purposes. The logical data values written to OP[7:0] shall have no effect on SDRAM operation, however timings need to be observed as for any other MR access command.

MR31_Bytemode Vref Selection (MA<7:0> = 1F_H):

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
Bytemode Vref Selection				(RFU)			

Function	Register Type	Operand	Data	Notes
Bytemode Vref Selection Lower Byte	Write-only	OP[6]	0_B : x16 device and no Byte mode selection (Default) 1_B : Disable to update MR12/MR14 for lower byte	1,2,3
Bytemode Vref Selection Upper Byte		OP[7]	0_B : x16 device and no Byte mode selection (Default) 1_B : Disable to update MR12/MR14 for upper byte	1,2,3

NOTE :

- The byte mode Vref selection is optional. Please consult with vendors for the availability to support feature.
- When Byte mode Vref selection is applied, the non-targeted byte is required to disable to update VrefCA and VrefDQ setting, assigned in MR12 and MR14 OP[6:0], for the other targeted byte.
 - In order to update MR12/MR14 setting only for upper byte, it is required to disable byte mode selection on lower byte, as applying MR31 OP[7:6] = 01_B.
 - In order to update MR12/MR14 setting only for lower byte, it is required to disable byte mode selection on upper byte, as applying MR31 OP[7:6] = 10_B.
 - When OP[7:6] = 00_B is applied, both lower byte and upper byte will be updated.
- When the configuration is not composed of byte mode device, MR31 OP[7:6] shall be the default value, 00_B.

MR32_DQ Calibration Pattern A (MA<7:0>=20_H):

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
DQ Calibration Pattern "A" (default = 5A _H)							

Function	Register Type	Operand	Data	Notes
Return DQ Calibration Pattern MR32 + MR40	Write	OP[7:0]	X_B : An MPC command with OP[6:0]=1000011 _B causes the device to return the DQ Calibration Pattern contained in this register and (followed by) the contents of MR40. A default pattern "5A _H " is loaded at power-up or RESET, or the pattern may be overwritten with a MRW to this register. The contents of MR15 and MR20 will invert the data pattern for a given DQ (See MR15 for more information)	

MR33:38_(Do Not Use) (MA<7:0> = 21_H-26_H):

IF THERE IS ANY OTHER OPERATION TO IMPLEMENT IN ADDITION TO SPECIFICATION IN THE DATASHEET OR JEDEC STANDARD, PLEASE CONTACT EACH BRANCH OFFICE OR HEADQUARTERS OF SAMSUNG ELECTRONICS.

Rev. 1.0

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MR39_Reserved for Testing (MA<7:0> = 27_H):

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
Valid 0 or 1							

Function	Register Type	Operand	Data	Notes
SDRAM will ignore	Write-only	OP[7:0]	Don't care	1

NOTE :

- 1) This register is reserved for testing purposes. The logical data values written to OP[7:0] shall have no effect on SDRAM operation, however timings need to be observed as for any other MR access command.

MR40_DQ Calibration Pattern B (MA<7:0>=28_H):

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
DQ Calibration Pattern "B" (default = 3C _H)							

Function	Register Type	Operand	Data	Notes
Return DQ Calibration Pattern MR32 + MR40	Write-only	OP[7:0]	X_B : A default pattern "3C _H " is loaded at power-up or RESET, or the pattern may be overwritten with a MRW to this register. See MR32,for more information.	1,2,3

NOTE :

- The pattern contained in MR40 is concatenated to the end of MR32 and transmitted on DQ[15:0] and DMI[1:0] when DQ Read Calibration is initiated via a MPC command. The pattern transmitted serially on each data lane, organized "little endian" such that the low-order bit in a byte is transmitted first. If the data pattern in MR40 is 27_H, then the first bit transmitted will be a '1', followed by '1', '1', '0', '0', '1', '0', and '0'. The bit stream will be 00100111_B.
- MR15 and MR20 may be used to invert the MR32/MR40 data patterns on the DQ pins. See MR15 and MR22 for more information. Data is never inverted on the DMI[1:0] pins.
- The data pattern is not transmitted on the DMI[1:0] pins if DBI-RD is disabled via MR3 OP[6].
- No Data Bus Inversion (DBI) function is enacted during DQ Read Calibration, even if DBI is enabled in MR3 OP[6].

MR41:47_ (Do Not Use)(MA<7:0> = 29_H-2F_H):

MR48:50_(RFU) (MA<7:0> = 30_H - 32_H) :

MR51_Single Ended RDQS, WDQS, Clock (MA<7:0> = 33_H) :

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
(RFU)				Single ended Clock	Single ended WDQS	Single ended RDQS	(RFU)

Function	Register Type	Operand	Data	Notes
Single ended RDQS	Write-only	OP[1]	0 _B : Differential Read DQS (Default) 1 _B : Single ended Read DQS	1,2,3,4,5
Single ended WDQS		OP[2]	0 _B : Differential Write DQS (Default) 1 _B : Single ended Write DQS	1,2,3,4,6
Single ended Clock		OP[3]	0 _B : Differential Clock (Default), CK _t /CK _c 1 _B : Single ended Clock, Only CK _t	1,2,3,4,7

NOTE :

- 1) The Single-ended mode maximum speed is 800MHz.
- 2) Device support for single ended mode features (MR51 OP[3:1]) is indicated in MR0 OP[5].
- 3) There are two physical registers assigned to each bit of this MR parameter, designated set point 0 and set point 1. Only the registers for the set point determined by the state of the FSP-WR bit (MR13 OP[6]) will be written to with an MRW command to this MR address.
- 4) There are two physical registers assigned to each bit of this MR parameter, designated set point 0 and set point 1. The device will operate only according to the values stored in the registers for the active set point, i.e., the set point determined by the state of the FSP-OP bit (MR13 OP[7]). The values in the registers for the inactive set point will be ignored by the device, and may be changed without affecting device operation.
- 5) When single ended RDQS mode is enabled (MR51 OP[1] =1B), DRAM drives Read DQSB low or Hi-Z.
- 6) When single ended WDQS mode is enabled (MR51 OP[2] =1B), Write DQSB is required to be at a valid logic level. A valid Write DQSB signal will meet this requirement.
- 7) When single ended Clock mode is enabled (MR51 OP[3] =1B), CK_c is required to be at a valid logic level. A valid CK_c signal will meet this requirement.

When DRAM is operating with single-ended mode, both CLKB and write DQSB shall be on "Low" state at all times whereas read DQSB is always on "Hi-Z" state. Refer to the table below.

		Differential Mode	Single-Ended Mode
CLK	CLK	Valid	Valid
	CLKB	Valid	0
Write DQS	DQS	Valid	Valid
	DQSB	Valid	0
Read DQS	DQS	Valid	Valid
	DQSB	Valid	Hi-Z

MR52:63_(RFU) (MA<7:0> = 34_H - 3F_H) :

3.0 TRUTH TABLES

Operation or timing that is not specified is illegal, and after such an event, in order to guarantee proper operation, the LPDDR4X device must be reset or power-cycled and then restarted through the specified initialization sequence before normal operation can continue.
CKE signal has to be held High when the commands listed in the command truth table input.

[Table 12] Command truth table

SDRAM Command	SDR Com- mand Pins	SDR CA pins (6)						CK_t edge	Notes
SDRAM Command	CS	CA0	CA1	CA2	CA3	CA4	CA5	CK_t edge	Notes
Deselect (DES)	L	X						R1	1,2
Multi-Purpose Command (MPC)	H	L	L	L	L	L	OP6	R1	1,9
	L	OP0	OP1	OP2	OP3	OP4	OP5	R2	
Precharge (PRE) (Per Bank, All Bank)	H	L	L	L	L	H	AB	R1	1,2,3,4
	L	BA0	BA1	BA2	V	V	V	R2	
Refresh (REF) (Per Bank, All Bank)	H	L	L	L	H	L	AB	R1	1,2,3,4
	L	BA0	BA1	BA2	V	V	V	R2	
Self Refresh Entry (SRE)	H	L	L	L	H	H	V	R1	1,2
	L	V						R2	
Write-1 (WR-1)	H	L	L	H	L	L	BL	R1	1,2,3,6,7,9
	L	BA0	BA1	BA2	V	C9	AP	R2	
Self Refresh Exit (SRX)	H	L	L	H	L	H	V	R1	1,2
	L	V						R2	
Mask Write-1 (MWR-1)	H	L	L	H	H	L	L	R1	1,2,3,5,6,9
	L	BA0	BA1	BA2	V	C9	AP	R2	
RFU	H	L	L	H	H	H	V	R1	1,2
	L	V						R2	
Read-1 (RD-1)	H	L	H	L	L	L	BL	R1	1,2,3,6,7,9
	L	BA0	BA1	BA2	V	C9	AP	R2	
CAS-2 (Write-2, Mask Write-2, Read-2, MRR-2, MPC)	H	L	H	L	L	H	C8	R1	1,8,9
	L	C2	C3	C4	C5	C6	C7	R2	
RFU	H	L	H	L	H	L	V	R1	1,2
	L	V						R2	
RFU	H	L	H	L	H	H	V	R1	1,2
	L	V						R2	
Mode Register Write-1 (MRW-1)	H	L	H	H	L	L	OP7	R1	1,11
	L	MA0	MA1	MA2	MA3	MA4	MA5	R2	
Mode Register Write-2 (MRW-2)	H	L	H	H	L	H	OP6	R1	1,11
	L	OP0	OP1	OP2	OP3	OP4	OP5	R2	
Mode Register Read-1 (MRR-1)	H	L	H	H	H	L	V	R1	1,2,12
	L	MA0	MA1	MA2	MA3	MA4	MA5	R2	
RFU	H	L	H	H	H	H	V	R1	1,2
	L	V						R2	
Activate-1 (ACT-1)	H	H	L	R12	R13	R14	R15	R1	1,2,3,10
	L	BA0	BA1	BA2	V	R10	R11	R2	
Activate-2 (ACT-2)	H	H	H	R6	R7	R8	R9	R1	1,10
	L	R0	R1	R2	R3	R4	R5	R2	

NOTE:

- 1) All LPDDR4X commands except for Deselect are 2 clock cycle long and defined by states of CS and CA[5:0] at the first rising edge of clock. Deselect command is 1 clock cycle long.
- 2) "V" means "H" or "L" (a defined logic level). "X" means don't care in which case CA[5:0] can be floated.
- 3) Bank addresses BA[2:0] determine which bank is to be operated upon.
- 4) AB "HIGH" during Precharge or Refresh command indicates that command must be applied to all banks and bank address is a don't care.
- 5) Mask Write-1 command supports only BL 16. For Mask Write-1 command, CA5 must be driven LOW on first rising clock cycle (R1).
- 6) AP "HIGH" during Write-1, Mask Write-1 or Read-1 commands indicates that an auto-precharge will occur to the bank associated with the Write, Mask Write or Read command.
- 7) If Burst Length on-the-fly is enabled, BL "HIGH" during Write-1 or Read-1 command indicates that Burst Length should be set on-the-fly to BL=32. BL "LOW" during Write-1 or Read-1 command indicates that Burst Length should be set on-the-fly to BL=16. If Burst Length on-the-fly is disabled, then BL must be driven to defined logic level "H" or "L".
- 8) For CAS-2 commands (Write-2 or Mask Write-2 or Read-2 or MRR-2 or MPC (Only Write FIFO, Read FIFO & Read DQ Calibration), C[1:0] are not transmitted on the CA[5:0] bus and are assumed to be zero. Note that for CAS-2 Write-2 or CAS-2 Mask Write-2 command, C[3:2] must be driven LOW.
- 9) Write-1 or Mask Write-1 or Read-1 or Mode Register Read-1 or MPC (Only Write FIFO, Read FIFO & Read DQ Calibration) command must be immediately followed by CAS-2 command consecutively without any other command in between. Write-1 or Mask Write-1 or Read-1 or mode register Read-1 or MPC (Only Write FIFO, Read FIFO & Read DQ Calibration) command must be issued first before issuing CAS-2 command. MPC (Only Start & Stop DQS Oscillator, Start & Latch ZQ Calibration) commands do not require CAS-2 command; they require two additional DES or NOP commands consecutively before issuing any other commands.
- 10) Activate-1 command must be immediately followed by Activate-2 command consecutively without any other command in between. Activate-1 command must be issued first before issuing Activate-2 command. Once Activate-1 command is issued, Activate-2 command must be issued before issuing another Activate-1 command.
- 11) MRW-1 command must be immediately followed by MRW-2 command consecutively without any other command in between. MRW-1 command must be issued first before issuing MRW-2 command.
- 12) MRR-1 command must be immediately followed by CAS-2 command consecutively without any other command in between. MRR-1 command must be issued first before issuing CAS-2 command.

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3.1 CKE Truth Tables

[Table 13] LPDDR4X : CKE Table 1), 2), 3), 4), 8)

Device Current State	CKE _{n-1}	CKE _n	Command n	Operation	Device Next State	Notes
Active Power Down	L	L	X	Maintain Active Power Down	Active Power Down	
	L	H	Deselect	Exit Active Power Down	Active	5,6
Idle Power Down	L	L	X	Maintain Idle Power Down	Idle Power Down	
	L	H	Deselect	Exit Idle Power Down	Idle	5,6
Self Refresh	L	L	X	Maintain power-down state within Self Refresh	Self Refresh	
	L	H	Deselect	Exit SREF power-down, enable command decode	Self Refresh	5,6,7
	H	L	Deselect	Enter SREF Power-Down, disable command decode	Self Refresh	5,7
	H	H	See Note 7	See Note 7	Self Refresh	7
Bank(s) Active	H	L	Deselect	Enter Active Power Down	Active Power Down	5
All Banks Idle	H	L	Deselect	Enter Idle Power Down	Idle Power Down	5, 8
Command Entry	H	H	Refer to the Command Truth Table			

NOTE :

- 1) CKE is a strictly asynchronous input, and as such, has no relationship to CK.
- 2) "X" means "don't care."
- 3) "Current State" is the state of the LPDDR4X-SDRAM prior to a toggle of CKE.
- 4) "CKEn-1" is the logic state of CKE prior to a CKE toggle event, and "CKEn" is the state of CKE after the toggle event.
- 5) "Deselect" is the only valid command that can be present on the bus when CKE is toggled.
- 6) Power-Down exit time (tXP) should elapse before a command other than Deselect is issued. The clock must toggle at least twice during the tXP period, and must be stable before issuing a command.
- 7) When the device is in Self.Refresh, only MRR, MRW, or MPC commands are allowed. Certain restrictions apply to changing register contents via a MRW command during SREF. See MRW section for more information.
- 8) In the case of ODT disabled, all DQ output shall be Hi-Z. In the case of ODT enabled, all DQ shall be terminated to VSSQ.

3.2 State Truth Table

The truth tables provide complementary information to the state diagram, they clarify the device behavior and the applied restrictions when considering the actual state of all banks.

[Table 14] Current State Bank n - Command to Bank n

Current State	Command	Operation	Next State	NOTES
Any	NOP	Continue previous operation	Current State	
Idle	ACTIVATE	Select and activate row	Active	
	Refresh (Per Bank)	Begin to refresh	Refreshing (Per Bank)	6
	Refresh (All Bank)	Begin to refresh	Refreshing (All Bank)	7
	MRW	Write value to Mode Register	MR Writing	7
	MRR	Read value from Mode Register	Idle MR Reading	
	Precharge	Deactivate row in bank or banks	Precharging	8, 13
Row Active	Read	Select column, and start read burst	Reading	10
	Write	Select column, and start write burst	Writing	10
	MRR	Read value from Mode Register	Active MR Reading	
	Precharge	Deactivate row in bank or banks	Precharging	8
Reading	Read	Select column, and start new read burst	Reading	9, 10
	Write	Select column, and start write burst	Writing	9, 10, 11
Writing	Write	Select column, and start new write burst	Writing	9, 10
	Read	Select column, and start read burst	Reading	9, 10, 12

NOTE :

- The table applies when both CKEn-1 and CKEn are HIGH, and after t_{XSR} or t_{XP} has been met if the previous state was Self Refresh or Power Down.
- All states and sequences not shown are illegal or reserved.
- Current State Definitions:
 - Idle: The bank or banks have been precharged, and tRP has been met.
 - Active: A row in the bank has been activated, and tRCD has been met. No data bursts / accesses and no register accesses are in progress.
 - Reading: A Read burst has been initiated, with Auto Precharge disabled.
 - Writing: A Write burst has been initiated, with Auto Precharge disabled.
- The following states must not be interrupted by a command issued to the same bank. NOP commands or allowable commands to the other bank should be issued on any clock edge occurring during these states. Allowable commands to the other banks are determined by its current state and LPDDR4 Pad definition and LPDDR4 SDRAM addressing.
 - Precharging: starts with the registration of a Precharge command and ends when tRP is met. Once tRP is met, the bank will be in the idle state.
 - Row Activating: starts with registration of an Activate command and ends when tRCD is met. Once tRCD is met, the bank will be in the 'Active' state.
 - Read with AP Enabled: starts with the registration of the Read command with Auto Precharge enabled and ends when tRP has been met. Once tRP has been met, the bank will be in the idle state.
 - Write with AP Enabled: starts with registration of a Write command with Auto Precharge enabled and ends when tRP has been met. Once tRP is met, the bank will be in the idle state.
- The following states must not be interrupted by any executable command; NOP commands must be applied to each positive clock edge during these states.
 - Refreshing (Per Bank): starts with registration of a Refresh (Per Bank) command and ends when tRFCpb is met. Once tRFCpb is met, the bank will be in an 'idle' state.
 - Refreshing (All Bank): starts with registration of an Refresh (All Bank) command and ends when tRFCab is met. Once tRFCab is met, the device will be in an 'all banks idle' state.
 - Idle MR Reading: starts with the registration of a MRR command and ends when tMRR has been met. Once tMRR has been met, the bank will be in the Idle state.
 - Active MR Reading: starts with the registration of a MRR command and ends when tMRR has been met. Once tMRR has been met, the bank will be in the Active state.
 - Idle MR Writing: starts with the registration of a MRW command and ends when tMRW has been met. Once tMRW has been met, the bank will be in the Idle state.
 - Active MR Writing: starts with the registration of a MRW command and ends when tMRW has been met. Once tMRW has been met, the bank will be in the Active state.
 - Precharging All: starts with the registration of a Precharge-All command and ends when tRP is met. Once tRP is met, the bank will be in the idle state.
- Bank-specific; requires that the bank is idle and no bursts are in progress.
- Not bank-specific; requires that all banks are idle and no bursts are in progress.
- This command may or may not be bank specific. If all banks are being precharged, they must be in a valid state for precharging.
- A command other than NOP should not be issued to the same bank while a Read or Write burst with Auto Precharge is enabled.
- The new Read or Write command could be Auto Precharge enabled or Auto Precharge disabled.
- A Write command may be applied after the completion of the Read burst; burst terminates are not permitted.
- A Read command may be applied after the completion of the Write burst, burst terminates are not permitted.
- If a Precharge command is issued to a bank in the Idle state, tRP shall still apply.

[Table 15] Current State Bank n - Command to Bank m

Current State of Bank n	Command for Bank m	Operation	Next State for Bank m	NOTES
Any	NOP	Continue previous operation	Current State of Bank m	
Idle	Any	Any command allowed to Bank m	-	
Row Activating, Active, or Precharging	Activate	Select and activate row in Bank m	Active	6
	Read	Select column, and start read burst from Bank m	Reading	7
	Write	Select column, and start write burst to Bank m	Writing	7
	Precharge	Deactivate row in bank or banks	Precharging	8
	MRR	Read value from Mode Register	Idle MR Reading or Active MR Reading	9,10
Reading (Autoprecharge disabled)	Read	Select column, and start read burst from Bank m	Reading	7
	Write	Select column, and start write burst to Bank m	Writing	7,12
	Activate	Select and activate row in Bank m	Active	
	Precharge	Deactivate row in bank or banks	Precharging	8
Writing/Masked Writing (Autoprecharge disabled)	Read	Select column, and start read burst from Bank m	Reading	7,14
	Write	Select column, and start write burst to Bank m	Writing	7
	Activate	Select and activate row in Bank m	Active	
	Precharge	Deactivate row in bank or banks	Precharging	8
Reading with Autoprecharge	Read	Select column, and start read burst from Bank m	Reading	7,13
	Write	Select column, and start write burst to Bank m	Writing	7,12,13
	Activate	Select and activate row in Bank m	Active	
	Precharge	Deactivate row in bank or banks	Precharging	8
Writing/Masked Writing with Autoprecharge	Read	Select column, and start read burst from Bank m	Reading	7,13,14
	Write	Select column, and start write burst to Bank m	Writing	7,13
	Activate	Select and activate row in Bank m	Active	
	Precharge	Deactivate row in bank or banks	Precharging	8

NOTE :

- The table applies when both CKEn-1 and CKEn are HIGH, and after t_{XSR} or t_{XP} has been met if the previous state was Self Refresh or Power Down.
- All states and sequences not shown are illegal or reserved.
- Current State Definitions:
 - Idle: The bank has been precharged, and t_{RP} has been met.
 - Active: A row in the bank has been activated, and t_{RCD} has been met. No data bursts/accesses and no register accesses are in progress.
 - Reading: A Read burst has been initiated, with Auto Precharge disabled.
 - Writing: A Write burst has been initiated, with Auto Precharge disabled.
- Refresh, Self-Refresh, and Mode register Write commands may only be issued when all bank are idle.
- The following states must not be interrupted by any executable command; NOP commands must be applied during each clock cycle while in these states:
 - Idle MR Reading: starts with the registration of a MRR command and ends when t_{MRR} has been met. Once t_{MRR} has been met, the bank will be in the Idle state.
 - Active MR Reading: starts with the registration of a MRR command and ends when t_{MRR} has been met. Once t_{MRR} has been met, the bank will be in the Active state.
 - Idle MR Writing: starts with the registration of a MRW command and ends when t_{MRW} has been met. Once t_{MRW} has been met, the bank will be in the Idle state.
 - Active MR Writing: starts with the registration of a MRW command and ends when t_{MRW} has been met. Once t_{MRW} has been met, the bank will be in the Active state.
- t_{RRD} must be met between Activate command to Bank n and a subsequent Activate command to Bank m. Additionally, in the case of multiple banks activated, t_{FAW} must be satisfied.
- Reads or Writes listed in the Command column include Reads and Writes with Auto Precharge enabled and Reads and Writes with Auto Precharge disabled.
- This command may or may not be bank specific. If all banks are being precharged, they must be in a valid state for precharging.
- MRR is allowed during the Row Activating state (Row Activating starts with registration of an Activate command and ends when t_{RCD} is met.)
- MRR is allowed during the Precharging state. (Precharging starts with registration of a Precharge command and ends when t_{RP} is met.)
- The next state for Bank m depends on the current state of Bank m (Idle, Row Activating, Precharging, or Active). The reader shall note that the state may be in transition when a MRR is issued. Therefore, if Bank m is in the Row Activating state and Precharging, the next state may be Active and Precharge dependent upon t_{RCD} and t_{RP} respectively.
- A Write command may be applied after the completion of the Read burst, burst terminates are not permitted.
- Read with auto precharge enabled or a Write with Auto Precharge enabled may be followed by any valid command to other banks provided that the timing restrictions described in the Precharge & Auto Precharge clarification table are followed.
- A Read command may be applied after the completion of the Write burst, burst terminates are not permitted.

4.0 ABSOLUTE MAXIMUM DC RATINGS

Stresses greater than those listed may cause permanent damage to the device.

This is a stress rating only, and functional operation of the device at these or any other conditions above those indicated in the operational sections of this specification is not implied. Exposure to absolute maximum rating conditions for extended periods may affect reliability.

[Table 16] Absolute Maximum DC Ratings

Parameter	Symbol	Min	Max	Units	Notes
V_{DD1} supply voltage relative to V_{SS}	V_{DD1}	-0.4	2.1	V	1
V_{DD2} supply voltage relative to V_{SS}	V_{DD2}	-0.4	1.4	V	1
V_{DDQ} supply voltage relative to V_{SSQ}	V_{DDQ}	-0.4	1.4	V	1
Voltage on any ball except V_{DD1} relative to V_{SS}	V_{IN}, V_{OUT}	-0.4	1.4	V	
Storage Temperature	T_{STG}	-55	125	°C	2

NOTE :

1) See Power Ramp for relationships between power supplies.

2) Storage Temperature is the case surface temperature on the center/top side of the LPDDR4X device. For the measurement conditions, please refer to JESD51-2 standard.

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5.0 AC & DC OPERATING CONDITIONS

5.1 Recommended DC Operating Conditions

[Table 17] Recommended DC Operating Conditions

Symbol	DRAM	LPDDR4X			Unit	Notes
		Min	Typ	Max		
VDD1	Core 1 Power	1.70	1.80	1.95	V	1,2
VDD2	Core 2 Power / Input Buffer Power	1.06	1.10	1.17	V	1,2,3
VDDQ	I/O Buffer Power	0.57	0.60	0.65	V	2,3

NOTE :
 1) VDD1 uses significantly less current than VDD2.
 2) The voltage range is for DC voltage only. DC is defined as the voltage supplied at the DRAM and is inclusive of all noise up to 20MHz at the DRAM package ball.
 3) The voltage noise tolerance from DC to 20MHz exceeding a pk-pk tolerance of 45mV at the DRAM ball is not included in the TdIVW.

5.2 Input Leakage Current

[Table 18] Input Leakage Current

Parameter/Condition	Symbol	Min	Max	Unit	Notes
Input Leakage current	I_L	-8	8	uA	1,2

NOTE :
 1) For CK_t, CK_c, CKE, CS, CA, ODT_CA and RESET_n. Any input $0V \leq V_{IN} \leq V_{DD2}$ (All other pins not under test = 0V).
 2) CA ODT is disabled for CK_t, CK_c, CS, and CA.

5.3 Input/Output Leakage Current

[Table 19] Input/Output Leakage Current

Parameter/Condition	Symbol	Min.	Max.	Unit	Notes
Input/Output Leakage current	I_{OZ}	-10	10	uA	1,2

NOTE :
 1) For DQ, DQS_t, DQS_c and DMI. Any I/O $0V \leq V_{OUT} \leq V_{DDQ}$.
 2) I/Os status are disabled: High Impedance and ODT Off.

5.4 Operating Temperature Range

[Table 20] Operating Temperature Range

Parameter/Condition	Symbol	Min	Max	Unit
Standard	T_{OPER}	-25	85	°C

NOTE :
 1) Operating Temperature is the case surface temperature on the center top side of the LPDDR4X device. For the measurement conditions, please refer to JESD51-2.
 2) Either the device case temperature rating or the temperature sensor (See "Temperature Sensor" on [Command Definition & Timing Diagram]) may be used to set an appropriate refresh rate, determine the need for AC timing de-rating and/or monitor the operating temperature. When using the temperature sensor, the actual device case temperature may be higher than the T_{OPER} rating that applies for the Standard or Extended Temperature Ranges. For example, T_{CASE} may be above 85°C when the temperature sensor indicates a temperature of less than 85°C.

6.0 AC AND DC INPUT/OUTPUT MEASUREMENT LEVELS

6.1 1.1V High speed LVCMOS (HS_LLVC MOS)

6.1.1 Standard specifications

All voltages are referenced to ground except where noted.

6.1.2 DC electrical characteristics

6.1.2.1 LPDDR4X Input Level for CKE

This definition applies to CKE_A/B.

[Table 21] LPDDR4X Input Level for CKE

Parameter	Symbol	Min.	Max.	Unit	Note
Input high level (AC)	$V_{IH(AC)}$	$0.75 \times V_{DD2}$	$V_{DD2} + 0.2$	V	1
Input low level (AC)	$V_{IL(AC)}$	-0.2	$0.25 \times V_{DD2}$	V	1
Input high level (DC)	$V_{IH(DC)}$	$0.65 \times V_{DD2}$	$V_{DD2} + 0.2$	V	
Input low level (DC)	$V_{IL(DC)}$	-0.2	$0.35 \times V_{DD2}$	V	

NOTE :

1) Refer LPDDR4X AC Over/Undershoot section.

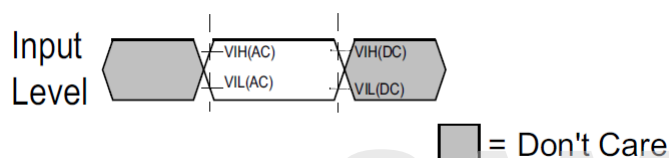


Figure 3. LPDDR4X Input AC timing definition for CKE

1-1). AC level is guaranteed transition point.

1-2). DC level is hysteresis.

6.1.2.2 LPDDR4X Input Level for Reset_n and ODT_CA

This definition applies to Reset_n and ODT_CA.

[Table 22] LPDDR4X Input Level for Reset_n and ODT_CA

Parameter	Symbol	Min.	Max.	Unit	Note
Input high level	V_{IH}	$0.80 \times V_{DD2}$	$V_{DD2} + 0.2$	V	1
Input low level	V_{IL}	-0.2	$0.20 \times V_{DD2}$	V	1

NOTE :

1) Refer LPDDR4X AC Over/Undershoot section.

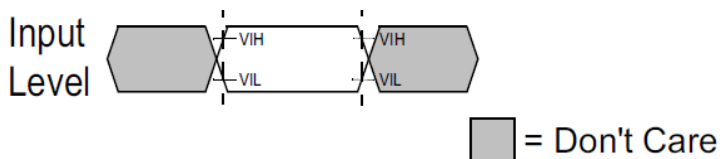


Figure 4. LPDDR4X Input AC timing definition for Reset_n and ODT_CA

6.1.3 AC Over/Undershoot

6.1.3.1 LPDDR4X AC Over/Undershoot

[Table 23] LPDDR4X AC Over/Undershoot

Parameter	Specification
Maximum peak amplitude allowed for overshoot area.	0.35V
Maximum peak amplitude allowed for undershoot area.	0.35V
Maximum overshoot area above V_{DD}/V_{DDQ} .	0.8V-ns
Maximum undershoot area below V_{SS}/V_{SSQ} .	0.8V-ns

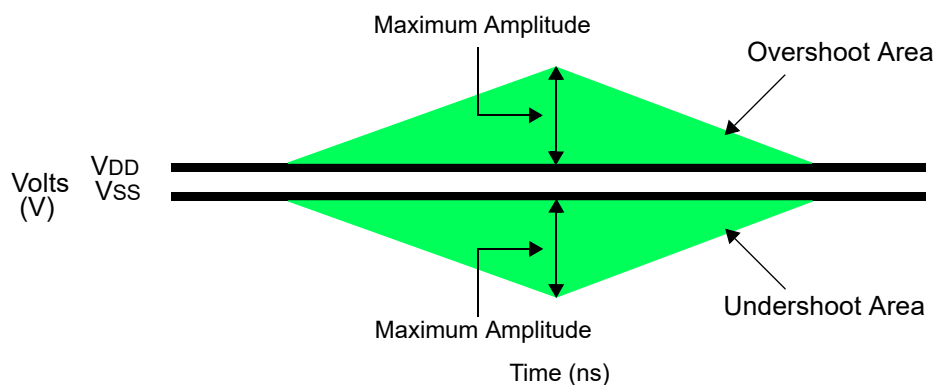


Figure 5. AC Overshoot and Undershoot Definition for Address and Control Pins

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6.2 Differential Input Voltage

6.2.1 Differential Input Voltage for CK

The minimum input voltage need to satisfy both $V_{\text{indiff_CK}}$ and $V_{\text{indiff_CK}}/2$ specification at input receiver and their measurement period is $1t_{\text{CK}}$. $V_{\text{indiff_CK}}$ is the peak to peak voltage centered on 0 volts differential and $V_{\text{indiff_CK}}/2$ is max and min peak voltage from 0V.

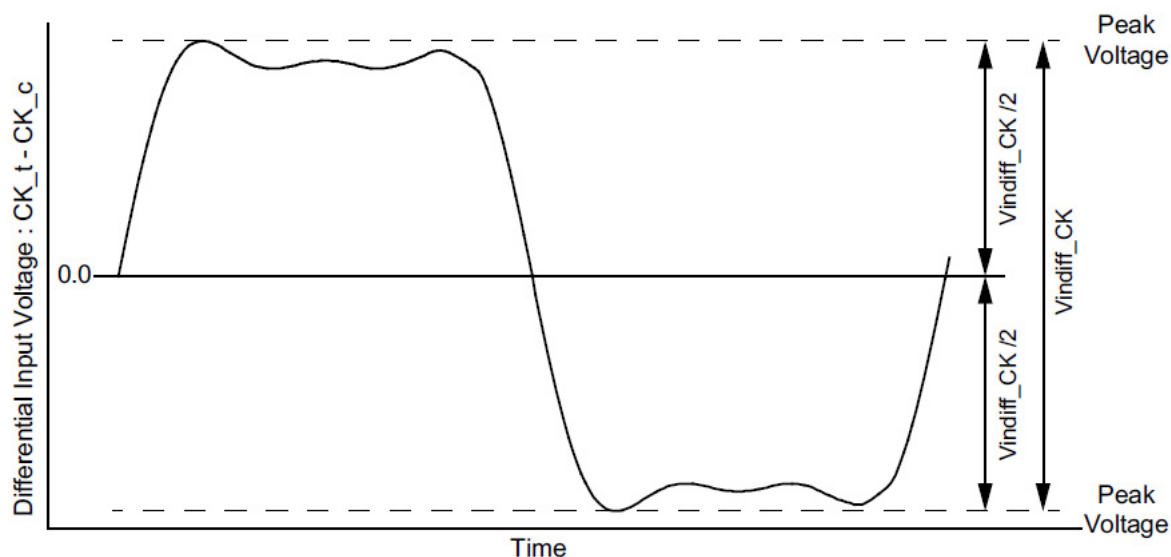


Figure 6. CK Differential Input Voltage

[Table 24] CK differential input voltage

Parameter	Symbol	Data Rate						Unit	Note
		1600/1866 ^{a)}		2133/2400/3200		3733/4266			
		Min	Max	Min	Max	Min	Max		
CK differential input voltage	Vindiff_CK	420	-	380	-	360	-	mV	1

a) The following requirements apply for DQ operating frequencies at or below 1333Mbps for all speed bins for the first column 1600/1866.

NOTE:

1) The peak voltage of Differential CK signals is calculated in a following equation.

$$V_{\text{indiff_CK}} = (\text{Max Peak Voltage}) - (\text{Min Peak Voltage})$$

$$\text{Max Peak Voltage} = \text{Max}(f(t))$$

$$\text{Min Peak Voltage} = \text{Min}(f(t))$$

$$f(t) = V_{\text{CK_t}} - V_{\text{CK_c}}$$

6.2.2 Peak voltage calculation method

The peak voltage of Differential Clock signals are calculated in a following equation.

$$VIH.DIFF.Peak\ Voltage = \text{Max}(f(t))$$

$$VIL.DIFF.Peak\ Voltage = \text{Min}(f(t))$$

$$f(t) = VCK_t - VCK_c$$

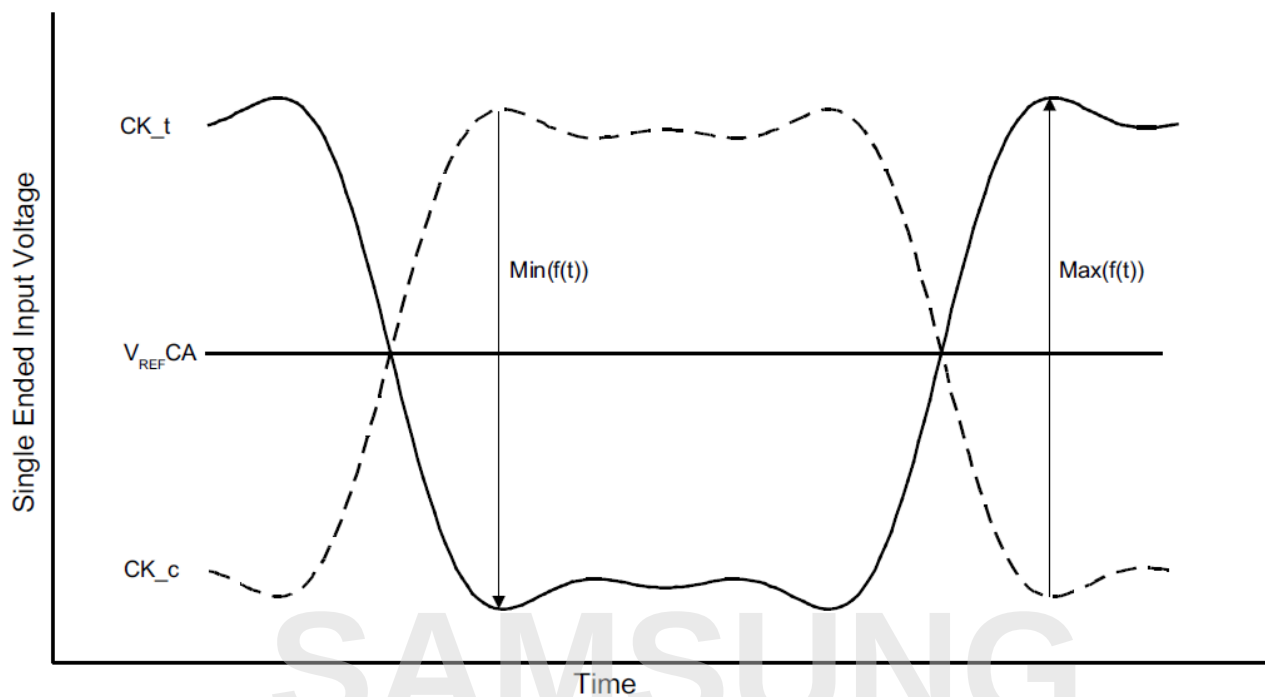


Figure 7. Definition of differential Clock Peak Voltage

NOTE:

1) VREFCA is LPDDR4X SDRAM internal setting value by VREF Training.

6.2.3 Single-Ended Input Voltage for Clock

The minimum input voltage need to satisfy both V_{inse_CK} , $V_{inse_CK_High/Low}$ specification at input receiver.

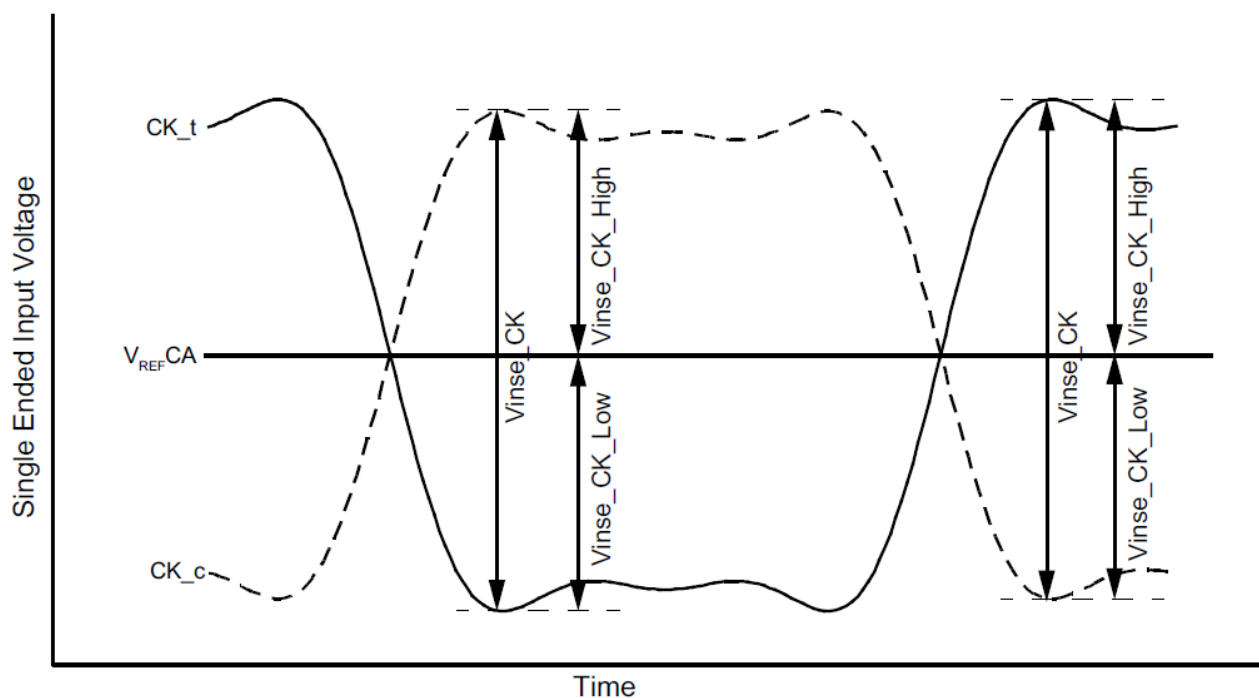


Figure 8. Clock Single-Ended Input Voltage

NOTE:

1) V_{REFCA} is LPDDR4X SDRAM internal setting value by VREF Training.

[Table 25] Clock Single-Ended input voltage

Parameter	Symbol	Data Rate						Unit
		1600/1866 ^{a)}		2133/2400/3200		3733/4266		
		Min	Max	Min	Max	Min	Max	
Clock Single-Ended input voltage	Vinse_CK	210	-	190	-	180	-	mV
Clock Single-Ended input voltage High from VREFDQ	Vinse_CK_High	105	-	95	-	90	-	mV
Clock Single-Ended input voltage Low from VREFDQ	Vinse_CK_Low	105	-	95	-	90	-	mV

a) The following requirements apply for DQ operating frequencies at or below 1333Mbps for all speed bins for the first column 1600/1866.

6.2.4 Differential Input Slew Rate Definition for Clock

Input slew rate for differential signals (CK_t, CK_c) are defined and measured as shown in Figure 9. and the following Tables.

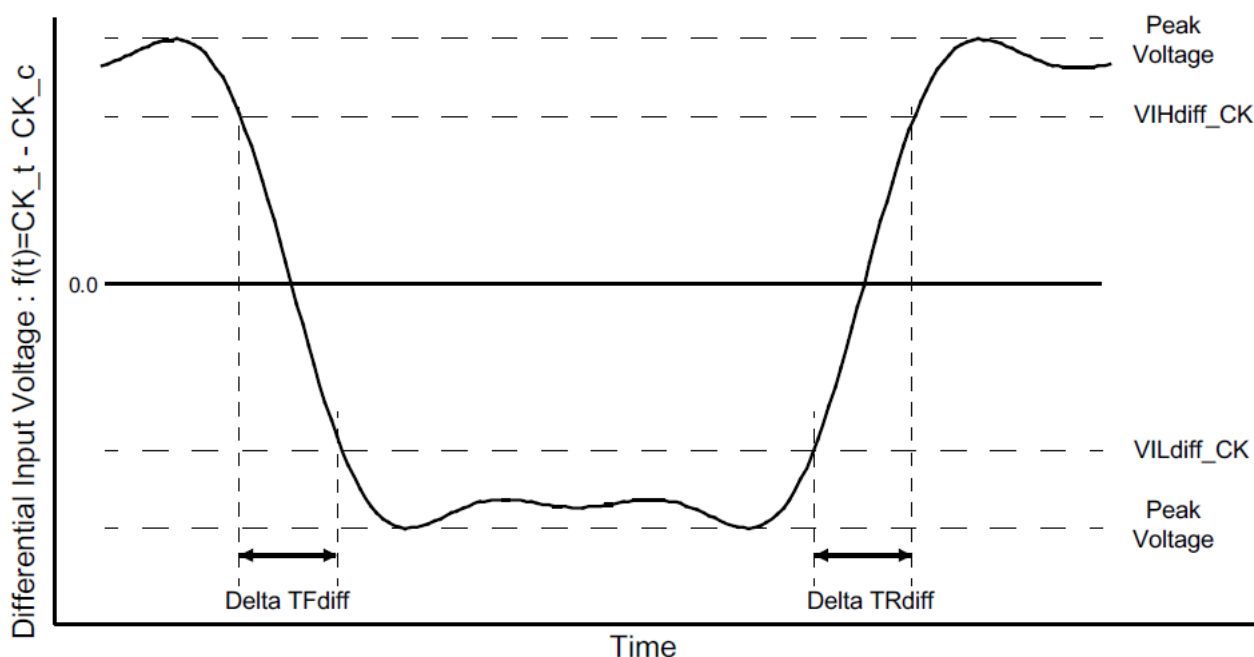


Figure 9. Differential Input Slew Rate Definition for CK_t, CK_c

NOTE :

- 1) Differential signal rising edge from VILdiff_CK to VIHdiff_CK must be monotonic slope.
- 2) Differential signal falling edge from VIHdiff_CK to VILdiff_CK must be monotonic slope.

[Table 26] Differential Input Slew Rate Definition for CK_t, CK_c

Description	From	To	Defined by
Differential input slew rate for rising edge (CK_t - CK_c)	VILdiff_CK	VIHdiff_CK	$ VILdiff_CK - VIHdiff_CK / \Delta TRdiff$
Differential input slew rate for falling edge (CK_t - CK_c)	VIHdiff_CK	VILdiff_CK	$ VILdiff_CK - VIHdiff_CK / \Delta TFdiff$

[Table 27] Differential Input Level for CK_t, CK_c

Parameter	Symbol	Data Rate						Unit	Note
		1600/1866 ^{a)}		2133/2400/3200		3733/4266			
		Min	Max	Min	Max	Min	Max		
Differential Input High	VIHdiff_CK	175	-	155	-	145	-	mV	
Differential Input Low	VILdiff_CK	-	-175	-	-155	-	-145	mV	

a) The following requirements apply for DQ operating frequencies at or below 1333Mbps for all speed bins for the first column 1600/1866.

[Table 28] Differential Input Slew Rate for CK_t, CK_c

Parameter	Symbol	Data Rate						Unit	Note
		1600/1866		2133/2400/3200		3733/4266			
		Min	Max	Min	Max	Min	Max		
Differential Input Slew Rate for Clock	SRIdiff_CK	2	14	2	14	2	14	V/ns	

6.2.5 Differential Input Cross Point Voltage for Clock

The cross point voltage of differential input signals (CK_t, CK_c) must meet the requirements in Table 29. The differential input cross point voltage VIX is measured from the actual cross point of true and complement signals to the mid level.

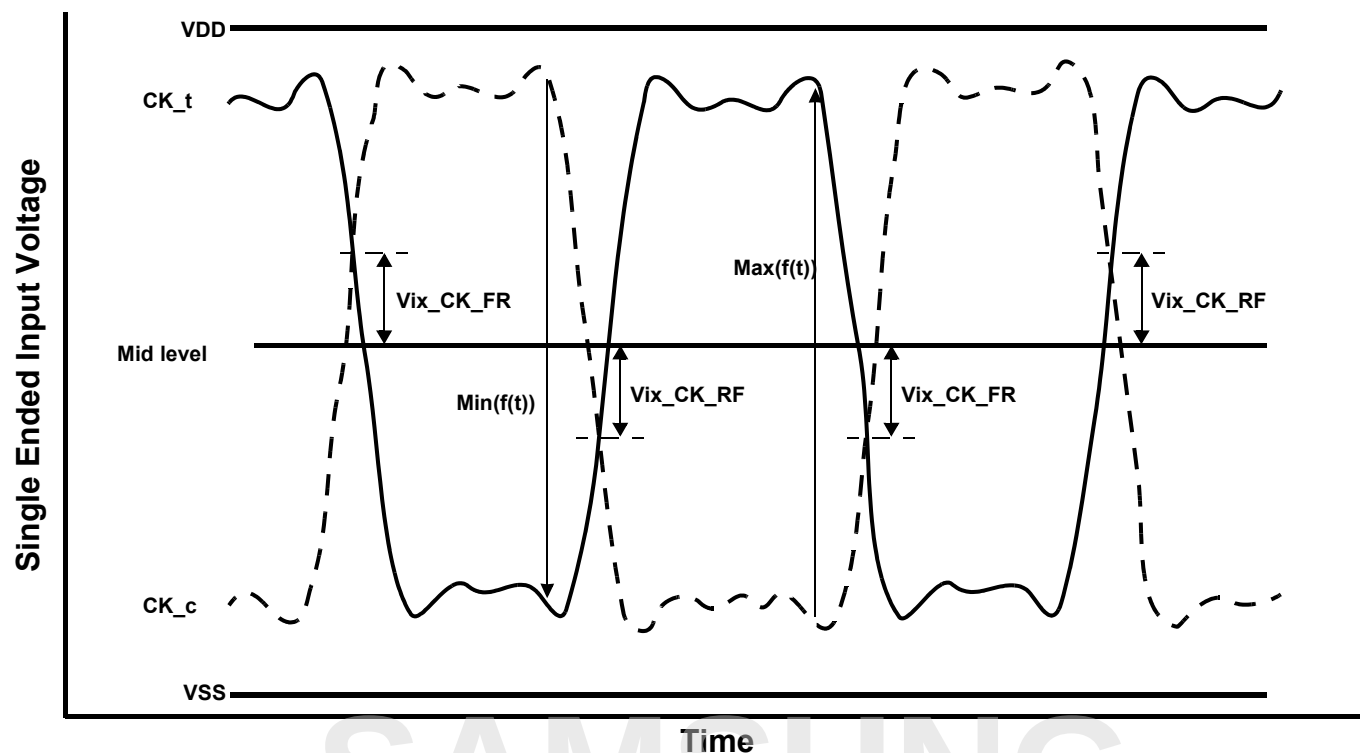


Figure 10. Vix Definition (Clock)

NOTE :

1) The base level of Vix_CK_FR/RF is VREFCA that is LPDDR4X SDRAM internal setting value by VREF Training.

[Table 29] Cross point voltage for differential input signals (Clock)

Parameter	Symbol	Data Rate						Unit	Note
		1600/1866 ^{a)}		2133/2400/3200		3733/4266			
		Min	Max	Min	Max	Min	Max		
Clock Differential input cross point voltage ratio	Vix_CK_ratio	-	25	-	25	-	25	%	1,2

a) The following requirements apply for DQ operating frequencies at or below 1333Mbps for all speed bins for the first column 1600/1866.

NOTE :

- 1) Vix_CK_Ratio is defined by this equation: $Vix_CK_Ratio = Vix_CK_FR / |Min(f(t))|$
- 2) Vix_CK_Ratio is defined by this equation: $Vix_CK_Ratio = Vix_CK_RF / Max(f(t))$
- 3) Vix_CK_FR is defined as delta between cross point (CK_t fall, CK_c rise) to $Min(f(t))/2$.
Vix_CK_RF is defined as delta between cross point (CK_t rise, CK_c fall) to $Max(f(t))/2$.
- 4) In LPDDR4X un-terminated case, CK mid-level is calculated as :
High level=VDDQ, Low level=VSS, Mid-level = $VDDQ/2$.
In LPDDR4 un-terminated case, Mid-level must be equal or lower than 369mV (33.6% of VDD2).

6.2.6 Differential Input Voltage for DQS

The minimum input voltage need to satisfy both $V_{\text{indiff_DQS}}$ and $V_{\text{indiff_DQS}}/2$ specification at input receiver and their measurement period is $1UI(t_{\text{CK}}/2)$. $V_{\text{indiff_DQS}}$ is the peak to peak voltage centered on 0 volts differential and $V_{\text{indiff_DQS}}/2$ is max and min peak voltage from 0V.

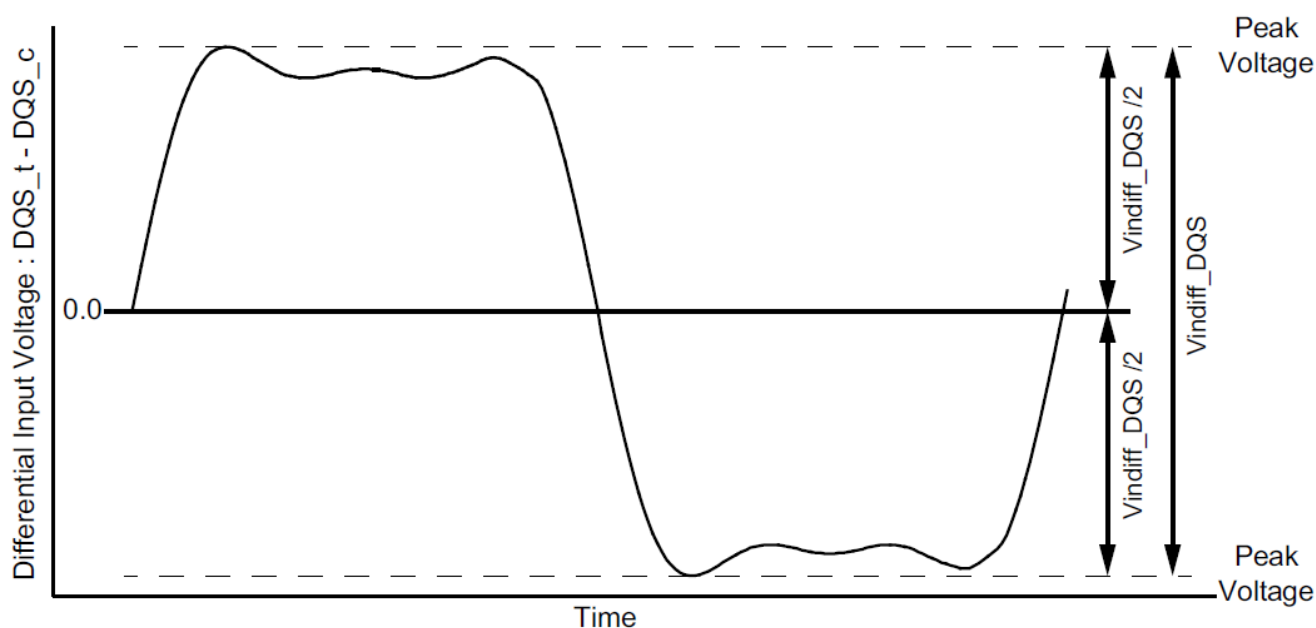


Figure 11. DQS Differential Input Voltage

[Table 30] DQS differential input voltage

Parameter	Symbol	Data Rate						Unit	Note
		1600/1866 ^{a)}		2133/2400/3200		3733/4266			
		Min	Max	Min	Max	Min	Max		
DQS differential input	Vindiff_DQS	360	-	360	-	340	-	mV	1

a) The following requirements apply for DQ operating frequencies at or below 1333Mbps for all speed bins for the first column 1600/1866.

NOTE :

1) The peak voltage of Differential DQS signals is calculated in a following equation.

$$V_{\text{indiff_DQS}} = (\text{Max Peak Voltage}) - (\text{Min Peak Voltage})$$

$$\text{Max Peak Voltage} = \text{Max}(f(t))$$

$$\text{Min Peak Voltage} = \text{Min}(f(t))$$

$$f(t) = VDQS_t - VDQS_c$$

6.2.7 Peak voltage calculation method

The peak voltage of Differential DQS signals are calculated in a following equation.

$$VIH.DIFF.Peak\ Voltage = \text{Max}(f(t))$$

$$VIL.DIFF.Peak\ Voltage = \text{Min}(f(t))$$

$$f(t) = VDQS_t - VDQS_c$$

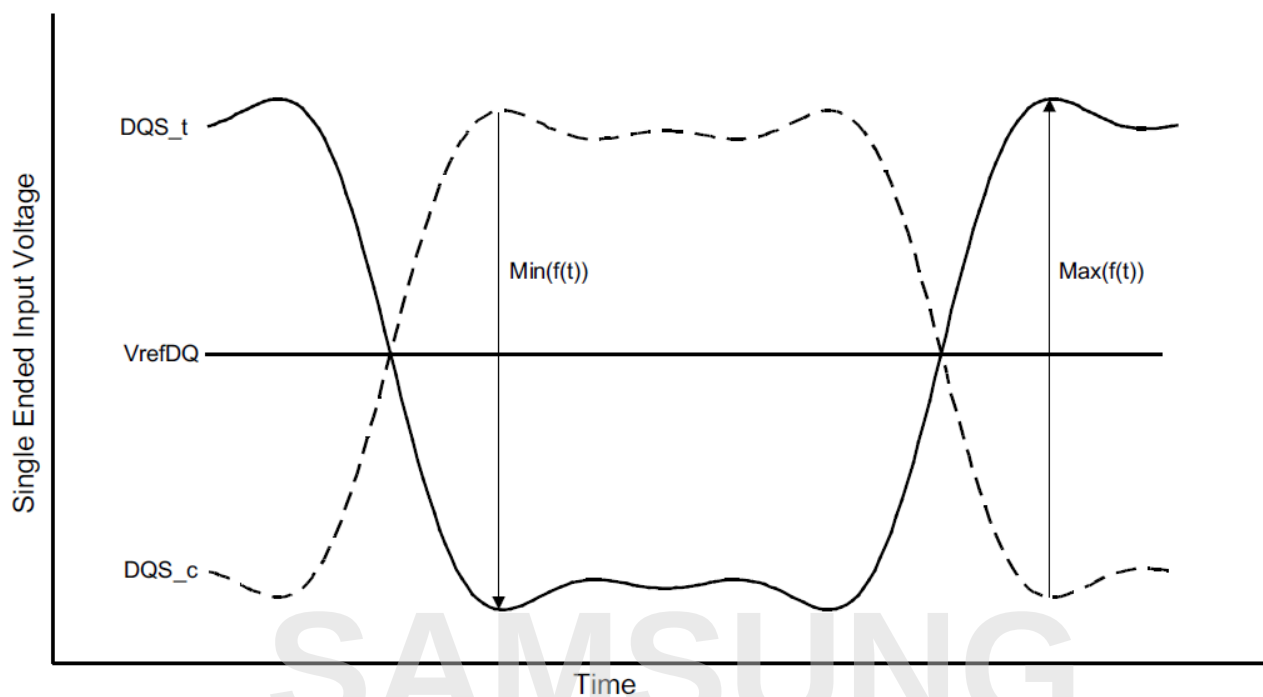


Figure 12. Definition of differential DQS Peak Voltage

NOTE :

1) VrefDQ is LPDDR4X SDRAM internal setting value by Vref Training.

6.2.8 Single-Ended Input Voltage for DQS

The minimum input voltage need to satisfy both V_{inse_DQS} , $V_{inse_DQS_High/Low}$ specification at input receiver.

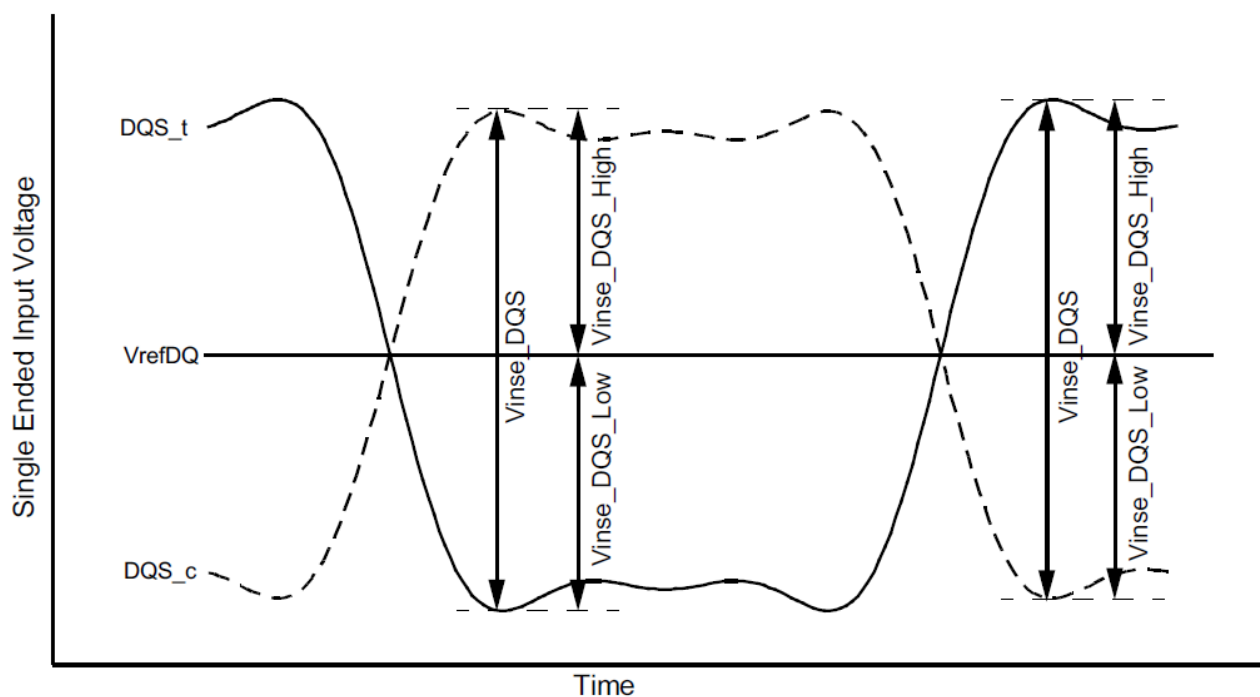


Figure 13. DQS Single-Ended Input Voltage

NOTE :

1) V_{refDQ} is LPDDR4X SDRAM internal setting value by V_{ref} Training.

[Table 31] DQS Single-Ended input voltage

Parameter	Symbol	Data Rate						Unit	Note
		1600/1866 a)		2133/2400/3200		3733/4266			
		Min	Max	Min	Max	Min	Max		
DQS Single-Ended input voltage	Vinse_DQS	180	-	180	-	170	-	mV	
DQS Single-Ended input voltage High from VrefDQ	Vinse_DQS_High	90	-	90	-	85	-	mV	
DQS Single-Ended input voltage Low from VrefDQ	Vinse_DQS_Low	90	-	90	-	85	-	mV	

a) The following requirements apply for DQ operating frequencies at or below 1333Mbps for all speed bins for the first column 1600/1866.

6.2.9 Differential Input Slew Rate Definition for DQS

Input slew rate for differential signals (DQS_t, DQS_c) are defined and measured as shown in Figure 14. and Table 32.

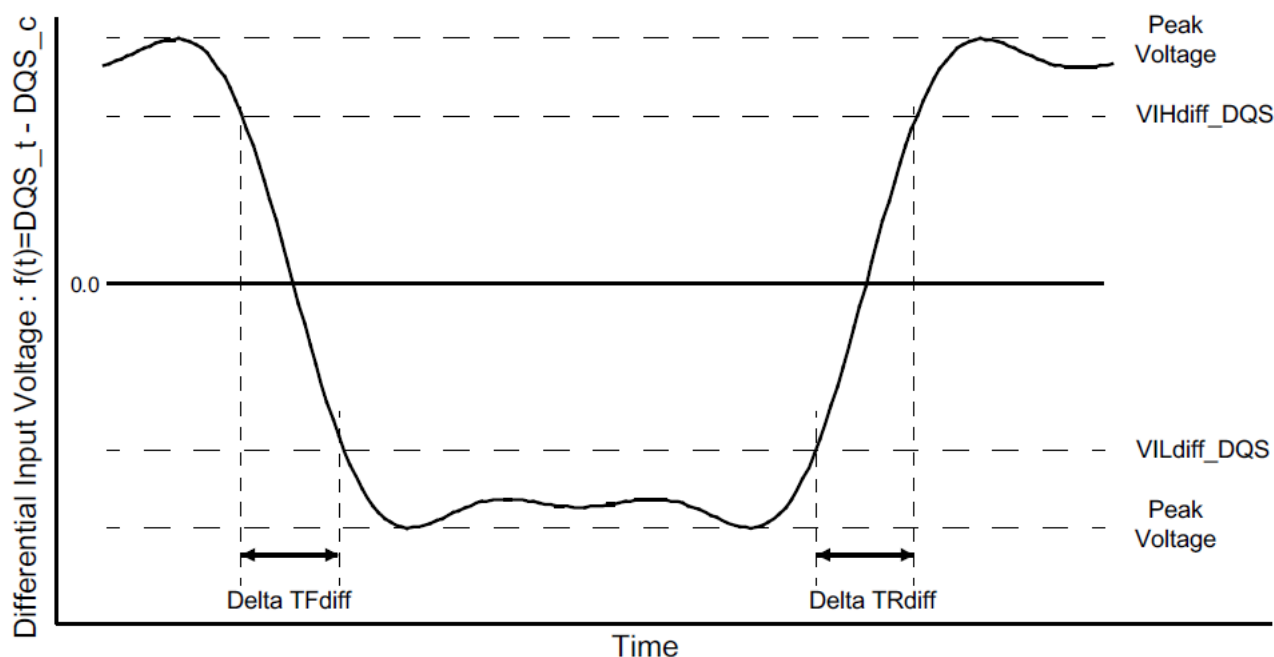


Figure 14. Differential Input Slew Rate Definition for DQS_t, DQS_c

NOTE :

- 1) Differential signal rising edge from VILdiff_DQS to VIHdiff_DQS must be monotonic slope.
- 2) Differential signal falling edge from VIHdiff_DQS to VILdiff_DQS must be monotonic slope.

[Table 32] Differential Input Slew Rate Definition for DQS_t, DQS_c

Description	From	To	Defined by
Differential input slew rate for rising edge (DQS _t - DQS _c)	VILdiff_DQS	VIHdiff_DQS	$ VILdiff_DQS - VIHdiff_DQS / \Delta TRdiff$
Differential input slew rate for falling edge (DQS _t - DQS _c)	VIHdiff_DQS	VILdiff_DQS	$ VILdiff_DQS - VIHdiff_DQS / \Delta TFdiff$

[Table 33] Differential Input Level for DQS_t, DQS_c

Parameter	Symbol	Data Rate						Unit	Note
		1600/1866 ^{a)}		2133/2400/3200		3733/4266			
		Min	Max	Min	Max	Min	Max		
Differential Input High	VIHdiff_DQS	140	-	140	-	120	-	mV	
Differential Input Low	VILdiff_DQS	-	-140	-	-140	-	-120	mV	

a) The following requirements apply for DQ operating frequencies at or below 1333Mbps for all speed bins for the first column 1600/1866.

[Table 34] Differential Input Slew Rate for DQS_t, DQS_c

Parameter	Symbol	Data Rate						Unit	Note
		1600/1866		2133/2400/3200		3733/4266			
		Min	Max	Min	Max	Min	Max		
Differential Input Slew Rate	SRIdiff	2	14	2	14	2	14	V/ns	

6.3 Differential Input Cross Point Voltage for DQS

The cross point voltage of differential input signals (DQS_t, DQS_c) must meet the requirements in Table 35. The differential input cross point voltage VIX is measured from the actual cross point of true and complement signals to the mid level that is V_{REFDQ}.

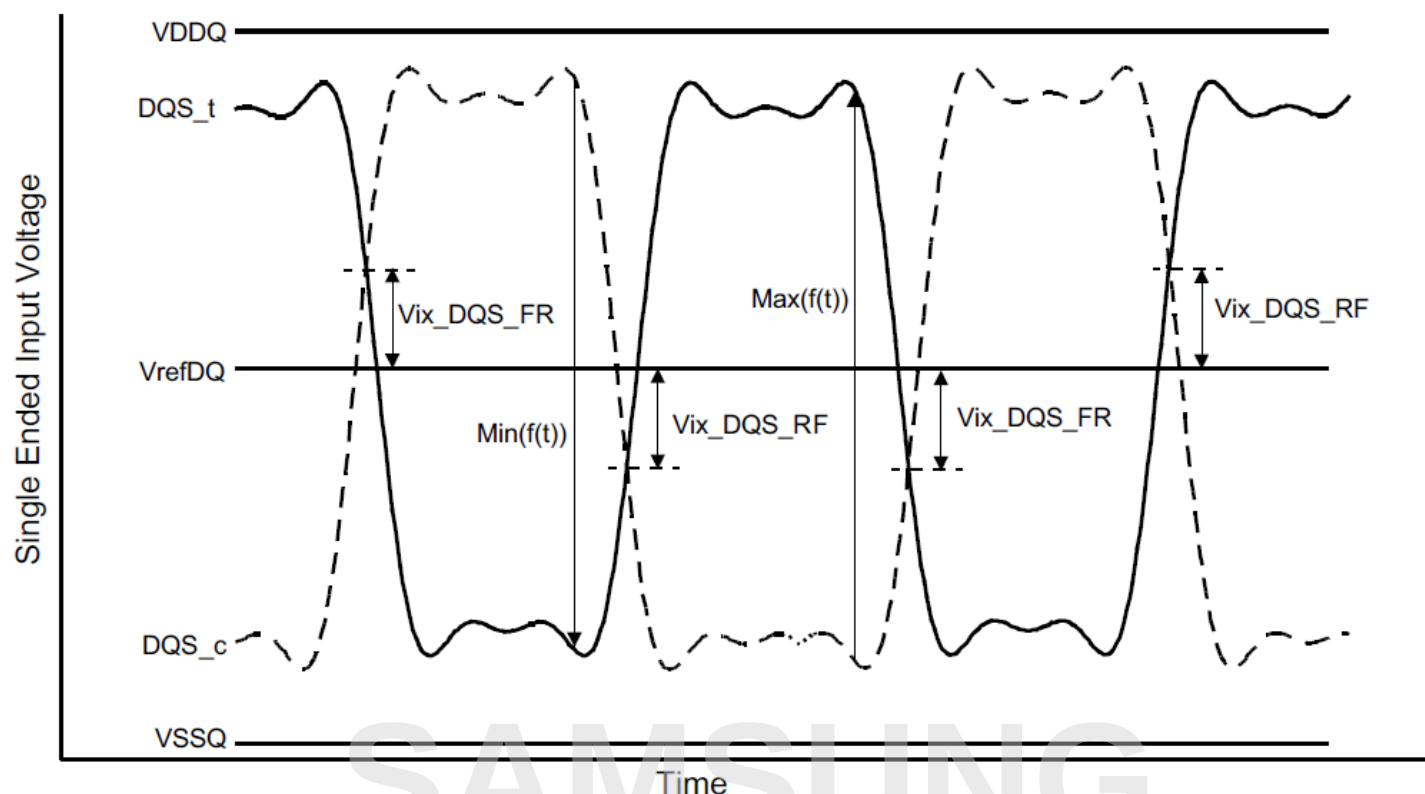


Figure 15. Vix Definition (DQS)

NOTE :

1) The base level of Vix_DQS_FR/RF is VrefDQ that is LPDDR4X SDRAM internal setting value by Vref Training.

[Table 35] Cross point voltage for differential input signals (DQS)

Parameter	Symbol	Data Rate						Units	Notes
		1600/1866 ^{a)}		2133/2400/3200		3733/4266			
		Min	Max	Min	Max	Min	Max		
DQS Differential input crosspoint voltage ratio	Vix_DQS_ratio	-	20	-	20	-	20	%	1,2

a) The following requirements apply for DQ operating frequencies at or below 1333Mbps for all speed bins for the first column 1600/1866.

NOTE :

1) Vix_DQS_Ratio is defined by this equation: $Vix_DQS_Ratio = Vix_DQS_FR / |Min(f(t))|$

2) Vix_DQS_Ratio is defined by this equation: $Vix_DQS_Ratio = Vix_DQS_RF / Max(f(t))$

6.4 Single Ended Output Slew Rate

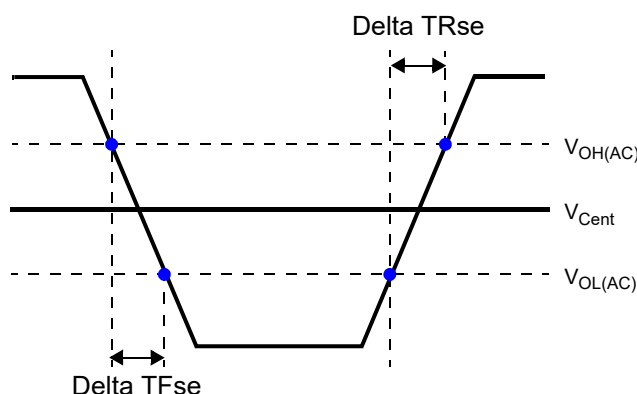


Figure 16. Single Ended Output Slew Rate Definition

[Table 36] Output Slew Rate (single-ended) for 0.6V VDDQ

Parameter	Symbol	Value		Units
		Min ¹⁾	Max ²⁾	
Single-ended Output Slew Rate ($V_{OH} = VDDQ \times 0.5$)	S_{RQse}	3.0	9.0	V/ns
Output slew-rate matching Ratio (Rise to Fall)		0.8	1.2	-

Description:

SR: Slew Rate

Q: Query Output (like in DQ, which stands for Data-in, Query-Output)

se: Single-ended Signals

NOTE :

- 1) Measured with output reference load.
- 2) The ratio of pull-up to pull-down slew rate is specified for the same temperature and voltage, over the entire temperature and voltage range. For a given output, it represents the maximum difference between pull-up and pull-down drivers due to process variation.
- 3) The output slew rate for falling and rising edges is defined and measured between $V_{OL(AC)} = 0.2 \times V_{OH(DC)}$ and $V_{OH(AC)} = 0.8 \times V_{OH(DC)}$.
- 4) Slew rates are measured under average SSO conditions, with 50% of DQ signals per data byte switching.

6.5 Differential Output Slew Rate

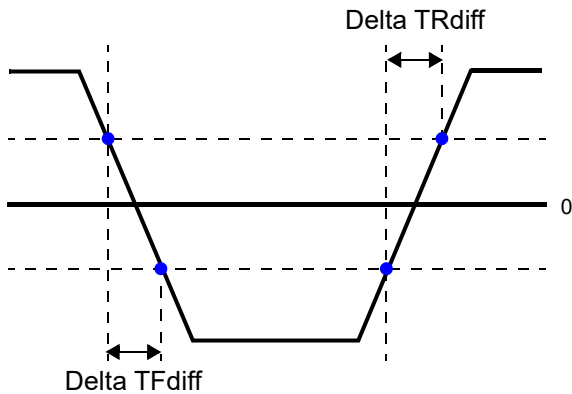


Figure 17. Differential Output Slew Rate Definition

[Table 37] Differential Output Slew Rate for 0.6V VDDQ

Parameter	Symbol	Value		Units
		Min	Max	
Differential Output Slew Rate ($V_{OH} = VDDQ \times 0.5$)	SRQdiff	6	18.0	V/ns

Description:

SR: Slew Rate

Q: Query Output (like in DQ, which stands for Data-in, Query-Output)

diff: Differential Signals

NOTE :

- 1) Measured with output reference load.
- 2) The output slew rate for falling and rising edges is defined and measured between $V_{OL(AC)} = -0.8 \times V_{OH(DC)}$ and $V_{OH(AC)} = 0.8 \times V_{OH(DC)}$.
- 3) Slew rates are measured under average SSO conditions, with 50% of DQ signals per data byte switching.

6.6 Overshoot and Undershoot for LVSTL

[Table 38] AC Overshoot/Undershoot Specification

Parameter		Data Rate					Units
		1600	1866	3200	3733	4266	
Maximum peak amplitude allowed for overshoot area. (See Figure 18.)	Max	0.3	0.3	0.3	0.3	0.3	V
Maximum peak amplitude allowed for undershoot area. (See Figure 18.)	Max	0.3	0.3	0.3	0.3	0.3	V
Maximum overshoot area above V_{DD} . (See Figure 18.)	Max	0.1	0.1	0.1	0.1	0.1	V-ns
Maximum undershoot area below V_{SS} . (See Figure 18.)	Max	0.1	0.1	0.1	0.1	0.1	V-ns

NOTE :

1) V_{DD2} stands for V_{DD} for CA[5:0], CK_t, CK_c, CS_n, CKE and ODT. V_{DD} stands for V_{DDQ} for DQ, DMI, DQS_t and DQS_c.

2) V_{SS} stands for V_{SS} for CA[5:0], CK_t, CK_c, CS_n, CKE and ODT. V_{SS} stands for V_{SSQ} for DQ, DMI, DQS_t and DQS_c.

3) Maximum peak amplitude values are referenced from actual V_{DD} and V_{SS} values.

4) Maximum area values are referenced from maximum operating V_{DD} and V_{SS} values.

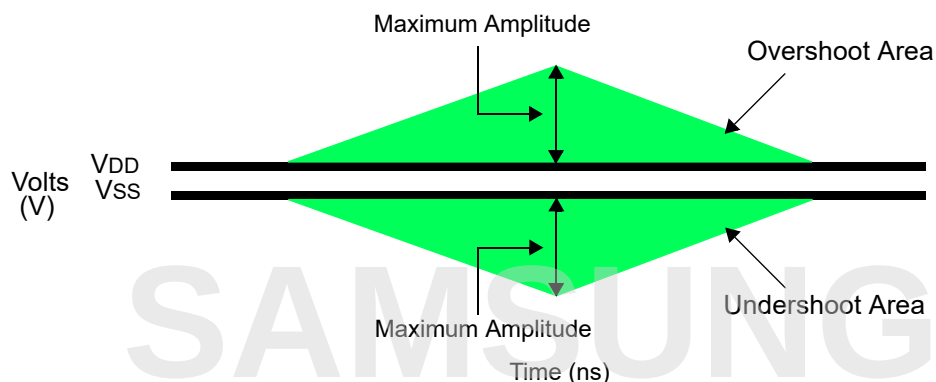


Figure 18. Overshoot and Undershoot Definition

6.7 LPDDR4X Driver Output Timing Reference Load

These 'Timing Reference Loads' are not intended as a precise representation of any particular system environment or a depiction of the actual load presented by a production tester. System designers should use IBIS or other simulation tools to correlate the timing reference load to a system environment. Manufacturers correlate to their production test conditions, generally one or more coaxial transmission lines terminated at the tester electronics.

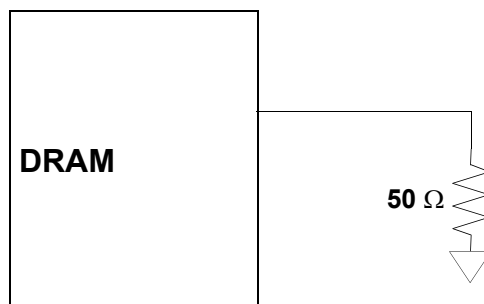


Figure 19. Driver Output Reference Load for Timing and Slew Rate

NOTE :

1) All output timing parameter values are reported with respect to this reference load. This reference load is also used to report slew rate.

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6.8 LVSTL (Low Voltage Swing Terminated Logic) IO System

LVSTL I/O cell is comprised of pull-up, pull-down driver and a terminator. The basic cell is shown in Figure 20 .

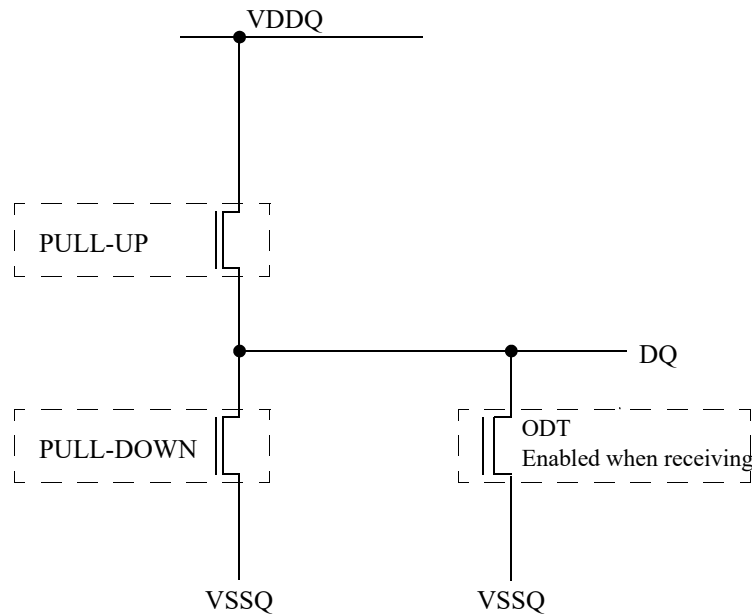


Figure 20. LVSTL I/O Cell

To ensure that the target impedance is achieved the LVSTL I/O cell is designed to calibrated as below procedure.

1) First calibrate the pull-down device against a 240 Ohm resistor to VDDQ via the ZQ pin

- Set Strength Control to minimum setting
- Increase drive strength until comparator detects data bit is less than $VDDQ/2$.
- NMOS pull-down device is calibrated to 240 Ohms

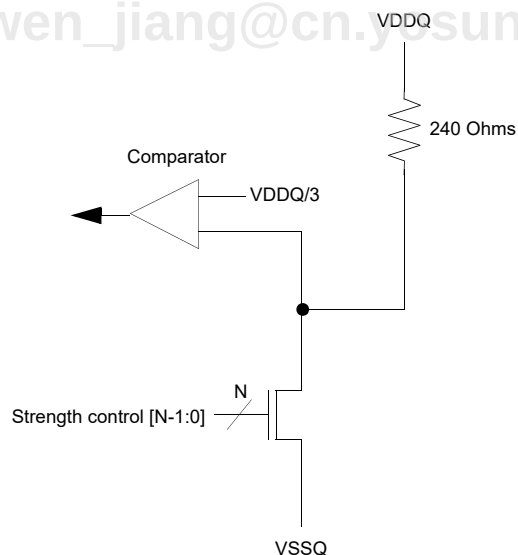


Figure 21. pull-down calibration

2) Then calibrate the pull-up device against the calibrated pull-down device.

- Set VOH target and NMOS controller ODT replica via MRS (VOH can be automatically controlled by ODT MRS)
- Set Strength Control to minimum setting
- Increase drive strength until comparator detects data bit is greater than VOH target
- NMOS pull-up device is now calibrated to VOH target

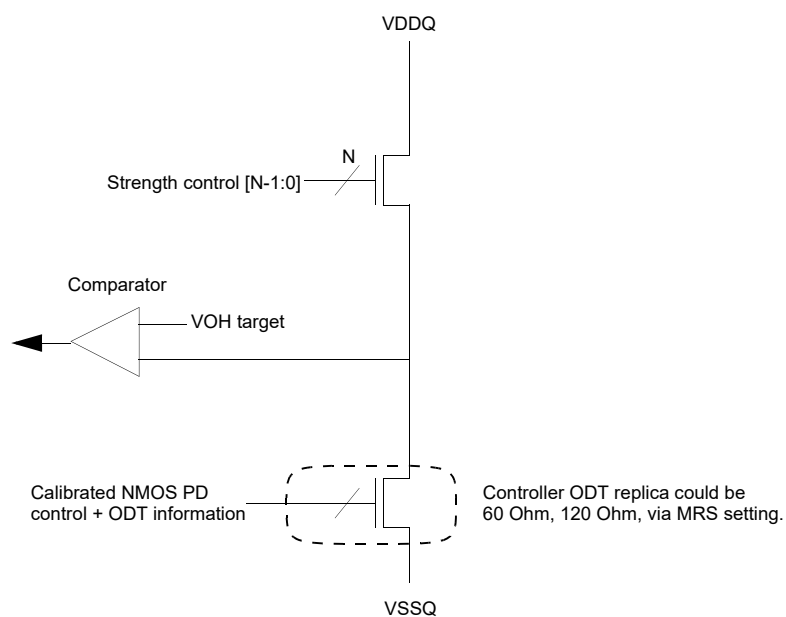


Figure 22. pull-up calibration

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7.0 INPUT/OUTPUT CAPACITANCE

[Table 39] Input/Output Capacitance

Parameter	Symbol	Min/Max	Value	Unit	Notes
Input capacitance, CK _t and CK _c	CCK	Min	0.5	pF	1,2
		Max	0.9		
Input capacitance delta, CK _t and CK _c	CDCK	Min	0.0	pF	1,2,3
		Max	0.09		
Input capacitance, All other input-only pins	CI	Min	0.5	pF	1,2,4
		Max	0.9		
Input capacitance delta, All other input-only pins	CDI	Min	-0.1	pF	1,2,5
		Max	0.1		
Input/output capacitance, DQ, DMI, DQS _t and DQS _c	CIO	Min	0.7	pF	1,2,6
		Max	1.3		
Input/output capacitance delta, DQS _t and DQS _c	CDDQS	Min	0.0	pF	1,2,7
		Max	0.1		
Input/output capacitance delta, DQ and DMI	CDIO	Min	-0.1	pF	1,2,8
		Max	0.1		
Input/output capacitance, ZQ pin	CZQ	Min	0.0	pF	1,2
		Max	5.0		

NOTE :

- 1) This parameter applies to die device only (does not include package capacitance).
- 2) This parameter is not subject to production test. It is verified by design and characterization. The capacitance is measured according to JEP147 (Procedure for measuring input capacitance using a vector network analyzer (VNA) with VDD1, VDD2, VDDQ, VSS, VSSQ applied and all other pins floating).
- 3) Absolute value of CCK_t - CCK_c.
- 4) CI applies to CS_n, CKE, CA0-CA5.
- 5) CDI = CI - 0.5 × (CCK_t + CCK_c)
- 6) DMI loading matches DQ and DQS.
- 7) Absolute value of CDQS_t and CDQS_c.
- 8) CDIO = CIO - 0.5 × (CDQS_t + CDQS_c) in byte-lane.

8.0 IDD SPECIFICATION PARAMETERS AND TEST CONDITIONS

8.1 IDD Measurement Conditions

The following definitions are used within the IDD measurement tables unless stated otherwise:

- LOW: $V_{IN} \leq V_{IL}(DC) \text{ MAX}$
- HIGH: $V_{IN} \geq V_{IH}(DC) \text{ MIN}$
- STABLE: Inputs are stable at a HIGH or LOW level
- SWITCHING: See Table 40 and Table 41.

[Table 40] Definition of Switching for CA Input Signals

Switching for CA								
CK_t edge	R1	R2	R3	R4	R5	R6	R7	R8
CKE	HIGH	HIGH	HIGH	HIGH	HIGH	HIGH	HIGH	HIGH
CS	LOW	LOW	LOW	LOW	LOW	LOW	LOW	LOW
CA0	HIGH	LOW	LOW	LOW	LOW	HIGH	HIGH	HIGH
CA1	HIGH	HIGH	HIGH	LOW	LOW	LOW	LOW	HIGH
CA2	HIGH	LOW	LOW	LOW	LOW	HIGH	HIGH	HIGH
CA3	HIGH	HIGH	HIGH	LOW	LOW	LOW	LOW	HIGH
CA4	HIGH	LOW	LOW	LOW	LOW	HIGH	HIGH	HIGH
CA5	HIGH	HIGH	HIGH	LOW	LOW	LOW	LOW	HIGH

NOTE :

- 1) CS must always be driven LOW.
- 2) 50% of CA bus is changing between HIGH and LOW once per clock for the CA bus.
- 3) The above pattern is used continuously during IDD measurement for IDD values that require switching on the CA bus.

[Table 41] CA pattern for IDD4R for BL=16

Clock Cycle Number	CKE	CS	Command	CA0	CA1	CA2	CA3	CA4	CA5
N	HIGH	HIGH	Read-1	L	H	L	L	L	L
N+1	HIGH	LOW		L	H	L	L	L	L
N+2	HIGH	HIGH		L	H	L	L	H	L
N+3	HIGH	LOW	CAS-2	L	L	L	L	L	L
N+4	HIGH	LOW	DES	L	L	L	L	L	L
N+5	HIGH	LOW	DES	L	L	L	L	L	L
N+6	HIGH	LOW	DES	L	L	L	L	L	L
N+7	HIGH	LOW	DES	L	L	L	L	L	L
N+8	HIGH	HIGH	Read-1	L	H	L	L	L	L
N+9	HIGH	LOW		L	H	L	L	H	L
N+10	HIGH	HIGH	CAS-2	L	H	L	L	H	H
N+11	HIGH	LOW		H	H	H	H	H	H
N+12	HIGH	LOW	DES	L	L	L	L	L	L
N+13	HIGH	LOW	DES	L	L	L	L	L	L
N+14	HIGH	LOW	DES	L	L	L	L	L	L
N+15	HIGH	LOW	DES	L	L	L	L	L	L

NOTE :

- 1) BA[2:0] = 010_B, CA[9:4] = 000000_B or 111111_B, Burst Order CA[3:2] = 00_B or 11_B (Same as LPDDR3 IDD4R Spec)
- 2) Difference from LPDDR3 Spec : CA pins are kept low with DES CMD to reduce ODT current.

[Table 42] CA pattern for IDD4W for BL=16

Clock Cycle Number	CKE	CS	Command	CA0	CA1	CA2	CA3	CA4	CA5
N	HIGH	HIGH	Write-1	L	L	H	L	L	L
N+1	HIGH	LOW		L	H	L	L	L	L
N+2	HIGH	HIGH	CAS-2	L	H	L	L	H	L
N+3	HIGH	LOW		L	L	L	L	L	L
N+4	HIGH	LOW	DES	L	L	L	L	L	L
N+5	HIGH	LOW	DES	L	L	L	L	L	L
N+6	HIGH	LOW	DES	L	L	L	L	L	L
N+7	HIGH	LOW	DES	L	L	L	L	L	L
N+8	HIGH	HIGH	Write-1	L	L	H	L	L	L
N+9	HIGH	LOW		L	H	L	L	H	L
N+10	HIGH	HIGH	CAS-2	L	H	L	L	H	H
N+11	HIGH	LOW		L	L	H	H	H	H
N+12	HIGH	LOW	DES	L	L	L	L	L	L
N+13	HIGH	LOW	DES	L	L	L	L	L	L
N+14	HIGH	LOW	DES	L	L	L	L	L	L
N+15	HIGH	LOW	DES	L	L	L	L	L	L

NOTE :

1) BA[2:0] = 010_B, CA[9:4] = 000000_B or 111111_B (Same as LPDDR3 IDD4W Spec.)

2) Difference from LPDDR3 Spec :

1-No burst ordering

2-CA pins are kept low with DES CMD to reduce ODT current.

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[Table 43] Data Pattern for IDD4W (DBI off) for BL=16

DBI OFF Case										No. of 1's
	DQ[7]	DQ[6]	DQ[5]	DQ[4]	DQ[3]	DQ[2]	DQ[1]	DQ[0]	DBI	
BL0	1	1	1	1	1	1	1	1	0	8
BL1	1	1	1	1	0	0	0	0	0	4
BL2	0	0	0	0	0	0	0	0	0	0
BL3	0	0	0	0	1	1	1	1	0	4
BL4	0	0	0	0	0	0	1	1	0	2
BL5	0	0	0	0	1	1	1	1	0	4
BL6	1	1	1	1	1	1	0	0	0	6
BL7	1	1	1	1	0	0	0	0	0	4
BL8	1	1	1	1	1	1	1	1	0	8
BL9	1	1	1	1	0	0	0	0	0	4
BL10	0	0	0	0	0	0	0	0	0	0
BL11	0	0	0	0	1	1	1	1	0	4
BL12	0	0	0	0	0	0	1	1	0	2
BL13	0	0	0	0	1	1	1	1	0	4
BL14	1	1	1	1	1	1	0	0	0	6
BL15	1	1	1	1	0	0	0	0	0	4
BL16	1	1	1	1	1	1	0	0	0	6
BL17	1	1	1	1	0	0	0	0	0	4
BL18	0	0	0	0	0	0	1	1	0	2
BL19	0	0	0	0	1	1	1	1	0	4
BL20	0	0	0	0	0	0	0	0	0	0
BL21	0	0	0	0	1	1	1	1	0	4
BL22	1	1	1	1	1	1	1	1	0	8
BL23	1	1	1	1	0	0	0	0	0	4
BL24	0	0	0	0	0	0	1	1	0	2
BL25	0	0	0	0	1	1	1	1	0	4
BL26	1	1	1	1	1	1	0	0	0	6
BL27	1	1	1	1	0	0	0	0	0	4
BL28	1	1	1	1	1	1	1	1	0	8
BL29	1	1	1	1	0	0	0	0	0	4
BL30	0	0	0	0	0	0	0	0	0	0
BL31	0	0	0	0	1	1	1	1	0	4
No. of 1's	16	16	16	16	16	16	16	16		

NOTE:

1) Simplified pattern compared with last showing.

Same data pattern was applied to DQ[4], DQ[5], DQ[6], DQ[7] for reducing complexity for IDD4W/R pattern programming.

IF THERE IS ANY OTHER OPERATION TO IMPLEMENT IN ADDITION TO SPECIFICATION
IN THE DATASHEET OR JEDEC STANDARD, PLEASE CONTACT EACH BRANCH OFFICE OR
HEADQUARTERS OF SAMSUNG ELECTRONICS.

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[Table 44] Data Pattern for IDD4R (DBI off) for BL=16

DBI OFF Case										No. of 1's
	DQ[7]	DQ[6]	DQ[5]	DQ[4]	DQ[3]	DQ[2]	DQ[1]	DQ[0]	DBI	
BL0	1	1	1	1	1	1	1	1	0	8
BL1	1	1	1	1	0	0	0	0	0	4
BL2	0	0	0	0	0	0	0	0	0	0
BL3	0	0	0	0	1	1	1	1	0	4
BL4	0	0	0	0	0	0	1	1	0	2
BL5	0	0	0	0	1	1	1	1	0	4
BL6	1	1	1	1	1	1	0	0	0	6
BL7	1	1	1	1	0	0	0	0	0	4
BL8	1	1	1	1	1	1	1	1	0	8
BL9	1	1	1	1	0	0	0	0	0	4
BL10	0	0	0	0	0	0	0	0	0	0
BL11	0	0	0	0	1	1	1	1	0	4
BL12	0	0	0	0	0	0	1	1	0	2
BL13	0	0	0	0	1	1	1	1	0	4
BL14	1	1	1	1	1	1	0	0	0	6
BL15	1	1	1	1	0	0	0	0	0	4
BL16	1	1	1	1	1	1	1	1	0	8
BL17	1	1	1	1	0	0	0	0	0	4
BL18	0	0	0	0	0	0	0	0	0	0
BL19	0	0	0	0	1	1	1	1	0	4
BL20	1	1	1	1	1	1	0	0	0	6
BL21	1	1	1	1	0	0	0	0	0	4
BL22	0	0	0	0	0	0	1	1	0	2
BL23	0	0	0	0	1	1	1	1	0	4
BL24	0	0	0	0	0	0	0	0	0	0
BL25	0	0	0	0	1	1	1	1	0	4
BL26	1	1	1	1	1	1	1	1	0	8
BL27	1	1	1	1	0	0	0	0	0	4
BL28	0	0	0	0	0	0	1	1	0	2
BL29	0	0	0	0	1	1	1	1	0	4
BL30	1	1	1	1	1	1	0	0	0	6
BL31	1	1	1	1	0	0	0	0	0	4
No. of 1's	16	16	16	16	16	16	16	16		

NOTE:

1) Same data pattern was applied to DQ[4], DQ[5], DQ[6], DQ[7] for reducing complexity for IDD4W/R pattern programming.

[Table 45] Data Pattern for IDD4W (DBI on) for BL=16

DBI ON Case										No. of 1's
	DQ[7]	DQ[6]	DQ[5]	DQ[4]	DQ[3]	DQ[2]	DQ[1]	DQ[0]	DBI	
BL0	0	0	0	0	0	0	0	0	1	1
BL1	1	1	1	1	0	0	0	0	0	4
BL2	0	0	0	0	0	0	0	0	0	0
BL3	0	0	0	0	1	1	1	1	0	4
BL4	0	0	0	0	0	0	1	1	0	2
BL5	0	0	0	0	1	1	1	1	0	4
BL6	0	0	0	0	0	0	1	1	1	3
BL7	1	1	1	1	0	0	0	0	0	4
BL8	0	0	0	0	0	0	0	0	1	1
BL9	1	1	1	1	0	0	0	0	0	4
BL10	0	0	0	0	0	0	0	0	0	0
BL11	0	0	0	0	1	1	1	1	0	4
BL12	0	0	0	0	0	0	1	1	0	2
BL13	0	0	0	0	1	1	1	1	0	4
BL14	0	0	0	0	0	0	1	1	1	3
BL15	1	1	1	1	0	0	0	0	0	4
BL16	0	0	0	0	0	0	1	1	1	3
BL17	1	1	1	1	0	0	0	0	0	4
BL18	0	0	0	0	0	0	1	1	0	2
BL19	0	0	0	0	1	1	1	1	0	4
BL20	0	0	0	0	0	0	0	0	0	0
BL21	0	0	0	0	1	1	1	1	0	4
BL22	0	0	0	0	0	0	0	0	1	1
BL23	1	1	1	1	0	0	0	0	0	4
BL24	0	0	0	0	0	0	1	1	0	2
BL25	0	0	0	0	1	1	1	1	0	4
BL26	0	0	0	0	0	0	1	1	1	3
BL27	1	1	1	1	0	0	0	0	0	4
BL28	0	0	0	0	0	0	0	0	1	1
BL29	1	1	1	1	0	0	0	0	0	4
BL30	0	0	0	0	0	0	0	0	0	0
BL31	0	0	0	0	1	1	1	1	0	4
No. of 1's	8	8	8	8	8	8	16	16	8	

 DBI enabled burst

[Table 46] Data Pattern for IDD4R (DBI on) for BL=16

DBI ON Case										No. of 1's
	DQ[7]	DQ[6]	DQ[5]	DQ[4]	DQ[3]	DQ[2]	DQ[1]	DQ[0]	DBI	
BL0	0	0	0	0	0	0	0	0	1	1
BL1	1	1	1	1	0	0	0	0	0	4
BL2	0	0	0	0	0	0	0	0	0	0
BL3	0	0	0	0	1	1	1	1	0	4
BL4	0	0	0	0	0	0	1	1	0	2
BL5	0	0	0	0	1	1	1	1	0	4
BL6	0	0	0	0	0	0	1	1	1	3
BL7	1	1	1	1	0	0	0	0	0	4
BL8	0	0	0	0	0	0	0	0	1	1
BL9	1	1	1	1	0	0	0	0	0	4
BL10	0	0	0	0	0	0	0	0	0	0
BL11	0	0	0	0	1	1	1	1	0	4
BL12	0	0	0	0	0	0	1	1	0	2
BL13	0	0	0	0	1	1	1	1	0	4
BL14	0	0	0	0	0	0	1	1	1	3
BL15	1	1	1	1	0	0	0	0	0	4
BL16	0	0	0	0	0	0	0	0	1	1
BL17	1	1	1	1	0	0	0	0	0	4
BL18	0	0	0	0	0	0	0	0	0	0
BL19	0	0	0	0	1	1	1	1	0	4
BL20	0	0	0	0	0	0	1	1	1	3
BL21	1	1	1	1	0	0	0	0	0	4
BL22	0	0	0	0	0	0	1	1	0	2
BL23	0	0	0	0	1	1	1	1	0	4
BL24	0	0	0	0	0	0	0	0	0	0
BL25	0	0	0	0	1	1	1	1	0	4
BL26	0	0	0	0	0	0	0	0	1	1
BL27	1	1	1	1	0	0	0	0	0	4
BL28	0	0	0	0	0	0	1	1	0	2
BL29	0	0	0	0	1	1	1	1	0	4
BL30	0	0	0	0	0	0	1	1	1	3
BL31	1	1	1	1	0	0	0	0	0	4
No. of 1's	8	8	8	8	8	8	16	16	8	

[Table 47] CA pattern for IDD4R for BL=32

Clock Cycle Number	CKE	CS	Command	CA0	CA1	CA2	CA3	CA4	CA5
N	HIGH	HIGH	Read-1	L	H	L	L	L	L
N+1	HIGH	LOW		L	H	L	L	L	L
N+2	HIGH	HIGH	CAS-2	L	H	L	L	H	L
N+3	HIGH	LOW		L	L	L	L	L	L
N+4	HIGH	LOW	DES	L	L	L	L	L	L
N+5	HIGH	LOW	DES	L	L	L	L	L	L
N+6	HIGH	LOW	DES	L	L	L	L	L	L
N+7	HIGH	LOW	DES	L	L	L	L	L	L
N+8	HIGH	LOW	DES	L	L	L	L	L	L
N+9	HIGH	LOW	DES	L	L	L	L	L	L
N+10	HIGH	LOW	DES	L	L	L	L	L	L
N+11	HIGH	LOW	DES	L	L	L	L	L	L
N+12	HIGH	LOW	DES	L	L	L	L	L	L
N+13	HIGH	LOW	DES	L	L	L	L	L	L
N+14	HIGH	LOW	DES	L	L	L	L	L	L
N+15	HIGH	LOW	DES	L	L	L	L	L	L
N+16	HIGH	HIGH	Read-1	L	H	L	L	L	L
N+17	HIGH	LOW		L	H	L	L	H	L
N+18	HIGH	HIGH	CAS-2	L	H	L	L	H	H
N+19	HIGH	LOW		H	H	L	H	H	H
N+20	HIGH	LOW	DES	L	L	L	L	L	L
N+21	HIGH	LOW	DES	L	L	L	L	L	L
N+22	HIGH	LOW	DES	L	L	L	L	L	L
N+23	HIGH	LOW	DES	L	L	L	L	L	L
N+24	HIGH	LOW	DES	L	L	L	L	L	L
N+25	HIGH	LOW	DES	L	L	L	L	L	L
N+26	HIGH	LOW	DES	L	L	L	L	L	L
N+27	HIGH	LOW	DES	L	L	L	L	L	L
N+28	HIGH	LOW	DES	L	L	L	L	L	L
N+29	HIGH	LOW	DES	L	L	L	L	L	L
N+30	HIGH	LOW	DES	L	L	L	L	L	L
N+31	HIGH	LOW	DES	L	L	L	L	L	L

NOTE :

1) BA[2:0] = 010B, CA[9:5] = 00000B or 11111B, Burst Order CA[4:2] = 000B or 111B.

[Table 48] CA pattern for IDD4W for BL=32

Clock Cycle Number	CKE	CS	Command	CA0	CA1	CA2	CA3	CA4	CA5
N	HIGH	HIGH	Write-1	L	L	H	L	L	L
N+1	HIGH	LOW		L	H	L	L	L	L
N+2	HIGH	HIGH	CAS-2	L	H	L	L	H	L
N+3	HIGH	LOW		L	L	L	L	L	L
N+4	HIGH	LOW	DES	L	L	L	L	L	L
N+5	HIGH	LOW	DES	L	L	L	L	L	L
N+6	HIGH	LOW	DES	L	L	L	L	L	L
N+7	HIGH	LOW	DES	L	L	L	L	L	L
N+8	HIGH	LOW	DES	L	L	L	L	L	L
N+9	HIGH	LOW	DES	L	L	L	L	L	L
N+10	HIGH	LOW	DES	L	L	L	L	L	L
N+11	HIGH	LOW	DES	L	L	L	L	L	L
N+12	HIGH	LOW	DES	L	L	L	L	L	L
N+13	HIGH	LOW	DES	L	L	L	L	L	L
N+14	HIGH	LOW	DES	L	L	L	L	L	L
N+15	HIGH	LOW	DES	L	L	L	L	L	L
N+16	HIGH	HIGH	Write-1	L	L	H	L	L	L
N+17	HIGH	LOW		L	H	L	L	H	L
N+18	HIGH	HIGH	CAS-2	L	H	L	L	H	H
N+19	HIGH	LOW		L	L	L	L	H	H
N+20	HIGH	LOW	DES	L	L	L	L	L	L
N+21	HIGH	LOW	DES	L	L	L	L	L	L
N+22	HIGH	LOW	DES	L	L	L	L	L	L
N+23	HIGH	LOW	DES	L	L	L	L	L	L
N+24	HIGH	LOW	DES	L	L	L	L	L	L
N+25	HIGH	LOW	DES	L	L	L	L	L	L
N+26	HIGH	LOW	DES	L	L	L	L	L	L
N+27	HIGH	LOW	DES	L	L	L	L	L	L
N+28	HIGH	LOW	DES	L	L	L	L	L	L
N+29	HIGH	LOW	DES	L	L	L	L	L	L
N+30	HIGH	LOW	DES	L	L	L	L	L	L
N+31	HIGH	LOW	DES	L	L	L	L	L	L

NOTE :

1) BA[2:0] = 010B, CA[9:5] = 00000B or 11111B.

[Table 49] Data Pattern for IDD4W (DBI off) for BL=32

DBI OFF Case										No. of 1's
	DQ[7]	DQ[6]	DQ[5]	DQ[4]	DQ[3]	DQ[2]	DQ[1]	DQ[0]	DBI	
BL0	1	1	1	1	1	1	1	1	0	8
BL1	1	1	1	1	0	0	0	0	0	4
BL2	0	0	0	0	0	0	0	0	0	0
BL3	0	0	0	0	1	1	1	1	0	4
BL4	0	0	0	0	0	0	1	1	0	2
BL5	0	0	0	0	1	1	1	1	0	4
BL6	1	1	1	1	1	1	0	0	0	6
BL7	1	1	1	1	0	0	0	0	0	4
BL8	1	1	1	1	1	1	1	1	0	8
BL9	1	1	1	1	0	0	0	0	0	4
BL10	0	0	0	0	0	0	0	0	0	0
BL11	0	0	0	0	1	1	1	1	0	4
BL12	0	0	0	0	0	0	1	1	0	2
BL13	0	0	0	0	1	1	1	1	0	4
BL14	1	1	1	1	1	1	0	0	0	6
BL15	1	1	1	1	0	0	0	0	0	4
BL16	1	1	1	1	1	1	0	0	0	6
BL17	1	1	1	1	0	0	0	0	0	4
BL18	0	0	0	0	0	0	1	1	0	2
BL19	0	0	0	0	1	1	1	1	0	4
BL20	0	0	0	0	0	0	0	0	0	0
BL21	0	0	0	0	1	1	1	1	0	4
BL22	1	1	1	1	1	1	1	1	0	8
BL23	1	1	1	1	0	0	0	0	0	4
BL24	0	0	0	0	0	0	1	1	0	2
BL25	0	0	0	0	1	1	1	1	0	4
BL26	1	1	1	1	1	1	0	0	0	6
BL27	1	1	1	1	0	0	0	0	0	4
BL28	1	1	1	1	1	1	1	1	0	8
BL29	1	1	1	1	0	0	0	0	0	4
BL30	0	0	0	0	0	0	0	0	0	0
BL31	0	0	0	0	1	1	1	1	0	4
BL32	1	1	1	1	1	1	1	1	0	8
BL33	1	1	1	1	0	0	0	0	0	4
BL34	0	0	0	0	0	0	0	0	0	0
BL35	0	0	0	0	1	1	1	1	0	4

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[Table 49] Data Pattern for IDD4W (DBI off) for BL=32

DBI OFF Case										No. of 1's
	DQ[7]	DQ[6]	DQ[5]	DQ[4]	DQ[3]	DQ[2]	DQ[1]	DQ[0]	DBI	
BL36	0	0	0	0	0	0	1	1	0	2
BL37	0	0	0	0	1	1	1	1	0	4
BL38	1	1	1	1	1	1	0	0	0	6
BL39	1	1	1	1	0	0	0	0	0	4
BL40	1	1	1	1	1	1	1	1	0	8
BL41	1	1	1	1	0	0	0	0	0	4
BL42	0	0	0	0	0	0	0	0	0	0
BL43	0	0	0	0	1	1	1	1	0	4
BL44	0	0	0	0	0	0	1	1	0	2
BL45	0	0	0	0	1	1	1	1	0	4
BL46	1	1	1	1	1	1	0	0	0	6
BL47	1	1	1	1	0	0	0	0	0	4
BL48	1	1	1	1	1	1	0	0	0	6
BL49	1	1	1	1	0	0	0	0	0	4
BL50	0	0	0	0	0	0	1	1	0	2
BL51	0	0	0	0	1	1	1	1	0	4
BL52	0	0	0	0	0	0	0	0	0	0
BL53	0	0	0	0	1	1	1	1	0	4
BL54	1	1	1	1	1	1	1	1	0	8
BL55	1	1	1	1	0	0	0	0	0	4
BL56	0	0	0	0	0	0	1	1	0	2
BL57	0	0	0	0	1	1	1	1	0	4
BL58	1	1	1	1	1	1	0	0	0	6
BL59	1	1	1	1	0	0	0	0	0	4
BL60	1	1	1	1	1	1	1	1	0	8
BL61	1	1	1	1	0	0	0	0	0	4
BL62	0	0	0	0	0	0	0	0	0	0
BL63	0	0	0	0	1	1	1	1	0	4
No. of 1's	32	32	32	32	32	32	32	32		

NOTE :

1) Simplified pattern compared with last showing.

Same data pattern was applied to DQ[4], DQ[5], DQ[6], DQ[7] for reducing complexity for IDD4W/R pattern programming.

[Table 50] Data Pattern for IDD4R (DBI off) for BL=32

DBI OFF Case										No. of 1's
	DQ[7]	DQ[6]	DQ[5]	DQ[4]	DQ[3]	DQ[2]	DQ[1]	DQ[0]	DBI	
BL0	1	1	1	1	1	1	1	1	0	8
BL1	1	1	1	1	0	0	0	0	0	4
BL2	0	0	0	0	0	0	0	0	0	0
BL3	0	0	0	0	1	1	1	1	0	4
BL4	0	0	0	0	0	0	1	1	0	2
BL5	0	0	0	0	1	1	1	1	0	4
BL6	1	1	1	1	1	1	0	0	0	6
BL7	1	1	1	1	0	0	0	0	0	4
BL8	1	1	1	1	1	1	1	1	0	8
BL9	1	1	1	1	0	0	0	0	0	4
BL10	0	0	0	0	0	0	0	0	0	0
BL11	0	0	0	0	1	1	1	1	0	4
BL12	0	0	0	0	0	0	1	1	0	2
BL13	0	0	0	0	1	1	1	1	0	4
BL14	1	1	1	1	1	1	0	0	0	6
BL15	1	1	1	1	0	0	0	0	0	4
BL16	1	1	1	1	1	1	0	0	0	6
BL17	1	1	1	1	0	0	0	0	0	4
BL18	0	0	0	0	0	0	1	1	0	2
BL19	0	0	0	0	1	1	1	1	0	4
BL20	0	0	0	0	0	0	0	0	0	0
BL21	0	0	0	0	1	1	1	1	0	4
BL22	1	1	1	1	1	1	1	1	0	8
BL23	1	1	1	1	0	0	0	0	0	4
BL24	0	0	0	0	0	0	1	1	0	2
BL25	0	0	0	0	1	1	1	1	0	4
BL26	1	1	1	1	1	1	0	0	0	6
BL27	1	1	1	1	0	0	0	0	0	4
BL28	1	1	1	1	1	1	1	1	0	8
BL29	1	1	1	1	0	0	0	0	0	4
BL30	0	0	0	0	0	0	0	0	0	0
BL31	0	0	0	0	1	1	1	1	0	4
BL32	0	0	0	0	0	0	1	1	0	2
BL33	0	0	0	0	1	1	1	1	0	4
BL34	1	1	1	1	1	1	0	0	0	6
BL35	1	1	1	1	0	0	0	0	0	4

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Rev. 1.0

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[Table 50] Data Pattern for IDD4R (DBI off) for BL=32

DBI OFF Case										No. of 1's
	DQ[7]	DQ[6]	DQ[5]	DQ[4]	DQ[3]	DQ[2]	DQ[1]	DQ[0]	DBI	
BL36	1	1	1	1	1	1	1	1	0	8
BL37	1	1	1	1	0	0	0	0	0	4
BL38	0	0	0	0	0	0	0	0	0	0
BL39	0	0	0	0	1	1	1	1	0	4
BL40	0	0	0	0	0	0	1	1	0	2
BL41	0	0	0	0	1	1	1	1	0	4
BL42	1	1	1	1	1	1	0	0	0	6
BL43	1	1	1	1	0	0	0	0	0	4
BL44	1	1	1	1	1	1	1	1	0	8
BL45	1	1	1	1	0	0	0	0	0	4
BL46	0	0	0	0	0	0	0	0	0	0
BL47	0	0	0	0	1	1	1	1	0	4
BL48	1	1	1	1	1	1	1	1	0	8
BL49	1	1	1	1	0	0	0	0	0	4
BL50	0	0	0	0	0	0	0	0	0	0
BL51	0	0	0	0	1	1	1	1	0	4
BL52	1	1	1	1	1	1	0	0	0	6
BL53	1	1	1	1	0	0	0	0	0	4
BL54	0	0	0	0	0	0	1	1	0	2
BL55	0	0	0	0	1	1	1	1	0	4
BL56	0	0	0	0	0	0	0	0	0	0
BL57	0	0	0	0	1	1	1	1	0	4
BL58	1	1	1	1	1	1	1	1	0	8
BL59	1	1	1	1	0	0	0	0	0	4
BL60	0	0	0	0	0	0	1	1	0	2
BL61	0	0	0	0	1	1	1	1	0	4
BL62	1	1	1	1	1	1	0	0	0	6
BL63	1	1	1	1	0	0	0	0	0	4
No. of 1's	32	32	32	32	32	32	32	32		

NOTE :

1) Same data pattern was applied to DQ[4], DQ[5], DQ[6], DQ[7] for reducing complexity for IDD4W/R pattern programming.

[Table 51] Data Pattern for IDD4W (DBI on) for BL=32

DBI ON Case										No. of 1's
	DQ[7]	DQ[6]	DQ[5]	DQ[4]	DQ[3]	DQ[2]	DQ[1]	DQ[0]	DBI	
BL0	0	0	0	0	0	0	0	0	1	1
BL1	1	1	1	1	0	0	0	0	0	4
BL2	0	0	0	0	0	0	0	0	0	0
BL3	0	0	0	0	1	1	1	1	0	4
BL4	0	0	0	0	0	0	1	1	0	2
BL5	0	0	0	0	1	1	1	1	0	4
BL6	0	0	0	0	0	0	1	1	1	3
BL7	1	1	1	1	0	0	0	0	0	4
BL8	0	0	0	0	0	0	0	0	1	1
BL9	1	1	1	1	0	0	0	0	0	4
BL10	0	0	0	0	0	0	0	0	0	0
BL11	0	0	0	0	1	1	1	1	0	4
BL12	0	0	0	0	0	0	1	1	0	2
BL13	0	0	0	0	1	1	1	1	0	4
BL14	0	0	0	0	0	0	1	1	1	3
BL15	1	1	1	1	0	0	0	0	0	4
BL16	0	0	0	0	0	0	1	1	1	3
BL17	1	1	1	1	0	0	0	0	0	4
BL18	0	0	0	0	0	0	1	1	0	2
BL19	0	0	0	0	1	1	1	1	0	4
BL20	0	0	0	0	0	0	0	0	0	0
BL21	0	0	0	0	1	1	1	1	0	4
BL22	0	0	0	0	0	0	0	0	1	1
BL23	1	1	1	1	0	0	0	0	0	4
BL24	0	0	0	0	0	0	1	1	0	2
BL25	0	0	0	0	1	1	1	1	0	4
BL26	0	0	0	0	0	0	1	1	1	3
BL27	1	1	1	1	0	0	0	0	0	4
BL28	0	0	0	0	0	0	0	0	1	1
BL29	1	1	1	1	0	0	0	0	0	4
BL30	0	0	0	0	0	0	0	0	0	0
BL31	0	0	0	0	1	1	1	1	0	4
BL32	0	0	0	0	0	0	0	0	1	1
BL33	1	1	1	1	0	0	0	0	0	4
BL34	0	0	0	0	0	0	0	0	0	0
BL35	0	0	0	0	1	1	1	1	0	4
BL36	0	0	0	0	0	0	1	1	0	2

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[Table 51] Data Pattern for IDD4W (DBI on) for BL=32

DBI ON Case										No. of 1's
	DQ[7]	DQ[6]	DQ[5]	DQ[4]	DQ[3]	DQ[2]	DQ[1]	DQ[0]	DBI	
BL37	0	0	0	0	1	1	1	1	0	4
BL38	0	0	0	0	0	0	1	1	1	3
BL39	1	1	1	1	0	0	0	0	0	4
BL40	0	0	0	0	0	0	0	0	1	1
BL41	1	1	1	1	0	0	0	0	0	4
BL42	0	0	0	0	0	0	0	0	0	0
BL43	0	0	0	0	1	1	1	1	0	4
BL44	0	0	0	0	0	0	1	1	0	2
BL45	0	0	0	0	1	1	1	1	0	4
BL46	0	0	0	0	0	0	1	1	1	3
BL47	1	1	1	1	0	0	0	0	0	4
BL48	0	0	0	0	0	0	1	1	1	3
BL49	1	1	1	1	0	0	0	0	0	4
BL50	0	0	0	0	0	0	1	1	0	2
BL51	0	0	0	0	1	1	1	1	0	4
BL52	0	0	0	0	0	0	0	0	0	0
BL53	0	0	0	0	1	1	1	1	0	4
BL54	0	0	0	0	0	0	0	0	1	1
BL55	1	1	1	1	0	0	0	0	0	4
BL56	0	0	0	0	0	0	1	1	0	2
BL57	0	0	0	0	1	1	1	1	0	4
BL58	0	0	0	0	0	0	1	1	1	3
BL59	1	1	1	1	0	0	0	0	0	4
BL60	0	0	0	0	0	0	0	0	1	1
BL61	1	1	1	1	0	0	0	0	0	4
BL62	0	0	0	0	0	0	0	0	0	0
BL63	0	0	0	0	1	1	1	1	0	4
No. of 1's	16	16	16	16	16	16	32	32	16	

 DBI enabled burst

[Table 52] Data Pattern for IDD4R (DBI on) for BL=32

DBI ON Case										No. of 1's
	DQ[7]	DQ[6]	DQ[5]	DQ[4]	DQ[3]	DQ[2]	DQ[1]	DQ[0]	DBI	
BL0	0	0	0	0	0	0	0	0	1	1
BL1	1	1	1	1	0	0	0	0	0	4
BL2	0	0	0	0	0	0	0	0	0	0
BL3	0	0	0	0	1	1	1	1	0	4
BL4	0	0	0	0	0	0	1	1	0	2
BL5	0	0	0	0	1	1	1	1	0	4
BL6	0	0	0	0	0	0	1	1	1	3
BL7	1	1	1	1	0	0	0	0	0	4
BL8	0	0	0	0	0	0	0	0	1	1
BL9	1	1	1	1	0	0	0	0	0	4
BL10	0	0	0	0	0	0	0	0	0	0
BL11	0	0	0	0	1	1	1	1	0	4
BL12	0	0	0	0	0	0	1	1	0	2
BL13	0	0	0	0	1	1	1	1	0	4
BL14	0	0	0	0	0	0	1	1	1	3
BL15	1	1	1	1	0	0	0	0	0	4
BL16	0	0	0	0	0	0	1	1	1	3
BL17	1	1	1	1	0	0	0	0	0	4
BL18	0	0	0	0	0	0	1	1	0	2
BL19	0	0	0	0	1	1	1	1	0	4
BL20	0	0	0	0	0	0	0	0	0	0
BL21	0	0	0	0	1	1	1	1	0	4
BL22	0	0	0	0	0	0	0	0	1	1
BL23	1	1	1	1	0	0	0	0	0	4
BL24	0	0	0	0	0	0	1	1	0	2
BL25	0	0	0	0	1	1	1	1	0	4
BL26	0	0	0	0	0	0	1	1	1	3
BL27	1	1	1	1	0	0	0	0	0	4
BL28	0	0	0	0	0	0	0	0	1	1
BL29	1	1	1	1	0	0	0	0	0	4
BL30	0	0	0	0	0	0	0	0	0	0
BL31	0	0	0	0	1	1	1	1	0	4
BL32	0	0	0	0	0	0	1	1	0	2
BL33	0	0	0	0	1	1	1	1	0	4
BL34	0	0	0	0	0	0	1	1	1	3
BL35	1	1	1	1	0	0	0	0	0	4
BL36	0	0	0	0	0	0	0	0	1	1
BL37	1	1	1	1	0	0	0	0	0	4
BL38	0	0	0	0	0	0	0	0	0	0
BL39	0	0	0	0	1	1	1	1	0	4
BL40	0	0	0	0	0	0	1	1	0	2
BL41	0	0	0	0	1	1	1	1	0	4

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[Table 52] Data Pattern for IDD4R (DBI on) for BL=32

DBI ON Case										No. of 1's
	DQ[7]	DQ[6]	DQ[5]	DQ[4]	DQ[3]	DQ[2]	DQ[1]	DQ[0]	DBI	
BL42	0	0	0	0	0	0	1	1	1	3
BL43	1	1	1	1	0	0	0	0	0	4
BL44	0	0	0	0	0	0	0	0	1	1
BL45	1	1	1	1	0	0	0	0	0	4
BL46	0	0	0	0	0	0	0	0	0	0
BL47	0	0	0	0	1	1	1	1	0	4
BL48	0	0	0	0	0	0	0	0	1	1
BL49	1	1	1	1	0	0	0	0	0	4
BL50	0	0	0	0	0	0	0	0	0	0
BL51	0	0	0	0	1	1	1	1	0	4
BL52	0	0	0	0	0	0	1	1	1	3
BL53	1	1	1	1	0	0	0	0	0	4
BL54	0	0	0	0	0	0	1	1	0	2
BL55	0	0	0	0	1	1	1	1	0	4
BL56	0	0	0	0	0	0	0	0	0	0
BL57	0	0	0	0	1	1	1	1	0	4
BL58	0	0	0	0	0	0	0	0	1	1
BL59	1	1	1	1	0	0	0	0	0	4
BL60	0	0	0	0	0	0	1	1	0	2
BL61	0	0	0	0	1	1	1	1	0	4
BL62	0	0	0	0	0	0	1	1	1	3
BL63	1	1	1	1	0	0	0	0	0	4
No. of 1's	16	16	16	16	16	16	32	32	16	

8.2 IDD Specifications

IDD values are for the entire operating voltage range, and all of them are for the entire standard range, with the exception of IDD6ET which is for the entire elevated temperature range.

[Table 53] LPDDR4X IDD Specification Parameters and Operating Conditions

Parameter/Condition	Symbol	Power Supply	Notes
Operating one bank active-precharge current: $t_{CK} = t_{CKmin}$; $t_{RC} = t_{RCmin}$; CKE is HIGH; CS is LOW between valid commands; CA bus inputs are switching; Data bus inputs are stable ODT disabled	IDD0 ₁	VDD1	1,10,11
	IDD0 ₂	VDD2	1,10,11
	IDD0 _Q	VDDQ	1,3,10,11
Idle power-down standby current: $t_{CK} = t_{CKmin}$; CKE is LOW; CS is LOW; All banks are idle; CA bus inputs are switching; Data bus inputs are stable ODT disabled	IDD2P ₁	VDD1	1,10,11
	IDD2P ₂	VDD2	1,10,11
	IDD2P _Q	VDDQ	1,3,10,11
Idle power-down standby current with clock stop: $CK_t = LOW$, $CK_c = HIGH$; CKE is LOW; CS is LOW; All banks are idle; CA bus inputs are stable; Data bus inputs are stable ODT disabled	IDD2PS ₁	VDD1	1,10,11
	IDD2PS ₂	VDD2	1,10,11
	IDD2PS _Q	VDDQ	1,3,10,11
Idle non power-down standby current: $t_{CK} = t_{CKmin}$; CKE is HIGH; CS is LOW; All banks are idle; CA bus inputs are switching; Data bus inputs are stable ODT disabled	IDD2N ₁	VDD1	1,10,11
	IDD2N ₂	VDD2	1,10,11
	IDD2N _Q	VDDQ	1,3,10,11
Idle non power-down standby current with clock stopped: $CK_t = LOW$; $CK_c = HIGH$; CKE is HIGH; CS is LOW; All banks are idle; CA bus inputs are stable; Data bus inputs are stable ODT disabled	IDD2NS ₁	VDD1	1,10,11
	IDD2NS ₂	VDD2	1,10,11
	IDD2NS _Q	VDDQ	1,3,10,11
Active power-down standby current: $t_{CK} = t_{CKmin}$; CKE is LOW; CS is LOW; One bank is active; CA bus inputs are switching; Data bus inputs are stable ODT disabled	IDD3P ₁	VDD1	1,10,11
	IDD3P ₂	VDD2	1,10,11
	IDD3P _Q	VDDQ	1,3,10,11
Active power-down standby current with clock stop: $CK_t = LOW$, $CK_c = HIGH$; CKE is LOW; CS is LOW; One bank is active; CA bus inputs are stable; Data bus inputs are stable ODT disabled	IDD3PS ₁	VDD1	1,10,11
	IDD3PS ₂	VDD2	1,10,11
	IDD3PS _Q	VDDQ	1,4,10,11

[Table 53] LPDDR4X IDD Specification Parameters and Operating Conditions

Parameter/Condition	Symbol	Power Supply	Notes
Active non-power-down standby current: $t_{CK} = t_{CKmin}$; CKE is HIGH; CS is LOW; One bank is active; CA bus inputs are switching; Data bus inputs are stable ODT disabled	IDD3N ₁	VDD1	1,10,11
	IDD3N ₂	VDD2	1,10,11
	IDD3N _Q	VDDQ	1,4,10,11
Active non-power-down standby current with clock stopped: CK _t =LOW, CK _c =HIGH; CKE is HIGH; CS is LOW; One bank is active; CA bus inputs are stable; Data bus inputs are stable ODT disabled	IDD3NS ₁	VDD1	1,10,11
	IDD3NS ₂	VDD2	1,10,11
	IDD3NS _Q	VDDQ	1,4,10,11
Operating burst READ current: $t_{CK} = t_{CKmin}$; CS is LOW between valid commands; One bank is active; BL = 16 or 32; RL = RL(MIN); CA bus inputs are switching; 50% data change each burst transfer ODT disabled	IDD4R ₁	VDD1	1,10,11
	IDD4R ₂	VDD2	1,10,11
	IDD4R _Q	VDDQ	1,5,10,11
Operating burst WRITE current: $t_{CK} = t_{CKmin}$; CS is LOW between valid commands; One bank is active; BL = 16 or 32; WL = WLmin; CA bus inputs are switching; 50% data change each burst transfer ODT disabled	IDD4W ₁	VDD1	1,10,11
	IDD4W ₂	VDD2	1,10,11
	IDD4W _Q	VDDQ	1,4,10,11
All-bank REFRESH Burst current: $t_{CK} = t_{CKmin}$; CKE is HIGH between valid commands; $t_{RC} = t_{RFCabmin}$; Burst refresh; CA bus inputs are switching; Data bus inputs are stable; ODT disabled	IDD5 ₁	VDD1	1,10,11
	IDD5 ₂	VDD2	1,10,11
	IDD5 _Q	VDDQ	1,4,10,11
All-bank REFRESH Average current: $t_{CK} = t_{CKmin}$; CKE is HIGH between valid commands; $t_{RC} = t_{REFI}$; CA bus inputs are switching; Data bus inputs are stable; ODT disabled	IDD5AB ₁	VDD1	1,10,11
	IDD5AB ₂	VDD2	1,10,11
	IDD5AB _Q	VDDQ	1,4,10,11
Per-bank REFRESH Average current: $t_{CK} = t_{CKmin}$; CKE is HIGH between valid commands; $t_{RC} = t_{REFI}/8$; CA bus inputs are switching; Data bus inputs are stable; ODT disabled	IDD5PB ₁	VDD1	1,10,11
	IDD5PB ₂	VDD2	1,10,11
	IDD5PB _Q	VDDQ	1,4,10,11
Power Down Self refresh current (-25°C to +85°C): CK _t =LOW, CK _c =HIGH; CKE is LOW; CA bus inputs are stable; Data bus inputs are stable; Maximum 1x Self-Refresh Rate; ODT disabled	IDD6 ₁	VDD1	6,7,9,10,11
	IDD6 ₂	VDD2	6,7,9,10,11
	IDD6 _Q	VDDQ	4,6,7,9,10,11

NOTE :

- 1) Published IDD values are the maximum of the distribution of the arithmetic mean.
- 2) ODT disabled: MR11[2:0] = 000_B.
- 3) IDD current specifications are tested after the device is properly initialized.
- 4) Measured currents are the summation of VDDQ and VDD2.
- 5) Guaranteed by design with output load = 5pF and RON = 40 ohm.
- 6) The 1x Self-Refresh Rate is the rate at which the LPDDR4X device is refreshed internally during Self-Refresh, before going into the elevated Temperature range.
- 7) This is the general definition that applies to full array Self Refresh.
- 8) For all IDD measurements, VIHCKE = 0.8 x VDD2, VILCKE = 0.2 x VDD2.
- 9) IDD6 25°C is guaranteed, IDD6 85°C is typical of the distribution of the arithmetic mean.
- 10) These specification values are the summation of all the channel current and both channels are under the same condition at the same time.
- 11) Dual Channel devices are specified in dual channel operation (both channels operating together).

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8.3 IDD Spec Table

[Table 54] IDD Specification for 16Gb LPDDR4X

Symbol		Power Supply	16Gb (x16/Ch, 2-Chip)	Units
			4266Mbps	
IDD0	IDD0 ₁	VDD1	10	mA
	IDD0 ₂	VDD2	65	mA
	IDD0 _Q	VDDQ	0.5	mA
IDD2P	IDD2P ₁	VDD1	2	mA
	IDD2P ₂	VDD2	6.25	mA
	IDD2P _Q	VDDQ	0.5	mA
IDD2PS	IDD2PS ₁	VDD1	2	mA
	IDD2PS ₂	VDD2	6.25	mA
	IDD2PS _Q	VDDQ	0.5	mA
IDD2N	IDD2N ₁	VDD1	3	mA
	IDD2N ₂	VDD2	26.5	mA
	IDD2N _Q	VDDQ	0.5	mA
IDD2NS	IDD2NS ₁	VDD1	3	mA
	IDD2NS ₂	VDD2	20.5	mA
	IDD2NS _Q	VDDQ	0.5	mA
IDD3P	IDD3P ₁	VDD1	2.8	mA
	IDD3P ₂	VDD2	13.5	mA
	IDD3P _Q	VDDQ	0.5	mA
IDD3PS	IDD3PS ₁	VDD1	2.8	mA
	IDD3PS ₂	VDD2	13.5	mA
	IDD3PS _Q	VDDQ	0.5	mA
IDD3N	IDD3N ₁	VDD1	3	mA
	IDD3N ₂	VDD2	34	mA
	IDD3N _Q	VDDQ	0.5	mA
IDD3NS	IDD3NS ₁	VDD1	3	mA
	IDD3NS ₂	VDD2	30	mA
	IDD3NS _Q	VDDQ	0.5	mA
IDD4R	IDD4R ₁	VDD1	46	mA
	IDD4R ₂	VDD2	465	mA
	IDD4R _Q	VDDQ	185	mA
IDD4W	IDD4W ₁	VDD1	41	mA
	IDD4W ₂	VDD2	460	mA
	IDD4W _Q	VDDQ	0.5	mA
IDD5	IDD5 ₁	VDD1	75	mA
	IDD5 ₂	VDD2	300	mA
	IDD5 _Q	VDDQ	0.5	mA
IDD5AB	IDD5AB ₁	VDD1	7	mA
	IDD5AB ₂	VDD2	40	mA
	IDD5AB _Q	VDDQ	0.5	mA

[Table 54] IDD Specification for 16Gb LPDDR4X

Symbol			Power Supply	16Gb (x16/Ch, 2-Chip)	Units
				4266Mbps	
IDD5PB	IDD5PB ₁		VDD1	7	mA
	IDD5PB ₂		VDD2	41	mA
	IDD5PB _Q		VDDQ	0.5	mA
IDD6	IDD6 ₁	25°C	VDD1	1	mA
		85°C		5	
	IDD6 ₂	25°C	VDD2	2.7	mA
		85°C		22	
	IDD6 _Q	25°C	VDDQ	0.4	mA
		85°C		0.5	

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9.0 AC AND DC OUTPUT MEASUREMENT LEVELS

9.1 Single Ended AC and DC Output Levels

Table 55 shows the output levels used for measurements of single ended signals.

[Table 55] Single-ended AC and DC Output Levels

Symbol	Parameter	Value			Unit	Notes
		Under LPDDR4X-TBD Un-term	TBD to 3200 VSSQ term	3200 to 4266 VSSQ term		
V_{OH} (DC)	AC, DC output high measurement level	VDDQ	VDDQ/2	VDDQ/2	V	1
V_{OL} (DC)	AC, DC output low measurement level	VSSQ	VSSQ	VSSQ	V	

NOTE :

1) 60ohm ODT value is assumed.

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9.2 Pull Up/Pull Down Driver Characteristics and Calibration

[Table 56] Pull-down Driver Characteristics, with ZQ Calibration

R _{ONPD,NOM}	Resistor	Min	Nom	Max	Unit
40 Ohm	R _{ON40PD}	0.9	1.0	1.1	R _{ZQ/6}
48 Ohm	R _{ON48PD}	0.9	1.0	1.1	R _{ZQ/5}
60 Ohm	R _{ON60PD}	0.9	1.0	1.1	R _{ZQ/4}
80 Ohm	R _{ON80PD}	0.9	1.0	1.1	R _{ZQ/3}
120 Ohm	R _{ON120PD}	0.9	1.0	1.1	R _{ZQ/2}
240 Ohm	R _{ON240PD}	0.9	1.0	1.1	R _{ZQ/1}

NOTE :

1) All value are after ZQ Calibration. Without ZQ Calibration R_{ONPD} values are $\pm 30\%$.

[Table 57] Terminated Pull-up Characteristics, with ZQ Calibration

VOH _{PU, nom}	VOH,nom (mV)	Min	Nor	Max	Unit
VDDQ $\times 0.5$	300	0.90	1.0	1.10	VOH,nom
VDDQ $\times 0.6$	360	0.90	1.0	1.10	VOH,nom

NOTE :

1) All values are after ZQ Calibration. Without ZQ Calibration VOH(nom) values are $\pm 30\%$

2) VOH,nom (mV) values are based on a nominal VDDQ = 0.6V.

[Table 58] Terminated Valid Calibration Points

VOH _{PU, nom}	ODT Value					
	240	120	80	60	48	40
VDDQ $\times 0.5$	VALID	VALID	VALID	VALID	VALID	VALID
VDDQ $\times 0.6$	DNU	VALID	DNU	VALID	DNU	DNU

NOTE :

1) Once the output is calibrated for a given VOH(nom) calibration point, the ODT value may be changed without recalibration.

2) If the VOH(nom) calibration point is changed, then re-calibration is required.

3) DNU = Do Not Use.

[Table 59] Pull-down Characteristics without ZQ Calibration

R _{ONPD,NOM}	Resistor	V _{out}	Min	Nom	Max	Unit	Notes
40.0 Ω	R _{ON40PD}	$0.5 \times V_{OH}$	0.70	1.00	1.30	R _{ZQ/6}	1
48.0 Ω	R _{ON48PD}	$0.5 \times V_{OH}$	0.70	1.00	1.30	R _{ZQ/5}	1

NOTE:

1) Across entire operating temperature range, without calibration.

10.0 ELECTRICAL CHARACTERISTICS AND AC TIMING

10.1 Clock Specification

The jitter specified is a random jitter meeting a Gaussian distribution. Input clocks violating the min/max values may result in malfunction of the LPDDR4X device.

10.1.1 Definition for tCK(avg) and nCK

tCK(avg) is calculated as the average clock period across any consecutive 200 cycle window, where each clock period is calculated from rising edge to rising edge.

$$tCK(avg) = \left(\sum_{j=1}^N tCK_j \right) / N$$

where $N = 200$

Unit 'tCK(avg)' represents the actual clock average tCK(avg) of the input clock under operation. Unit 'nCK' represents one clock cycle of the input clock, counting the actual clock edges.

tCK(avg) may change by up to +/-1% within a 100 clock cycle window, provided that all jitter and timing specs are met.

10.1.2 Definition for tCK(abs)

tCK(abs) is defined as the absolute clock period, as measured from one rising edge to the next consecutive rising edge.

tCK(abs) is not subject to production test.

10.1.3 Definition for tCH(avg) and tCL(avg)

tCH(avg) is defined as the average high pulse width, as calculated across any consecutive 200 high pulses.

$$tCH(avg) = \left(\sum_{j=1}^N tCH_j \right) / (N \times tCK(avg))$$

where $N = 200$

tCL(avg) is defined as the average low pulse width, as calculated across any consecutive 200 low pulses.

$$tCL(avg) = \left(\sum_{j=1}^N tCL_j \right) / (N \times tCK(avg))$$

where $N = 200$

10.1.4 Definition for tCH(abs) and tCL(abs)

tCH(abs) is the absolute instantaneous clock high pulse width, as measured from one rising edge to the following falling edge.

tCL(abs) is the absolute instantaneous clock low pulse width, as measured from one falling edge to the following rising edge.

Both tCH(abs) and tCL(abs) are not subject to production test.

10.1.5 Definition for tJIT(per)

tJIT(per) is the single period jitter defined as the largest deviation of any signal tCK from tCK(avg).

tJIT(per) = Min/max of {tCK_i - tCK(avg) where i = 1 to 200}.

tJIT(per)_{act} is the actual clock jitter for a given system.

tJIT(per)_{allowed} is the specified allowed clock period jitter.

tJIT(per) is not subject to production test.

IF THERE IS ANY OTHER OPERATION TO IMPLEMENT IN ADDITION TO SPECIFICATION IN THE DATASHEET OR JEDEC STANDARD, PLEASE CONTACT EACH BRANCH OFFICE OR HEADQUARTERS OF SAMSUNG ELECTRONICS.

Rev. 1.0

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10.1.6 Definition for tJIT(cc)

tJIT(cc) is defined as the absolute difference in clock period between two consecutive clock cycles.

$t_{JIT(cc)} = \text{Max of } \{t_{CK(i+1)} - t_{CK(i)}\}$.

tJIT(cc) defines the cycle to cycle jitter.

tJIT(cc) is not subject to production test.

10.1.7 Definition for tERR(nper)

tERR(nper) is defined as the cumulative error across n multiple consecutive cycles from tCK(avg).

tERR(nper)_{act} is the actual clock jitter over n cycles for a given system.

tERR(nper)_{allowed} is the specified allowed clock period jitter over n cycles.

tERR(nper) is not subject to production test.

$$tERR(nper) = \left(\sum_{j=i}^{i+n-1} tCK_j \right) - n \times tCK(avg)$$

tERR(nper)_{min} can be calculated by the formula shown below:

$$tERR(nper), min = (1 + 0.68 LN(n)) \times tJIT(per), min$$

tERR(nper)_{max} can be calculated by the formula shown below

$$tERR(nper), max = (1 + 0.68 LN(n)) \times tJIT(per), max$$

Using these equations, tERR(nper) tables can be generated for each tJIT(per)_{act} value.

10.1.8 Definition for duty cycle jitter tJIT(duty)

tJIT(duty) is defined with absolute and average specification of tCH / tCL.

$$tJIT(duty), min = MIN((tCH(abs), min - tCH(avg), min), (tCL(abs), min - tCL(avg), min)) \times tCK(avg)$$

$$tJIT(duty), max = MAX((tCH(abs), max - tCH(avg), max), (tCL(abs), max - tCL(avg), max)) \times tCK(avg)$$

10.1.9 Definition for tCK(abs), tCH(abs) and tCL(abs)

These parameters are specified per their average values, however it is understood that the following relationship between the average timing and the absolute instantaneous timing holds at all times.

[Table 60] Definition for tCK(abs), tCH(abs), and tCL(abs)

Parameter	Symbol	Min	Unit
Absolute Clock Period	t _{CK} (abs)	tCK(avg) _{min} + tJIT(per) _{min}	ps
Absolute Clock HIGH Pulse Width	t _{CH} (abs)	tCH(avg) _{min} + tJIT(duty) _{min} / tCK(avg) _{min}	tCK(avg)
Absolute Clock LOW Pulse Width	t _{CL} (abs)	tCL(avg) _{min} + tJIT(duty) _{min} / tCK(avg) _{min}	tCK(avg)

NOTE :

1) tCK(avg)_{min} is expressed in ps for this table.

2) tJIT(duty)_{min} is a negative value.

10.2 Period Clock Jitter

LPDDR4X devices can tolerate some clock period jitter without core timing parameter de-rating. This section describes device timing requirements in the presence of clock period jitter (tJIT(per)) in excess of the values found in Table 10.4, AC Timing and how to determine cycle time de-rating and clock cycle de-rating.

10.2.1 Clock period jitter effects on core timing parameters

(tRCD, tRP, tRTP, tWR, tWRA, tWTR, tRC, tRAS, tRRD, tFAW)

Core timing parameters extend across multiple clock cycles. Period clock jitter will impact these parameters when measured in numbers of clock cycles. When the device is operated with clock jitter within the specification limits, the LPDDR4X device is characterized and verified to support $tnPARAM = RU\{tPARAM / tCK(avg)\}$.

When the device is operated with clock jitter outside specification limits, the number of clocks or tCK(avg) may need to be increased based on the values for each core timing parameter.

10.2.1.1 Cycle time de-rating for core timing parameters

For a given number of clocks (tnPARAM), for each core timing parameter, average clock period (tCK(avg)) and actual cumulative period error (tERR(tnPARAM),act) in excess of the allowed cumulative period error (tERR(tnPARAM),allowed), the equation below calculates the amount of cycle time de-rating (in ns) required if the equation results in a positive value for a core timing parameter.

$$CycleTimeDerating = MAX\left\{\left(\frac{tPARAM + tERR(tnPARAM), act - tERR(tnPARAM), allowed}{tnPARAM} - tCK(avg)\right), 0\right\}$$

A cycle time derating analysis should be conducted for each core timing parameter. The amount of cycle time derating required is the maximum of the cycle time de-ratings determined for each individual core timing parameter.

10.2.1.2 Clock Cycle de-rating for core timing parameters

For a given number of clocks (tnPARAM) for each core timing parameter, clock cycle de-rating should be specified with amount of period jitter (tJIT(per)).

For a given number of clocks (tnPARAM), for each core timing parameter, average clock period (tCK(avg)) and actual cumulative period error (tERR(tnPARAM),act) in excess of the allowed cumulative period error (tERR(tnPARAM),allowed), the equation below calculates the clock cycle derating (in clocks) required if the equation results in a positive value for a core timing parameter.

$$ClockCycleDerating = RU\left\{\frac{tPARAM + tERR(tnPARAM), act - tERR(tnPARAM), allowed}{tCK(avg)}\right\} - tnPARAM$$

A clock cycle de-rating analysis should be conducted for each core timing parameter.

10.2.2 Clock jitter effects on Command/Address timing parameters

Command/address timing parameters (tIS, tIH, tISb, tIHb) are measured from a command/address signal (CS or CA[5:0]) transition edge to its respective clock signal (CK_t/ CK_c) crossing. The specification values are not affected by the tJIT(per) applied, because the setup and hold times are relative to the clock signal crossing that latches the command/address. Regardless of clock jitter values, these values must be met.

10.2.3 Clock jitter effects on Read timing parameters

10.2.3.1 tRPRE

When the device is operated with input clock jitter, tRPRE needs to be de-rated by the actual period jitter (tJIT(per),act,max) of the input clock in excess of the allowed period jitter (tJIT(per),allowed,max). Output de-ratings are relative to the input clock.

$$tRPRE(min, derated) = 0.9 - \left(\frac{tJIT(per), act, max - tJIT(per), allowed, max}{tCK(avg)} \right)$$

For example,

if the measured jitter into a LPDDR4X device has tCK(avg) = 625ps, tJIT(per),act,min = -xx, and tJIT(per),act,max = +xx ps, then tRPRE,min,derated = 0.9 - (tJIT(per),act,max - tJIT(per),allowed,max)/tCK(avg) = 0.9 - (xx - xx)/xx = yy tCK(avg).

10.2.3.2 tLZ(DQ), tHZ(DQ), tDQSCK, tLZ(DQS), tHZ(DQS)

These parameters are measured from a specific clock edge to a data signal (DMn, DQm.: n=0,1,2,3. m=0 –31) transition and will be met with respect to that clock edge. Therefore, they are not affected by the amount of clock jitter applied (i.e. tJIT(per)).

10.2.3.3 tQSH, tQSL

These parameters are affected by duty cycle jitter which is represented by tCH(abs)min and tCL(abs)min.

These parameters determine absolute Data-Valid window(DVW) at the LPDDR4X device pin.

Absolute min DVW @LPDDR4X device pin = min { (tQSH(abs)min – tDQSQmax), (tQSL(abs)min – tDQSQmax) }

This minimum DVW shall be met at the target frequency regardless of clock jitter.

10.2.3.4 tRPST

tRPST is affected by duty cycle jitter which is represented by tCL(abs). Therefore tRPST(abs)min can be specified by tCL(abs)min.

tRPST(abs)min = tCL(abs)min – 0.05 = tQSL(abs)min

10.2.4 Clock jitter effects on Write timing parameters

10.2.4.1 tDS, tDH

These parameters are measured from a data signal (DMn, DQm.: n=0,1,2,3. m=0 –31) transition edge to its respective data strobe signal (DQSn_t, DQSn_c : n=0,1,2,3) crossing. The spec values are not affected by the amount of clock jitter applied (i.e. tJIT(per)), as the setup and hold are relative to the data strobe signal crossing that latches the data. Regardless of clock jitter values, these values shall be met.

10.2.4.2 tDSS, tDSH

These parameters are measured from a data strobe signal (DQSx_t, DQSx_c) crossing to its respective clock signal (CK_t/CK_c) crossing. The spec values are not affected by the amount of clock jitter applied (i.e. tJIT(per)), as the setup and hold of the data strobes are relative to the corresponding clock signal crossing. Regardless of clock jitter values, these values shall be met.

10.2.4.3 tDQSS

This parameter is measured from a data strobe signal (DQSx_t, DQSx_c) crossing to the subsequent clock signal (CK_t/CK_c) crossing. When the device is operated with input clock jitter, this parameter needs to be de-rated by the actual period jitter tJIT(per),act of the input clock in excess of the allowed period jitter tJIT(per),allowed.

$$tDQSS(min, derated) = 0.75 - \frac{tJIT(per), act, min - tJIT(per), allowed, min}{tCK(avg)}$$

$$tDQSS(max, derated) = 1.25 - \frac{tJIT(per), act, max - tJIT(per), allowed, max}{tCK(avg)}$$

For example,

if the measured jitter into an LPDDR4X device has tCK(avg) = 625ps, tJIT(per),act,min = -xxps, and tJIT(per),act,max = +xx ps, then:

tDQSS,(min,derated) = 0.75 - (-xx + yy)/625 = xxxx tCK(avg)

tDQSS,(max,derated) = 1.25 - (xx . yy)/625 = xxxx tCK(avg)

10.3 LPDDR4X Refresh Requirement

[Table 61] LPDDR4X Refresh Requirement Parameters per die for Dual Channel SDRAM devices

Parameter		Symbol	16Gb	Unit
Density per Channel			8Gb	
Number of Banks per Channel			8	
Refresh Window Tcase ≤ 85°C		t _{REFW}	32	ms
Refresh Window 1/2-Rate Refresh		t _{REFW}	16	ms
Refresh Window 1/4-Rate Refresh		t _{REFW}	8	ms
Required number of REFRESH commands in a t _{REFW} window (min)		R	8,192	-
Average Refresh Interval	REFab	t _{REFI} ³⁾	3.904	us
	REFpb	t _{REFIpb}	488	ns
Refresh Cycle time (All Banks)		t _{RFCab}	280	ns
Refresh Cycle time (Per Bank)		t _{RFCpb}	140	ns
Per-bank Refresh to Per-bank Refresh different bank Time		t _{pbR2pbR}	90	ns

NOTE :

1) Refresh for each channel is independent of the other channel on the die, or other channels in a package. Power delivery in the user's system should be verified to make sure the DC operating conditions are maintained when multiple channels are refreshed simultaneously.

2) Self refresh abort feature is available for higher density devices starting with 12Gb dual channel device and 6Gb single channel device and tXSR_abort(min) is defined as t_{RFCpb} + 17.5ns.

3) t_{REFI} values for all bank refresh is Tc = -25~85°C, Tc means Operating Case Temperature.

10.4 AC Timing

[Table 62] LPDDR4X AC Timing Table

Parameter	Symbol	Min/ Max	LPDDR4X			Unit
			3200Mbps	3733Mbps	4266Mbps	
Maximum clock frequency		~	1600	1866	2133	MHz
Clock Timing						
Average Clock Period	$t_{CK(avg)}$	MIN	0.625	0.536	0.468	ns
		MAX	100			
Average HIGH pulse width	$t_{CH(avg)}^{35)}$	MIN	0.45			$t_{CK(avg)}$
		MAX	0.55			
Average LOW pulse width	$t_{CL(avg)}^{35)}$	MIN	0.45			$t_{CK(avg)}$
		MAX	0.55			
Absolute clock period	$t_{CK(abs)}$	MIN	$t_{CK(avg)} \text{ MIN} + t_{JIT(per)} \text{ MIN}$			ns
Absolute HIGH clock pulse width	$t_{CH(abs)}$	MIN	0.43			$t_{CK(avg)}$
		MAX	0.57			
Absolute LOW clock pulse width	$t_{CL(abs)}$	MIN	0.43			$t_{CK(avg)}$
		MAX	0.57			
Clock period jitter	$t_{JIT(per)}$	MIN	-40	-36	-30	ps
		MAX	40	36	30	
Maximum Clock Jitter between two consecutive cycles	$t_{JIT(cc)}$	MAX	80	72	60	ps
Duty cycle jitter (with supported jitter)	$t_{JIT(duty), allowed}$	MIN	$\min((t_{CH(abs),min} - t_{CH(avg),min}), (t_{CL(abs),min} - t_{CL(avg),min})) \times t_{CK(avg)}$			ps
		MAX	$\max((t_{CH(abs),max} - t_{CH(avg),max}), (t_{CL(abs),max} - t_{CL(avg),max})) \times t_{CK(avg)}$			
Core AC Parameters ¹⁷⁾						
READ latency (no DBI)	RL	MIN	28	32	36	$t_{CK(avg)}$
WRITE latency (set A)	WL	MIN	14	16	18	$t_{CK(avg)}$
ACTIVATE-to-ACTIVATE command period (same bank)	t_{RC}	MIN	$t_{RAS} + t_{RPab}$ (with all-bank precharge) $t_{RAS} + t_{RPpb}$ (with per-bank precharge)			ns
Minimum Self-Refresh Time (Entry to Exit)	t_{SR}	MIN	$\max(15ns, 3t_{CK})$			ns
SELF REFRESH exit to next valid command delay	t_{XSR}	MIN	$\text{Max}(t_{RFCab} + 7.5ns, 2t_{CK})$			ns
Exit power down to next valid command delay	t_{XP}	MIN	$\text{Max}(7.5ns, 5t_{CK})$			ns
CAS-to-CAS delay	t_{CCD}	MIN	BL/2			$t_{CK(avg)}$
CAS to CAS delay Masked Write	$t_{CCDMW}^{31)}$	MIN	$4 \times t_{CCD}$			$t_{CK(avg)}$
Internal READ to PRECHARGE command delay	t_{RTP}	MIN	$\text{Max}(7.5ns, 8t_{CK})$			ns
RAS-to-CAS delay	t_{RCD}	MIN	$\text{Max}(18ns, 4t_{CK})$			ns
Row Precharge Time (single bank)	t_{RPpb}	MIN	$\text{Max}(18ns, 4t_{CK})$			ns
Row Precharge Time (all banks)	t_{RPab}	MIN	$\text{Max}(21ns, 4t_{CK})$			ns
Row active time	t_{RAS}	MIN	$\text{Max}(42ns, 3t_{CK})$			ns
		MAX	$\min(9 \times t_{REFI} \times \text{Refresh Rate}^{19}), 70.2)$			us
WRITE recovery time	t_{WR}	MIN	$\text{Max}(18ns, 6t_{CK})$			ns
WRITE-to-READ delay	t_{WTR}	MIN	$\text{Max}(10ns, 8t_{CK})$			ns
Active bank-A to Active bank-B	t_{RRD}	MIN	$\text{Max}(10ns, 4t_{CK})$		$\text{Max}(7.5ns, 4t_{CK})^{32)}$	ns
Precharge to Precharge Delay	$t_{PPD}^{34)}$	MIN	4			tCK
Four-bank ACTIVATE Window	t_{FAW}	MIN	40		$30^{32)}$	ns
CKE minimum pulse width during SELF REFRESH (low pulse width during SELF REFRESH)	t_{CKELPD}	MIN	$\text{Max}(7.5ns, 3t_{CK})$			ns

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[Table 62] LPDDR4X AC Timing Table

Parameter	Symbol	Min/ Max	LPDDR4X			Unit
			3200Mbps	3733Mbps	4266Mbps	
READ AC Parameters ⁴⁾						
Read preamble	t _{RPRE} ^{5), 8)}	MIN	2.0			t _{CK} (avg)
0.5 tCK Read postamble	t _{RPST} ^{5), 9)}	MIN	0.5			t _{CK} (avg)
1.5 tCK Read postamble	t _{RPST}	MIN	1.5			t _{CK} (avg)
DQ low-impedance time from CK_t, CK_c	t _{LZ} (DQ) ⁵⁾	MIN	(RL × t _{CK}) + t _{DQSCK} (Min) - 200ps			ps
DQ high impedance time from CK_t, CK_c	t _{HZ} (DQ) ⁵⁾	MAX	(RL × t _{CK}) + t _{DQSCK} (Max) + t _{DQSQ} (Max) + (BL/2 × t _{CK}) - 100ps			ps
DQS_c low-impedance time from CK_t, CK_c	t _{LZ} (DQS) ⁵⁾	MIN	(RL × t _{CK}) + t _{DQSCK} (Min) - (t _{PRE} (Max) × t _{CK}) - 200ps			ps
DQS_c high impedance time from CK_t, CK_c	t _{HZ} (DQS) ⁵⁾	MAX	(RL × t _{CK}) + t _{DQSCK} (Max) + (BL/2 × t _{CK}) - (RPST(Max) × t _{CK}) - 100ps			ps
DQS-DQ skew	t _{DQSQ}	MAX	0.18			UI
tDQSCK AC Parameters						
DQS output access time from CK_t/CK_c	t _{DQSCK} ¹⁴⁾	MIN	1500			ps
		MAX	3500			
DQS output access time from CK_t/CK_c temperature variation	t _{DQSCK_temp} ¹⁵⁾	MAX	4			ps/°C
DQS output access time from CK_t/CK_c voltage variation	t _{DQSCK_volt} ¹⁶⁾	MAX	7			ps/mV
CK to DQS rank to rank variation	t _{DQSCK_rank2rank} ^{22), 23)}	MAX	1.0			ns
Self Refresh Parameters						
Delay from SRE command to CKE Input low	t _{ESCKE} ²⁴⁾	MIN	Max(1.75ns, 3tCK)			ns
Minimum Self Refresh Time	t _{SR} ²⁴⁾	MIN	Max(15ns, 3tCK)			ns
Exit Self Refresh to Valid commands	t _{XSR} ^{24),25)}	MIN	Max(tRFCab + 7.5ns, 2tCK)			ns
WRITE AC Parameters ⁴⁾						
Write command to 1 st DQS latching	t _{DQSS}	MIN	0.75			t _{CK} (avg)
		MAX	1.25			
DQS input high-level width	t _{DQSH} ³⁶⁾	MIN	0.4			t _{CK} (avg)
DQS input low-level width	t _{DQSL} ³⁶⁾	MIN	0.4			t _{CK} (avg)
DQS falling edge to CK setup time	t _{DSS}	MIN	0.2			t _{CK} (avg)
DQS falling edge hold time from CK	t _{DSH}	MIN	0.2			t _{CK} (avg)
Write preamble	t _{WPRE}	MIN	2.0			t _{CK} (avg)
0.5 tCK Write postamble	t _{WPST} ²¹⁾	MIN	0.5			t _{CK} (avg)
1.5 tCK Write postamble	t _{WPST} ²¹⁾	MIN	1.5			t _{CK} (avg)
ZQ Calibration Parameters						
ZQ Calibration	t _{ZQCAL}	MIN	1			us
ZQ Calibration Values Latch Time	t _{ZQLAT}	MN	Max (30ns, 8tCK)			ns
ZQ Calibration RESET time	t _{ZQRESET}	MIN	Max (50ns, 3tCK)			ns
Power Down Parameters						
CKE minimum pulse width (HIGH and LOW pulse width)	t _{CKE}	MIN	max(7.5ns, 4tCK)			-
Delay from Valid command to CKE Input low	t _{CMDCKE} ²⁶⁾	MIN	Max(1.75ns, 3tCK)			ns
Valid Clock Requirement after CKE Input Low	t _{CKELCK} ²⁶⁾	MIN	Max(5ns, 5tCK)			ns
Valid CS Requirement before CKE Input Low	t _{CSCKE}	MIN	1.75			ns
Valid CS Requirement after CKE Input Low	t _{CKELCS}	MIN	Max(5ns,5tCK)			ns

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[Table 62] LPDDR4X AC Timing Table

Parameter	Symbol	Min/ Max	LPDDR4X			Unit
			3200Mbps	3733Mbps	4266Mbps	
Valid Clock Requirement before CKE Input High	t _{CKCKEH} ²⁶⁾	MIN	Max(1.75ns, 3tCK)			-ns
Exit power- down to next valid command delay	t _{XP} ²⁶⁾	MIN	Max(7.5ns, 5tCK)			ns
Valid CS Requirement before CKE Input High	t _{CSCKEH}	MIN	1.75			ns
Valid CS Requirement after CKE Input High	t _{CKEHCS}	MIN	Max(7.5ns,5tCK)			ns
Valid Clock and CS Requirement after CKE Input low after MRW Command	t _{MRWCKEL} ²⁶⁾	MIN	Max(14ns,10tCK)			ns
Valid Clock and CS Requirement after CKE Input low after ZQ Calibration Start Command	t _{ZQCKE} ²⁶⁾	MIN	Max(1.75ns,3tCK)			ns
Command Address Input Parameters ⁴⁾						
Rx Mask voltage - p-p	VcIVW	MAX	155	150	145	mV
Rx timing window	TcIVW	MAX	0.3			UI*
CA AC input pulse amplitude pk-pk	VIHL_AC	MIN	190	180	180	mV
CA input pulse width	TcIPW	MIN	0.6			UI*
Input Slew Rate over VcIVW	SRIN_cIVW	MIN	1			V/ns
		MAX	7			
Mode Register Read/Write AC Timing						
Additional time after tXP has expired until MRR command	t _{MRRi}	MIN	t _{RCD} + 3nCK			-
MODE REGISTER READ command period	t _{MRR}	MIN	8			nCK
MODE REGISTER WRITE command period	t _{MRW}	MIN	Max(10ns, 10nCK)			-
Mode register set command delay	t _{MRD}	MIN	Max(14ns, 10tCK)			-
Boot Parameters (10 MHz - 55 MHz) ^{11), 12), 13)}						
Clock Cycle Time	t _{CKb}	max	100			ns
		MIN	18			
Address & Control Input Setup Time	t _{ISb}	MIN	1150			ps
Address & Control Input Hold Time	t _{IHb}	MIN	1150			ps
DQS Output Data Access Time from CK_t/CK_c	t _{DQSCb}	MIN	2.0			ns
		MAX	10.0			
Data Strobe Edge to Output Data Edge	t _{DQSQb}	MAX	1.2			ns
Command Bus Training AC Parameters						
Valid Clock Requirement after CKE Input low	t _{CKELCK}	MIN	Max(5ns, 5nCK)			tCK
Data Setup for VREF Training Mode	t _{DStrain}	MIN	2			ns
Data Hold for VREF Training Mode	t _{DHtrain}	MIN	2			ns
Asynchronous Data Read	t _{ADR}	MAX	20			ns
CA Bus Training command to CA Bus Training command delay	t _{CACD} ²⁹⁾	MIN	RU(t _{ADR} /t _{CK})			tCK
Valid Strobe Requirement before CKE Low	t _{DQSCKE} ³⁰⁾	MIN	10			ns
First CA Bus Training Command Following CKE LOW	t _{CAENT}	MIN	250			ns
VREF Step Time-multiple steps	t _{VREFCA_LONG}	MAX	250			ns
VREF Step Time-one step	t _{VREFCA_SHORT}	MAX	80			ns
Valid Clock Requirement before CS High	t _{CKPRECS}	MIN	2t _{CK} + t _{XP} (t _{XP} = max(7.5ns, 5nCK))			-
Valid Clock Requirement after CS High	t _{CKPSTCS}	MIN	max(7.5ns, 5nCK)			-
Minimum delay from CS to DQS toggle in command bus training	t _{CS_VREF}	MIN	2			tCK
Minimum delay from CKE High to Strobe High Impedance	t _{CKEHDQS}	-	10			ns
Valid Clock Requirement before CKE Input High	t _{CKCKEH}	MIN	Max(1.75ns, 3tCK)			
CA Bus Training CKE High to DQ Tri-state	t _{MRZ}	MIN	1.5			ns

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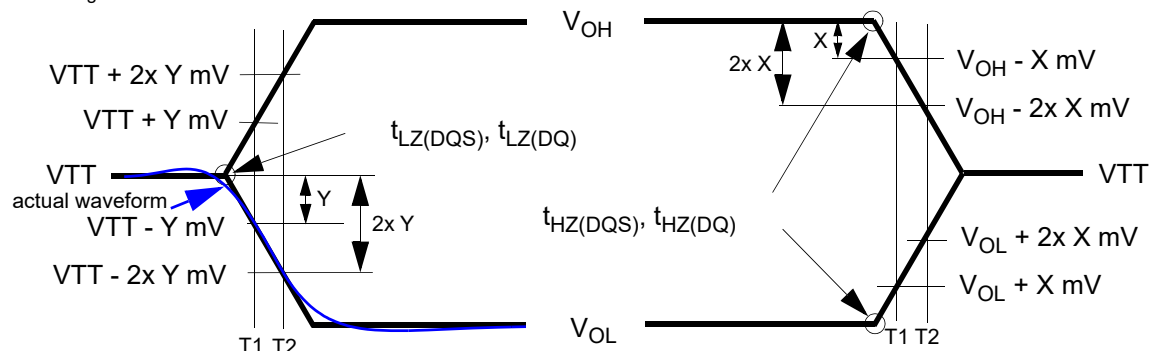
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[Table 62] LPDDR4X AC Timing Table

Parameter	Symbol	Min/ Max	LPDDR4X			Unit
			3200Mbps	3733Mbps	4266Mbps	
ODT turn-on Latency from CKE	t _{CKELODOn}	MIN	20			ns
ODT turn-off Latency from CKE	t _{CKELODOff}	MIN	20			ns
Exit Command Bus Training Mode to next valid command delay 33)	t _{XCBT_Short}	MIN	Max(5nCK, 200ns)			-
	t _{XCBT_Middle}	MIN	Max(5nCK, 200ns)			-
	t _{XCBT_Long}	MIN	Max(5nCK, 250ns)			-
Write Leveling Parameters						
DQS_t/DQS_c delay after write leveling mode is programmed	t _{WLDQSEN}	MIN	20			tCK
Write preamble for Write Leveling	t _{WLWPRE}	MIN	20			tCK
First DQS_t/DQS_c edge after write leveling mode is programmed	t _{WLMRD}	MIN	40			tCK
Write leveling output delay	t _{WLO}	MAX	20			ns
Mode register set command delay	t _{MRD}	MIN	Max(14ns, 10tCK)			ns
Valid Clock Requirement before DQS Toggle	t _{CKPRDQS}	MIN	Max(7.5ns, 4tCK)			-
Valid Clock Requirement after DQS Toggle	t _{CKPSTDQS}	MIN	Max(7.5ns, 4tCK)			-
Write leveling hold time	t _{WLH} ²⁷⁾	MIN	75	60	50	ps
Write leveling setup time	t _{WLS} ²⁷⁾	MIN	75	60	50	ps
Write leveling input valid window	t _{WLIVW} ²⁸⁾	MIN	120	100	90	ps
Temperature De-Rating AC Timing ²⁰⁾						
DQS output access time from CK_t/CK_c (derated)	t _{DQSK}	MAX	3600			ps
RAS-to-CAS delay (derated)	t _{RCD}	MIN	t _{RCD} + 1.875			ns
ACTIVATE-to- ACTIVATE command period (derated)	t _{RC}	MIN	t _{RC} + 3.75			ns
Row active time (derated)	t _{RAS}	MIN	t _{RAS} + 1.875			ns
Row precharge time (derated)	t _{RP}	MIN	t _{RP} + 1.875			ns
Active bank A to active bank B (derated)	t _{RRD}	MIN	t _{RRD} + 1.875			ns

NOTE :

- Frequency values are for reference only. Clock cycle time (tCK) is used to determine device capabilities.
- All AC timings assume an input slew rate of TBDV/ns.
- Measured with 4 V/ns differential CK_ t/CK_ c slew rate and nominal VIX.
- READ, WRITE, and Input setup and hold values are referenced to V_{REF} .
- For LOW-to-HIGH and HIGH-to-LOW transitions, the timing reference is at the point when the signal crosses the transition threshold (V_{TT}). tHZ and tLZ transitions occur in the same access time (with respect to clock) as valid data transitions. These parameters are not referenced to a specific voltage level but to the time when the device output is no longer driving (for tRPST, tHZ(DQS) and tHZ(DQ)), or begins driving (for tRPRE, tLZ(DQS), tLZ(DQ)). Operating and Timing [Burst Read:RL=12, BL=8, tDQSK<tCK] shows a method to calculate the point when device is no longer driving tHZ(DQS) and tHZ(DQ), or begins driving tLZ(DQS), tLZ(DQ) by measuring the signal at two different voltages. The actual voltage measurement points are not critical as long as the calculation is consistent.
- Output Transition Timing

Start driving point = $2 \times T1 - T2$ End driving point = $2 \times T1 - T2$

- The parameters tLZ(DQS), tLZ(DQ), tHZ(DQS), and tHZ(DQ) are defined as single-ended. The timing parameters tRPRE and tRPST are determined from the differential signal DQS_ t-DQS_ c.
- Measured from the point when DQS_ t/DQS_ c begins driving the signal to the point when DQS_ t/DQS_ c begins driving the first rising strobe edge. See Pre and Post-amble section in Operating & Timing spec
- Measured from the last falling strobe edge of DQS_ t/DQS_ c to the point when DQS_ t/DQS_ c finishes driving the signal.
- Input set-up/hold time for signal (CA[9:0], CS).
- To ensure device operation before the device is configured, a number of AC boot-timing parameters are defined in this table. Boot parameter symbols have the letter b

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- appended (for example, tCK during boot is tCKb).
- 12) The LPDDR4X device will set some default values upon receiving a RESET (MRW) command as specified in "Definition".
 - 13) The output skew parameters are measured with default output impedance settings using the reference load.
 - 14) Includes DRAM process, voltage and temperature variation. It includes the AC noise impact for frequencies > 20 MHz and max voltage of 45 mV pk-pk from DC-20 MHz at a fixed temperature on the package. The voltage supply noise must comply to the component Min-Max DC Operating conditions.
 - 15) tDQCKCK_temp max delay variation as a function of Temperature.
 - 16) tDQCKCK_volt max delay variation as a function of DC voltage variation for VDDQ and VDD2. tDQCKCK_volt should be used to calculate timing variation due to VDDQ and VDD2 noise < 20 MHz. Host controller do not need to account for any variation due to VDDQ and VDD2 noise > 20 MHz. The voltage supply noise must comply to the component Min-Max DC Operating conditions. The voltage variation is defined as the $\text{Max}\{|\text{abs}\{tDQCKCK_{min}@V1 - tDQCKCK_{max}@V2\}|, |\text{abs}\{tDQCKCK_{max}@V1 - tDQCKCK_{min}@V2\}|\}$.
 - 17) Precharge to precharge timing restriction does not apply to Auto-Precharge commands.
 - 18) tXSR/tXP/tZQLAT are defined as "to the first rising clock edge next valid command".
 - 19) Refresh Rate is specified by MR4, OP[2:0].
 - 20) Timing derating applies for operation at 85°C to 105°C.
 - 21) The length of Write Postamble depends on MR3 OP1 setting.
 - 22) The same voltage and temperature are applied to tDQCKCK_rank2rank.
 - 23) tDQCKCK_rank2rank parameter is applied to multi-ranks per byte lane within a package consisting of the same design dies.
 - 24) Delay time has to satisfy both analog time(ns) and clock count(tCK). It means that tESCKE will not expire until CK has toggled through at least 3 full cycles (3 * tCK) and 1.75ns has transpired. The case which 3tCK is applied to is shown below.

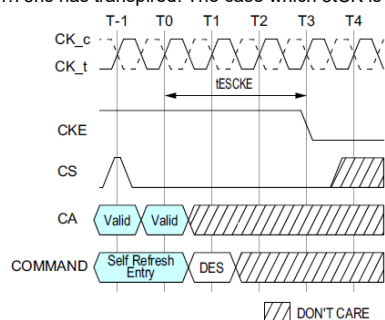


Figure 23. tESCKE Timing

- 25) MRR-1, CAS-2, DES, MPC, MRW-1 and MRW-2 except PASR Bank/Segment setting are only allowed during this period.
- 26) Delay time has to satisfy both analog time(ns) and clock count(nCK). For example, tCMDCKE will not expire until CK has toggled through at least 3 full cycles (3 * tCK) and 1.75ns has transpired. The case which 3nCK is applied to is shown below.

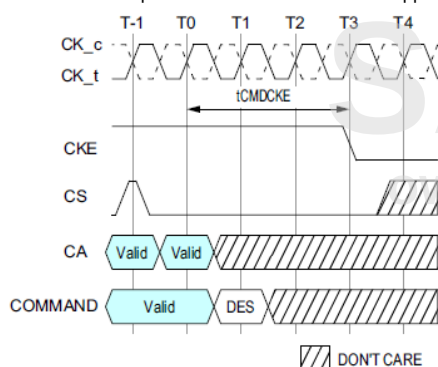


Figure 24. tCMDCKE Timing

- 27) In addition to the traditional setup and hold time specifications above, there is value in a input valid window based specification for write-leveling training. As the training is based on each device, worst case process skews for setup and hold do not make sense to close timing between CK and DQS.
- 28) tWLIVW is defined in a similar manner to tDIVW_Total, except that here it is a DQS input valid window with respect to CK. This would need to account for all VT (voltage and temperature) drift terms between CK and DQS within the DRAM that affect the write-leveling input valid window. The DQS input mask for timing with respect to CK is shown in Figure 25. The "total" mask (tWLIVW) defines the time the input signal must not encroach in order for the DQS input to be successfully captured by CK with a BER of lower than tbd. The mask is a receiver property and it is not the valid data-eye.

DQS_t/DQS_c and CK_t/CK_c at DRAM Latch

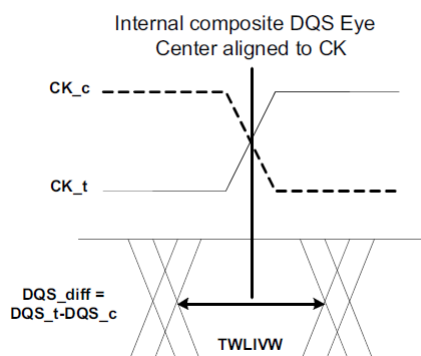


Figure 25. DQS_t/DQS_c to CK_t/CK_c timings at the DRAM pins referenced from the internal latch

- 29) If tACD is violated, the data for samples which violate tACD will not be available, except for the last sample (where tACD after this sample is met). Valid data for the last sample will be available after tADR.
- 30) DQS_t has to retain a low level during tDQCKE period, as well as DQS_c has to retain a high level.
- 31) See Masked Write Operation for detail.

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- 32) Devices supporting 4266Mbps specification shall support these timings at lower data rates.
- 33) Exit Command Bus Training Mode to next valid command delay Time depends on value of $V_{REF}(CA)$ setting: MR12 OP[5:0] and $V_{REF}(CA)$ Range: MR12 OP[6] of FSP-OP 0 and 1. The details are shown in tFC value mapping table. Additionally exit Command Bus Training Mode to next valid command delay Time may affect $V_{REF}(DQ)$ setting. Settling time of $V_{REF}(DQ)$ level is same as $V_{REF}(CA)$ level.
- 34) Precharge to precharge timing restriction does not apply to Auto-Precharge commands.
- 35) This value is also valid for 1600Mbps Single-Ended CK ODT-off case.
- 36) This value is also valid for 1600Mbps Single-Ended WDQS ODT-off case.

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10.5 CA Rx Voltage and Timing

The command and address(CA) including CS input receiver compliance mask for voltage and timing is shown in the figure below. All CA, CS signals apply the same compliance mask and operate in single data rate mode.

The CA input receiver mask for voltage and timing is shown in the figure below is applied across all CA pins. The receiver mask (Rx Mask) defines the area that the input signal must not encroach in order for the DRAM input receiver to be expected to be able to successfully capture a valid input signal; it is not the valid data-eye.

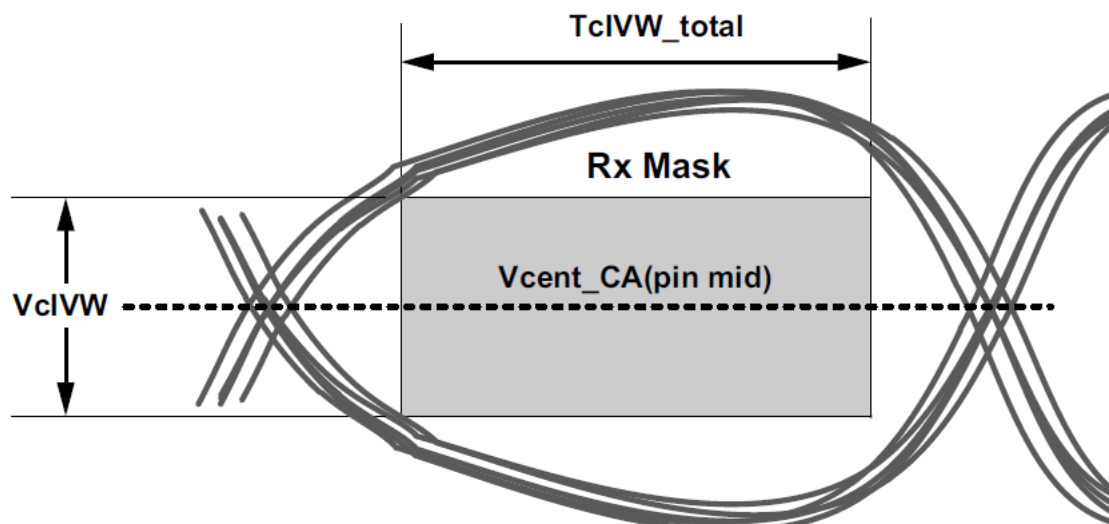


Figure 26. CA Receiver (Rx) mask

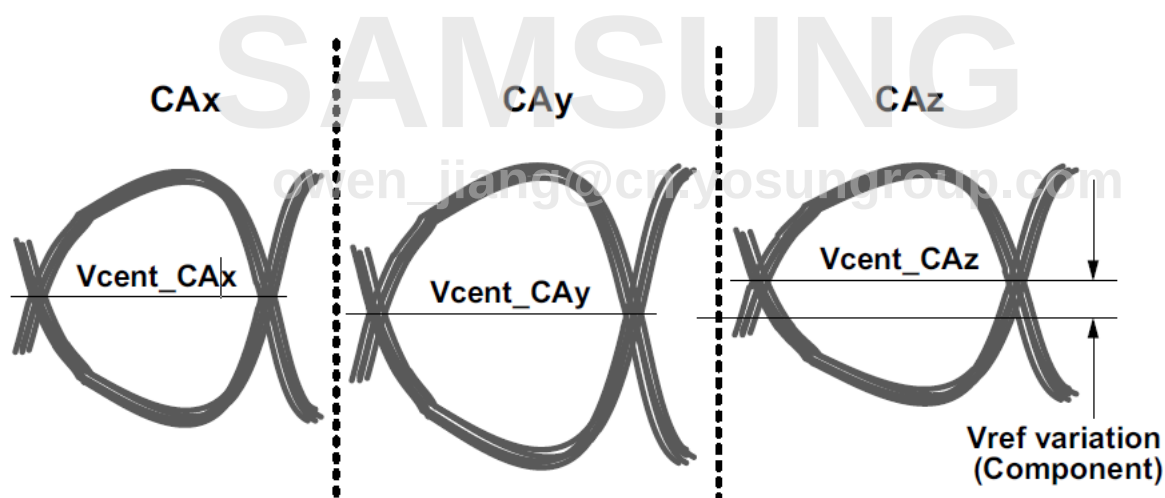


Figure 27. Across pin V_{REFCA} voltage variation

$V_{cent_CA}(\text{pin avg})$ is defined as the midpoint between the largest V_{cent_CA} voltage level and the smallest V_{cent_CA} voltage level across all CA and CS pins for a given DRAM component. Each CA V_{cent} level is defined by the center, i.e. widest opening, of the cumulative data input eye as depicted in Figure 27. This clarifies that any DRAM component level variation must be accounted for within the DRAM CA Rx mask. The component level V_{ref} will be set by the system to account for R_{on} and ODT settings.

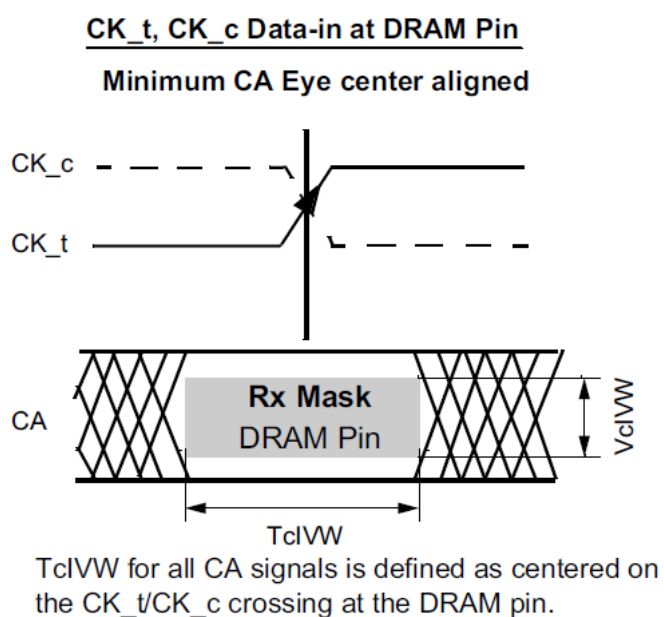


Figure 28. CA Timings at the DRAM pins

All of the timing terms in Figure 28. are measured from the CK_t/CK_c to the center(midpoint) of the TcIVW window taken at the VcIVW_total voltage levels centered around Vcent_CA(pin mid).

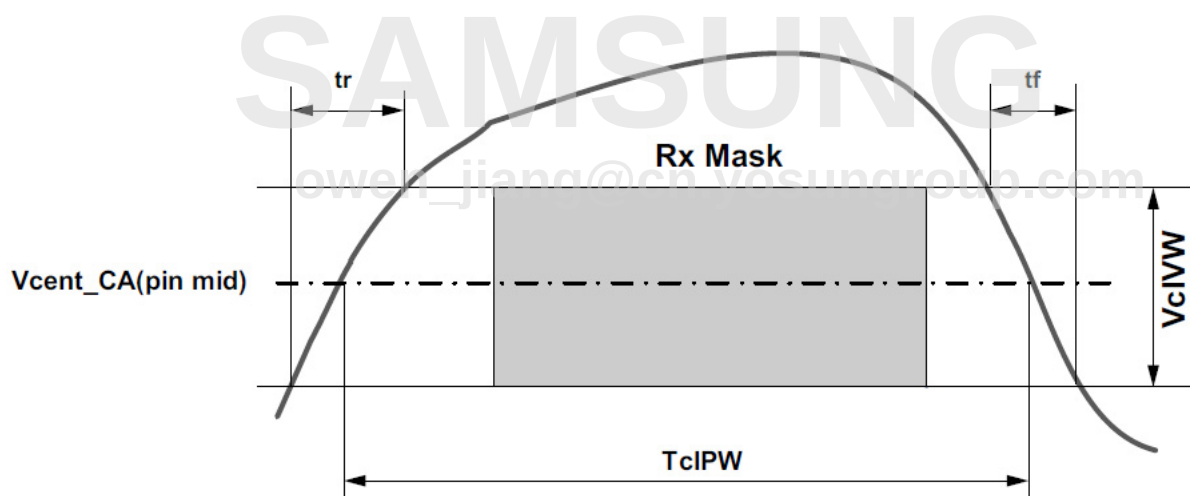


Figure 29. CA TcIPW and SRIN_cIVW definition (for each input pulse)

NOTE :

1) $SRIN_cIVW = VcIVW_Total / (tr \text{ or } tf)$, signal must be monotonic within tr and tf range.

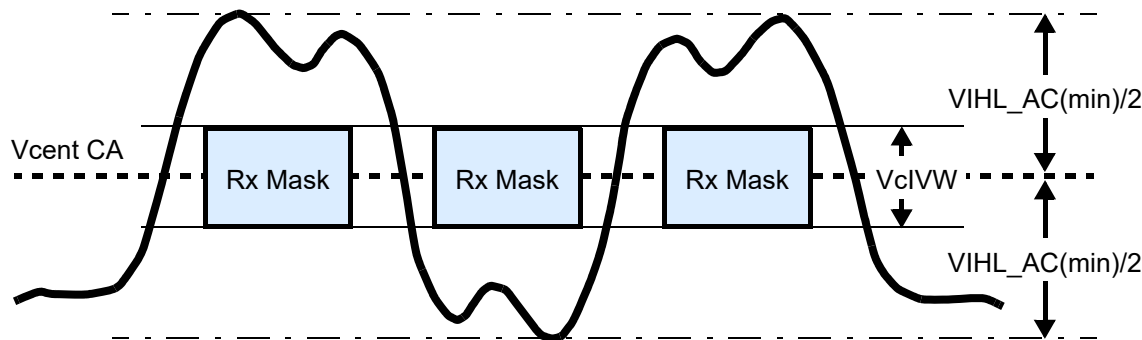


Figure 30. CA VIH_L_AC definition (for each input pulse)

[Table 63] DRAM CMD/ADR, CS

Symbol	Parameter	DQ-1333 ^{A)}		DQ-1600				DQ-1866		DQ-3200		DQ-3733		DQ-4266		Unit	NOTE
				Clock SE ⁸⁾		Clock Diff. ⁹⁾											
		min	max	min	max	min	max	min	max	min	max	min	max				
VcIVW	Rx Mask voltage - p-p	-	175	-	220	-	175	-	175	-	155	-	150	-	145	mV	1,2,3
TcIVW	Rx timing window	-	0.3	-	0.35	-	0.3	-	0.3	-	0.3	-	0.3	-	0.3	UI*	1,2,3
VIHL_AC	CA AC input pulse amplitude pk-pk	210	-	210	-	210	-	210	-	190	-	180	-	180	-	mV	4,7
TcIPW	CA input pulse width	0.55	-	0.55	-	055	-	0.55	-	0.6	-	0.6	-	0.6	-	UI*	5
SRIN_cIVW	Input Slew Rate over VcIVW	1	7	1	7	1	7	1	7	1	7	1	7	1	7	V/ns	6

* UI=tCK(avg)min

NOTE :

- 1) CA Rx mask voltage and timing parameters at the pin including voltage and temperature drift.
- 2) Rx mask voltage VcIVW total(max) must be centered around Vcent_CA (pin_mid).
- 3) Vcent_CA must be within the adjustment range of the CA internal Vref.
- 4) CA only input pulse signal amplitude into the receiver must meet or exceed VIH_L_AC at any point over the total UI. No timing requirement above level. VIH_L_AC is the peak to peak voltage centered around Vcent_CA(pin_mid) such that VIH_L_AC/2 min must be met both above and below Vcent_CA.
- 5) CA only minimum input pulse width defined at the Vcent_CA (pin_mid).
- 6) Input slew rate over VcIVW Mask centered at Vcent_CA (pin_mid).
- 7) VIH_L_AC does not have to be met when no transitions are occurring.
- 8) MR51 OP[3] = 1B: Single ended Clock, Only CK_t.
- 9) MR51 OP[3] = 0B: Differential Clock (Default), CK_t/CK_c.

A) The following Rx voltage and absolute timing requirements apply for DQ operating frequencies at or below 1333 for all speed bins. For example the TcIVW(ps) = 450ps at or below 1333 operating frequencies.

10.6 DRAM Data Timing

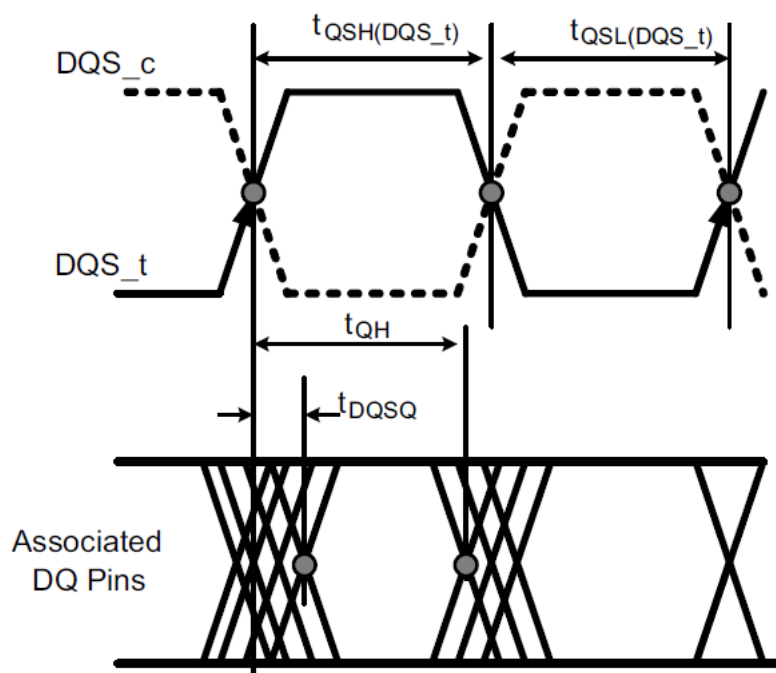
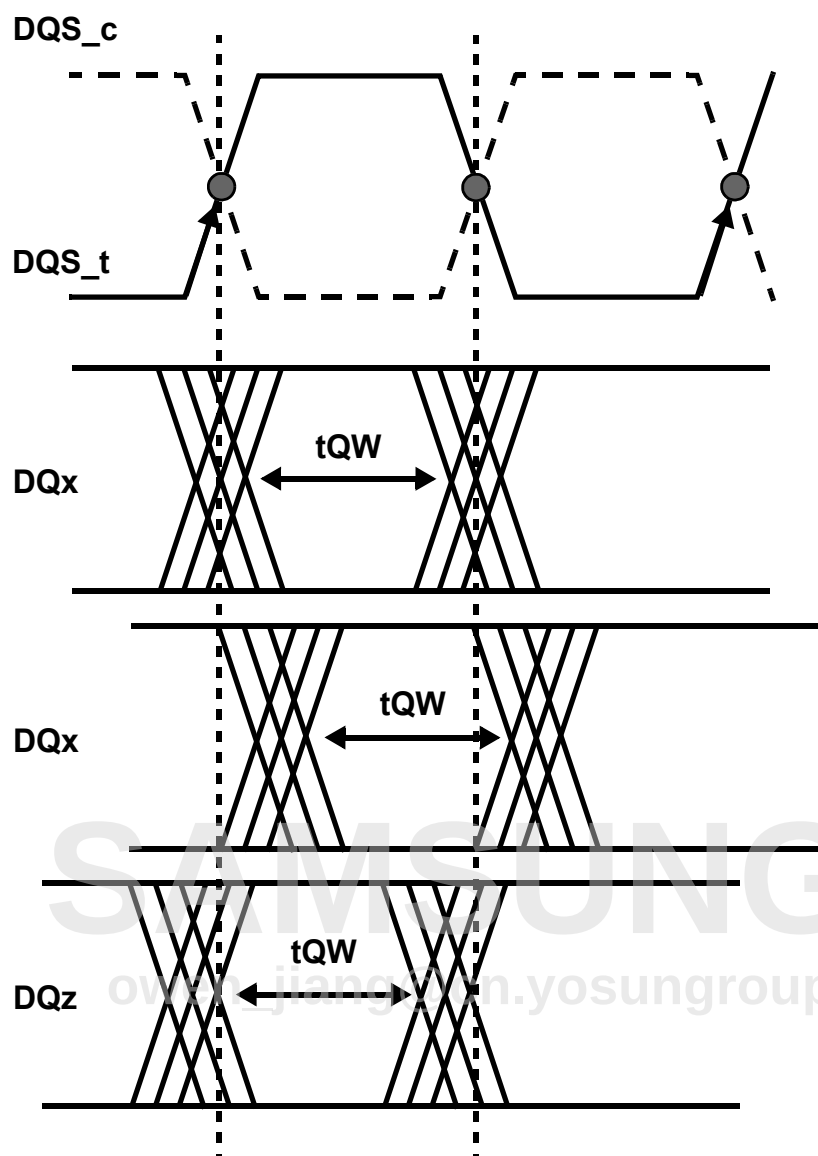


Figure 31. Read data timing definitions t_{QH} and t_{DQSQ} across on DQ signals per DQS group

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Figure 32. Read data timing t_{QW} valid window defined per DQ signal

[Table 64] Read output timings

Parameter	Symbol	LPDDR4X-1600/1866		LPDDR4X-2133/2400		LPDDR4X-3200		LPDDR4X-3733		LPDDR4X-4266		Units	Notes
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
Data Timing													
DQS_t, DQS_c to DQ Skew total, per group, per access (DBI-Disabled)	t _{DQSQ}	-	0.18	-	0.18	-	0.18	-	0.18	-	0.18	UI	
DQ output hold time total from DQS_t, DQS_c (DBI-Disabled)	t _{QH}	min(t _{QS_H} , t _{QSL})	-	min(t _{QS_H} , t _{QSL})	-	min(t _{QS_H} , t _{QSL})	-	min(t _{QS_H} , t _{QSL})	-	min(t _{QS_H} , t _{QSL})	-	UI	
DQ output window time total, per pin (DBI-Disabled)	t _{QW_{total}}	0.75	-	0.73	-	0.7	-	0.7	-	0.7	-	UI	3
DQ output window time deterministic, per pin (DBI-Disabled)	t _{QW_{dj}}	-	TBD	-	TBD	-	TBD	-	TBD	-	TBD	UI	2,3
DQS_t, DQS_c to DQ Skew total, per group, per access (DBI-Enabled)	t _{DQSQ_{_D_{BI}}}	-	0.18	-	0.18	-	0.18	-	0.18	-	0.18	UI	
DQ output hold time total from DQS_t, DQS_c (DBI-Enabled)	t _{QH_{_DBI}}	min(t _{QSH_{_DBI}} , t _{QSL_{_DBI}})	-	min(t _{QSH_{_DBI}} , t _{QSL_{_DBI}})	-	min(t _{QSH_{_DBI}} , t _{QSL_{_DBI}})	-	min(t _{QSH_{_DBI}} , t _{QSL_{_DBI}})	-	min(t _{QSH_{_DBI}} , t _{QSL_{_DBI}})	-	UI	
DQ output window time total, per pin (DBI-Enabled)	t _{QW_{total}_{_DBI}}	0.75	-	0.73	-	0.7	-	0.7	-	0.7	-	UI	3
Data Strobe Timing													
DQS_t, DQS_c differential output low time (DBI-Disabled)	t _{QSL}	t _{CL(abs)} -0.05	-	t _{CL(abs)} -0.05	-	t _{CL(abs)} -0.05	-	t _{CL(abs)} -0.05	-	t _{CL(abs)} -0.05	-	t _{CK(avg)}	3,4
DQS_t, DQS_c differential output high time (DBI-Disabled)	t _{QSH}	t _{CH(abs)} -0.05	-	t _{CH(abs)} -0.05	-	t _{CH(abs)} -0.05	-	t _{CH(abs)} -0.05	-	t _{CH(abs)} -0.05	-	t _{CK(avg)}	3,5
DQS_t, DQS_c differential output low time (DBI-Enabled)	t _{QSL_{_DBI}}	t _{CL(abs)} -0.045	-	t _{CL(abs)} -0.045	-	t _{CL(abs)} -0.045	-	t _{CL(abs)} -0.045	-	t _{CL(abs)} -0.045	-	t _{CK(avg)}	4,6
DQS_t, DQS_c differential output high time (DBI-Enabled)	t _{QSH_{_DBI}}	t _{CH(abs)} -0.045	-	t _{CH(abs)} -0.045	-	t _{CH(abs)} -0.045	-	t _{CH(abs)} -0.045	-	t _{CH(abs)} -0.045	-	t _{CK(avg)}	5,6

*Unit UI = t_{CK(avg)}/min/2**NOTE :**

- 1) The deterministic component of the total timing. Measurement method tbd.
- 2) This parameter will be characterized and guaranteed by design.
- 3) This parameter is function of input clock jitter. These values assume the min t_{CH(abs)} and t_{CL(abs)}. When the input clock jitter min t_{CH(abs)} and t_{CL(abs)} is 0.44 or greater of t_{CK(avg)} the min value of t_{QSL} will be t_{CL(abs)}-0.04 and t_{QSH} will be t_{CH(abs)}-0.04.
- 4) t_{QSL} describes the instantaneous differential output low pulse width on DQS_t - DQS_c, as it measured the next rising edge from an arbitrary falling edge.
- 5) t_{QSH} describes the instantaneous differential output high pulse width on DQS_t - DQS_c, as it measured the next rising edge from an arbitrary falling edge.
- 6) This parameter is function of input clock jitter. These values assume the min t_{CH(abs)} and t_{CL(abs)}. When the input clock jitter min t_{CH(abs)} and t_{CL(abs)} is 0.44 or greater of t_{CK(avg)} the min value of t_{QSL} will be t_{CL(abs)}-0.04 and t_{QSH} will be t_{CH(abs)}-0.04.

10.7 DQ Rx Voltage And Timing

The DQ input receiver mask for voltage and timing is shown Figure 33. is applied per pin. The “total” mask (V_{dIVW_total} , T_{dIVW_total}) defines the area the input signal must not encroach in order for the DQ input receiver to successfully capture an input signal with a BER of lower than tbd. The mask is a receiver property and it is not the valid data-eye.

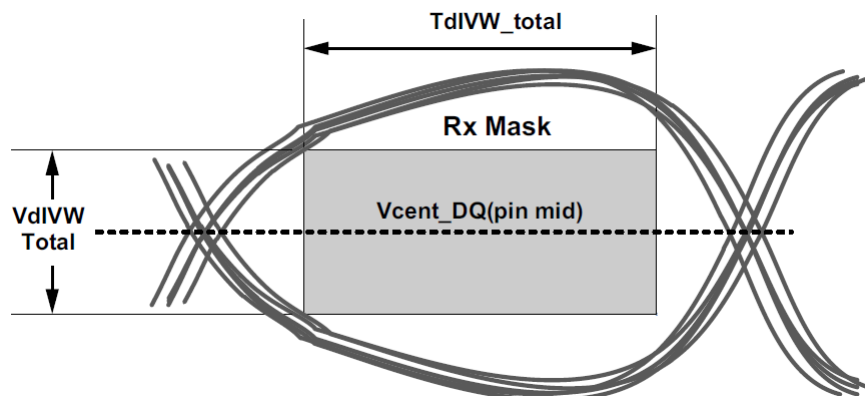


Figure 33. DQ Receiver (Rx) mask

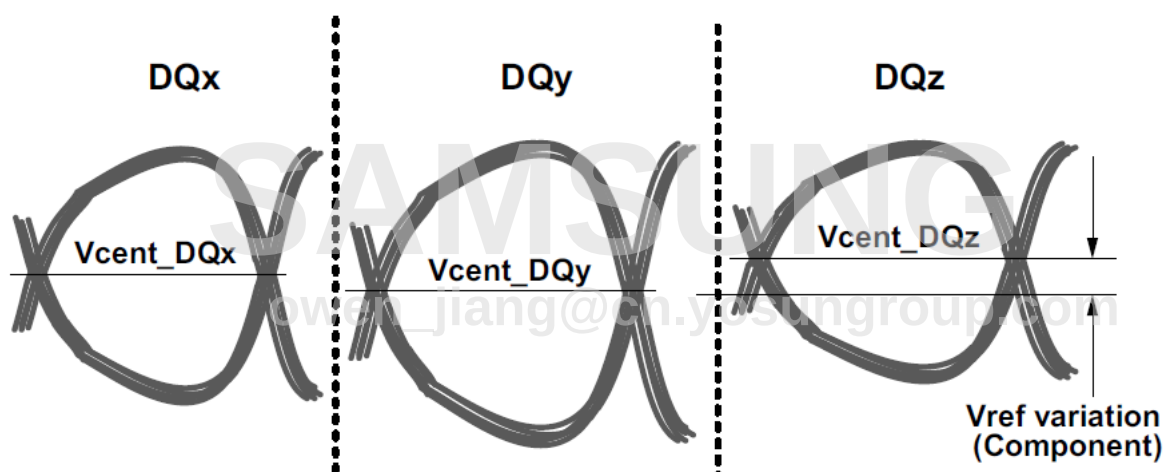


Figure 34. Across pin V_{REFDQ} voltage variation

$V_{cent_DQ(pin\ mid)}$ is defined as the midpoint between the largest V_{cent_DQ} voltage level and the smallest V_{cent_DQ} voltage level across all DQ pins for a given DRAM component. Each DQ V_{cent} is defined by the center, i.e., widest opening, of the cumulative data input eye as depicted in Figure 34. This clarifies that any DRAM component level variation must be accounted for within the DRAM Rx mask. The component level V_{ref} will be set by the system to account for R_{on} and ODT settings.

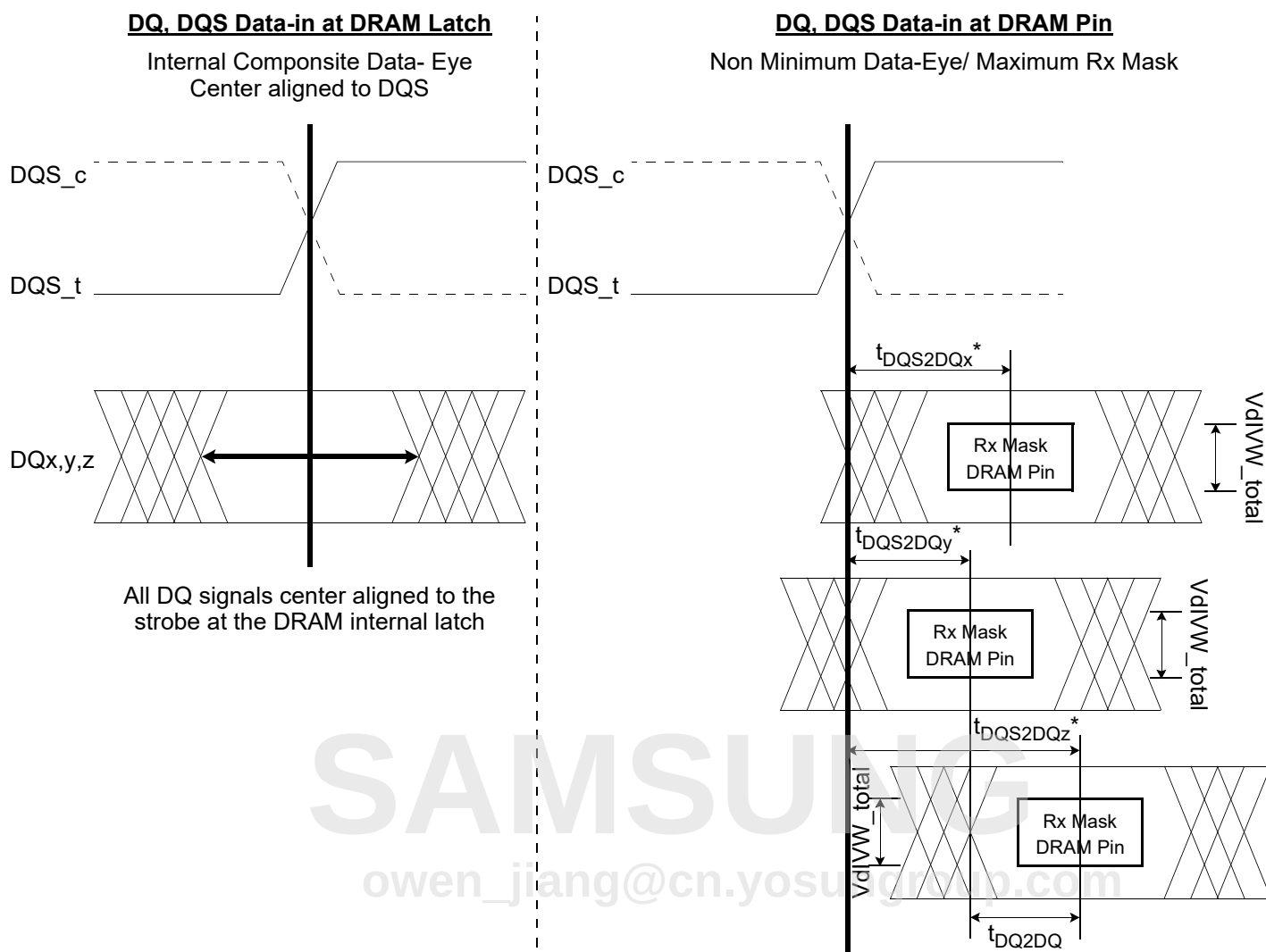


Figure 35. DQ to DQS t_{DQS2DQ} & t_{DQDQ} Timings at the DRAM pins referenced from the internal latch

NOTE :

- 1) t_{DQS2DQ} is measured at the center(midpoint) of the TdVW window.
- 2) DQz represents the max t_{DQS2DQ} in this example.
- 3) DQy represents the min t_{DQS2DQ} in this example.

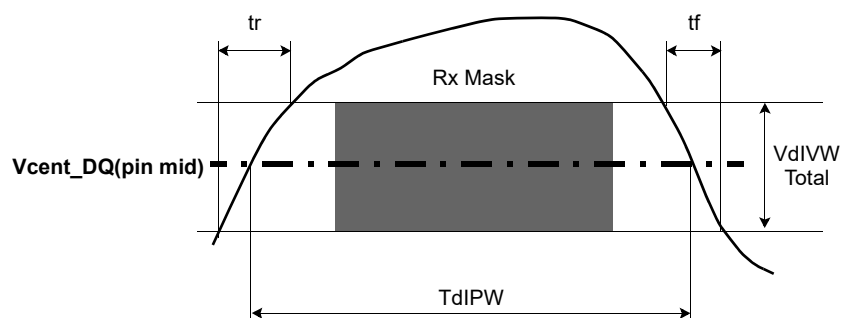


Figure 36. DQ TdIPW and SRIN_dIVW definition (for each input pulse)

NOTE :

1) $SRIN_dIVW = VdIVW_Total / (tr \text{ or } tf)$, signal must be monotonic within tr and tf range.

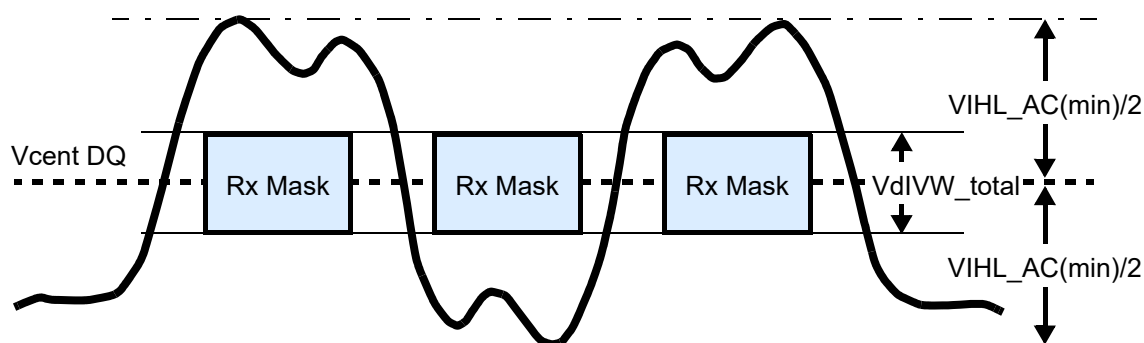


Figure 37. DQ VIH_L_AC definition (for each input pulse)

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[Table 65] DRAM DQs In Receive Mode;

Symbol	Parameter	1600 ^{A)}				1866/2133/ 2400		3200		3733		4266		Unit	NOTE
		WDQS SE ¹⁷⁾		WDQS Diff ¹⁸⁾											
		min	max	min	max	min	max	min	max	min	max	min	max		
VdIVW_total	Rx Mask voltage - p-p total	-	180	-	140	-	140	-	140	-	130	-	120	mV	1,2,3,4
TdIVW_total	Rx timing window total (At VdIVW voltage levels)	-	0.35	-	0.22	-	0.22	-	0.25	-	0.25	-	0.25	UI*	1,2,4
TdIVW_1bit	Rx timing window 1 bit toggle (At VdIVW voltage levels)	-	TBD	-	TBD	-	TBD	-	TBD	-	TBD	-	TBD	UI*	1,2,4,12
VIHL_AC	DQ AC input pulse amplitude pk-pk	180	-	180	-	180	-	180	-	180	-	170	-	mV	5,13
TdIPW DQ	Input pulse width (At Vcent_DQ)	0.45	-	0.45	-	0.45	-	0.45	-	0.45	-	0.45	-	UI*	6
t _{DQS2DQ}	DQ to DQS offset	250	700	250	700	250	700	250	700	250	700	250	700	ps	7
t _{DQ2DQ}	DQ to DQ offset	-	30	-	30	-	30	-	30	-	30	-	30	ps	8
t _{DQS2DQ_temp}	DQ to DQS offset temperature variation	-	0.4	-	0.4	-	0.4	-	0.4	-	0.4	-	0.4	ps/°C	9
t _{DQS2DQ_volt}	DQ to DQS offset voltage variation	-	25	-	25	-	25	-	25	-	25	-	25	ps/ 50mV	10
SRIN_dIVW	Input Slew Rate over VdIVW_total	1	7	1	7	1	7	1	7	1	7	1	7	V/ns	11
t _{DQS2DQ_rank2rank}	DQ to DQS offset rank to rank variation	-	200	-	200	-	200	-	200	-	200	-	200	ps	14,15,16

* UI=tck(avg)/min/2

NOTE :

- 1) Data Rx mask voltage and timing parameters are applied per pin and includes the DRAM DQ to DQS voltage AC noise impact for frequencies >20MHz and max voltage of 45mV pk-pk from DC-20MHz at a fixed temperature on the package. The voltage supply noise must comply to the component Min-Max DC operating conditions.
- 2) The design specification is a BER <td. The BER will be characterized and extrapolated if necessary using a dual dirac method.
- 3) Rx mask voltage VdIVW total(max) must be centered around Vcent_DQ(pin_mid).
- 4) Vcent_DQ must be within the adjustment range of the DQ internal Vref.
- 5) DQ only input pulse amplitude into the receiver must meet or exceed VIHL AC at any point over the total UI. No timing requirement above level. VIHL AC is the peak to peak voltage centered around Vcent_DQ(pin_mid) such that VIHL_AC/2 min must be met both above and below Vcent_DQ.
- 6) DQ only minimum input pulse width defined at the Vcent_DQ(pin_mid).
- 7) DQ to DQS offset is within byte from DRAM pin to DRAM internal latch. Includes all DRAM process, voltage and temperature variation.
- 8) DQ to DQ offset defined within byte from DRAM pin to DRAM internal latch for a given component.
- 9) TDQS2DQ max delay variation as a function of temperature.
- 10) TDQS2DQ max delay variation as a function of the DC voltage variation for VDDQ and VDD2. It includes the VDDQ and VDD2 AC noise impact for frequencies > 20MHz and max voltage of 45mV pk-pk from DC-20MHz at a fixed temperature on the package.
- 11) Input slew rate over VdIVW Mask centered at Vcent_DQ(pin_mid).
- 12) Rx mask defined for a one pin toggling with other DQ signals in a steady state.
- 13) VIHL_AC does not have to be met when no transitions are occurring.
- 14) The same voltage and temperature are applied to t_{DQS2DQ_rank2rank}.
- 15) t_{DQS2DQ_rank2rank} parameter is applied to multi-ranks per byte lane within a package consisting of the same design dies.
- 16) t_{DQS2DQ_rank2rank} support was added to JESD209-4B, some older devices designed to support JESD209-4 and JESD209-4A may not support this parameter. Refer to vendor datasheet.
- 17) MR51 OP[2]=1B : Single ended Write DQS.
- 18) MR51 OP[2]=0B : Differential Write DQS.

A) The Rx voltage and absolute timing requirements apply for all DQ operating frequencies at or below 1600 for all speed bins. For example TdIVW_total(ps) = 137.5ps at or below 1600 operating frequencies.